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Design and development of the control PCB for a drone with a study of potential space applications

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ABSTRACT

This project focuses on the development of a dedicated flight control PCB specifically optimized for multirotor systems.

The TFM begins with the design of a customized hardware platform using Altium Designer, WEBENCH and STM32CubeIDE. It explores design capabilities and alternatives by comparing against commercial and hobbyist flight controllers. The project then aims to perform PCB layer stack-up, impedance control, and routing strategies to ensure signal integrity, with the objective of generating industry-standard fabrication files.

Following the hardware development phase, the project moves into software integration. This consists of configuring the hardware and selecting pinouts to support the ChibiOS. Allowing the porting and execution of the ArduPilot firmware onto the custom hardware, using STM32CubeProgrammer for deployment.

Finally, the project evaluates the developed PCB components, with attention to the steps that could make the hardware suitable for space applications based on current space standards. This involves adaptations such as radiation tolerance, thermal management, and the use of components certified for operation in harsh environments.

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INTRODUCTION

Since the appearance of the first unmanned vehicle in 1849 in the form of a balloon and its formal introduction in conflicts between 1917-1942, drones have changed the paradigm of interacting with the environment indirectly.

During this early start of the XXI century, drones have taken a very important role in society in multiple aspects. Such as the transport of goods, reconnaissance activities, mapping, firefighting and more. The development of IoT and cheap electronics impulses and leads drones into a promising insight of a better future.

Motivation and objectives

This TFM is devoted to learning, exploring and expanding knowledge of the basic components essential for creating autonomous devices.

Control and power distribution boards are critical subsystems in unmanned vehicles. This project focuses on the complete design process of a dedicated drone control PCB. Furthermore, it includes a study of potential space applications and their specific technical requirements.

The **principal objectives** of this TFM are:

1. Develop a functional drone control PCB.
2. Create the drivers to manage the drone control PCB with ArduPilot.
3. Evaluate the current design and expose space requirements.

Structure

The project is composed of 5 chapters and 3 annexes:

- **Chapters:**
 - **Chapter 1 and 2:** Hardware design and routing of a flight controller.
 - **Chapter 3:** ArduPilot software porting into designed hardware.
 - **Chapter 4:** Space applications and hardware analysis.
 - **Chapter 5:** Tests, issues, cost, environmental and social impact.
 - **Conclusions.**
- **Annexes:**
 - **Annex-1:** Basic PCB analysis of flight controller components and several market flight controllers.
 - **Annex-2:** Space environment and electronics space standards overview.
 - **Annex-3:** Testing, project issues, comparison and cost.

CHAPTER 1. PCB SCHEMATIC DESIGN

This chapter details the design of the PCB schematics. It begins with an introduction to the requirements and applicable standards, followed by a brief explanation of the composition of each system, balancing reliability, cost and performance.

1.1. Introduction to the basics of the architecture of a flight controller

The flight controller acts as the brain of the aircraft, executing a continuous feedback loop to maintain stability and follow pilot commands. The MCU (Microcontroller Unit) serves as the central processor, managing real-time data from an IMU (Inertial Measurement Unit), which typically combines an accelerometer and a gyroscope, to determine the drone's attitude. To ensure precise flight, the architecture includes barometers for altitude hold and various voltage regulators to provide clean, stable power to sensitive electronics. Furthermore, the flight controller can integrate storage units and specialized communication buses such as UART, CAN, SPI and I2C, to interact with external peripherals like GPS, ESCs, and receivers. This forms a cohesive ecosystem that acquires data, computes control corrections, and generates precise actuator commands to achieve physical movement.

Referencing ANNEX-1 is highly recommended to understand the electronic components used in current hobbyist and commercial drone control boards, as well as their specific manufacturing requirements.

1.2. Pixhawk Standards and Software

1.2.1. Pixhawk Standards

There is a wide variety of drone controller boards and systems on the market. While the vast majority do not follow any specific standards, professional grade controllers typically follow established conventions.

One of the most famous standards for professional open source controllers is the Pixhawk standards [1]. In fact, Pixhawk is known by the flight controllers open-source family based on Pixhawk Standard. These flight controllers can be programmed with PX4 and ArduPilot operating firmwares.

Regarding Pixhawk Standards, there are many varieties, but these are divided into:

- **Pixhawk Autopilot Reference Standards:**
 - **Autopilot v6x Standard (FMUv6):** Describes the hardware architecture requirements. FMU stands for Flight Management Unit.
 - **Autopilot v5x Standard (FMUv5)**
 - **Autopilot v6C Standard (FMUv6)**
 - **Autopilot v6U Standard (FMUv6)**
 - **Autopilot Bus Standard:** Addresses bus protection and configuration, specifically between PCBs and peripherals.

- **Autopilot Connector Standard:** Defines the hardware and pin selection required to provide connections, mainly to peripherals.
- **Pixhawk Payload Bus Standards:**
 - This standard covers design considerations, as well as software and hardware interfaces for cameras, gimbals, actuators and general payloads.
- **Pixhawk Smart Battery Standard** (Coming soon)

Standard autopilot architectures typically use a main processor for computation and a coprocessor for I/O, sometimes involving up to three processor units. This type of configuration is too complex for the current project. Therefore, the developed PCB uses a single high performance processor, following the Autopilot Connector-Bus Standard.

1.2.2. Software

Drone firmware can be present in multiple forms. Generally most flight controllers can be programmed with open source firmwares, with common C language, Arduino or HAL (Hardware Abstraction Layer). Some developers even choose to program unmanned vehicles on bare metal, which is possible but difficult.

Some open-source firmwares are:

- **ArduPilot:** Professional grade firmware, accurate and robust. Its main GCS (Ground Control Station) is Mission Planner. [2]
- **PX4:** Devoted for investigation (ROS, Robot Operating System) and autonomous operations; created by Pixhawk. It uses QGroundControl. [3]
- **Betaflight:** Devoted to FPV drones. Basic firmware that works well on small boards. The same Betaflight provides the GCS. [4]
- **INAV:** In the middle path between ArduPilot and Betaflight. Used on FPV with positioning systems that seek more precision than Betaflight. [5]

Before creating a controller board, firmware has to be selected beforehand. Programs like Betaflight work with most standard sensors and MCUs (Microcontroller Units). However, ArduPilot and PX4 generally expect specific buses and components.

This does not mean you can't use different chips, but using supported hardware makes programming the bootloader much easier. Advanced users can modify the source code to support any MCU or sensor, but this is complex and not recommended for beginners.

In this project as it tries to develop a professional grade board, ArduPilot was chosen.

1.3. PCB Schematics

The PCB design was carried out using Altium Designer. All the schematics, footprints and steps files were obtained from Altium itself [6], EasyEDA [7], UltraLibrarian [8] and Component Search Engine [9], (SamacSystem in other words). To mention that not always all the footprints, schematics are the identical between these sources; some of them can contain errors, so a previous datasheet and manufacturer capabilities search has to be done before manufacturing.

The chosen manufacturer was JLCPCB, due to its capabilities, material selection, market status and cost, [10].

Since the hardware is designed with the intention of being ported to ArduPilot, strict adherence to ArduPilot's data and library requirements was mandatory. Otherwise, some buses or components may not work correctly or might need special drivers. If the reader wants to know more about this, please take a look at the Porting chapter of this TFM.

1.3.1. MCU system

The principal component of a drone controller is the MCU (Microcontroller Unit). Since the objective of this project is to produce a professional-grade system, the STM32 family was selected as the optimal architecture. Some typical MCUs used on drones and available to use with ArduPilot are:

- **STM32F405:** One of the most common MCUs for drones, only one core ARM Cortex-M4, max frequency of 168 MHz, 1 MB Flash, 192 KB RAM, FPU (Single-precision of 32 bits), and the ART Accelerator (which functions similarly to a cache). Its cost is around 3-8 €. Found on boards around <100 €.
- **STM32H743:** Commonly found in some high end drones, only one core ARM Cortex-M7, max frequency of 480 MHz, 2 MB Flash, 1 MB RAM, FPU (Double-precision of 64 bits) and with cache L1. Its cost is around 6-15 €. Found on boards around 70-400 €. [11], [12]
- **STM32H757:** Not so common in drones, however Pixhawk and high end controllers like Pixhawk Cube Orange+ use this MCU in parallel with another MCU, do not confuse it with Standard Cube Orange that uses a version similar to the H743 processor. H757 uses dual core (ARM Cortex-M7 + ARM Cortex-M4), it can achieve frequencies of M7-480 MHz and M4-240 MHz, 2 MB Flash, 1 MB RAM, FPU (Double and Single precision), and cache L1. Its cost is around 15-25 €. Found on boards around +450 €.

ArduPilot struggles with 1 MB of storage, as the firmware is almost 1 MB. This is because it contains multiple lines of code and complex drivers such as the use of DMA (Direct Memory Access).

This project uses the STM32H743 as it is a good MCU with enough Flash. Another option is to provide external memory but it is better to keep things simple. Specifically the package LQFP100 (100 pins) in STM32H743VIT6 version, with the silicon

revision V. It is a small package, ideal for small applications, and it can work within a wide range of temperatures.

1.3.1.1. Conditioning

This section outlines the essential conditioning and support circuitry for the MCU. This processor is powered by 3.3 V with a max current of 620 mA. The maximum voltage admitted by the MCU is around 4 V, although specific pins are 5 V tolerant. All ceramic capacitors must have a maximum tolerance of 20% and be placed as close as possible to the MCU pins. These MCU power pins require a specific conditioning:

- **VDD and VDDA pins:** The VDD power line requires a 100 nF decoupling capacitor at each pin, plus a single bulk 4.7 μ F capacitor. Similarly, VDDA and VREF require 100 nF per pin and a 1 μ F bulk capacitor.
- **VBAT and VCAP pins:** VCAP only needs a 2.2 μ F capacitor connection with less than 100 m Ω of ESR (Equivalent Series Resistor). While VBAT has to be connected directly to the power supply though a 100 nF capacitor can be useful to apply for stability.
- **NRST pin:** Pin used to reset the MCU, external pull up 10 k Ω resistor often used in design though not necessary.

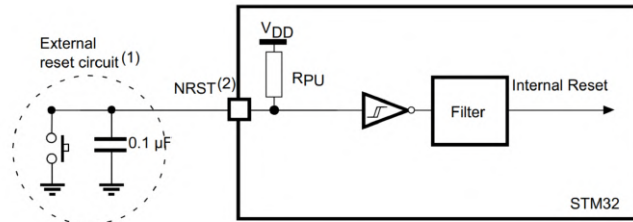


Fig. 1.1 NRST generic scheme

- **BOOT pin:** Used to place the MCU into bootloader mode, enabling programming via interfaces such as USB or UART. By default, it requires a 10 k Ω pull-down resistor. To activate bootloader mode, the pin is pulled to 3.3 V (typically via a resistor or jumper). A 100 nF capacitor is added for noise reduction.

The ESD (ElectroStatic Discharge) voltage protection of the pins using HBM (Human Body Model) is around 1 kV while the CDM (Charge Device Model) is around 250 V.

External protection to data lines has been added with TVS diodes, specifically TPD2E2U06DRLR (+25 kV) IC; with a breakdown of 6.5 V, not close to the operating voltage but with a low capacitance of 1.5 pF, ideal for high speed data transmission.

1.3.1.2. Clocks

The internal clock for these processors is a charged switched capacitor, which is not ideal. 2 External clocks are needed. ArduPilot admits these clocks in direct or bypass mode. The applied option has been bypass mode for both clocks: HSE (High Speed External oscillator) and LSE (Low Speed External oscillator or known as the real clock).

- HSE:** The MCU supports an external input frequency range of 4-50 MHz. The chosen HSE has been SIT1602AI-22-33E-26MHz [13]. This IC is a MEMS (Micro ElectroMechanical System) without quartz, it vibrates with a special silicon structure. It is more reliable for systems like drones as it is more resistant to vibrations and impacts. Its stability is around 25 ppm. Only a 100 nF decoupling capacitor and a <10 kΩ pull up resistors are required for the conditioning of this circuit.

During the development of this board, due to software compatibility issues with ArduPilot this initial chip was substituted by ASEMB-24.000MHZ-LC-T [14]. The difference between them is minimal, but this new clock runs at 24 MHz instead of 26 MHz, making it ArduPilot compliant. However, it has a lower precision of 50 ppm.

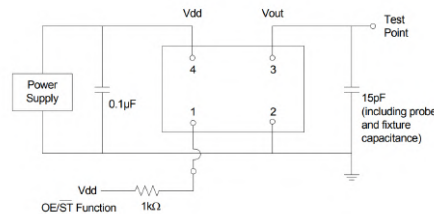


Fig. 1.2 Clock generic scheme

- LSE:** Named as RTC (Real Time Clock), the MCU standard frequency is 32.768 kHz, the SIT1552AI-JE-DCC-32.768D was chosen [15]. This programmable IC is extremely small, smaller than a grain of rice and provides precise timing with a deviation of only 5 ppm. Furthermore, it is considered the world's smallest TCXO (Temperature Compensated Crystal Oscillator). No external conditioning circuitry was required.

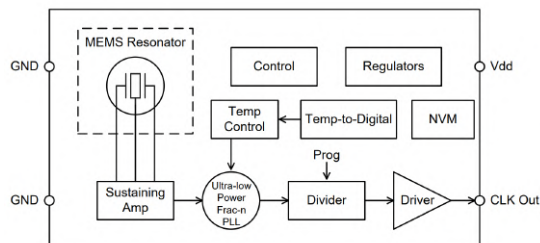


Fig. 1.3 LSE scheme

1.3.1.3. Pins and Schematics

The pin assignment and configuration were performed using STM32CubeIDE [16]. This environment allows for the selection and distribution of communication buses and peripherals according to the project's specific requirements.

Pins devoted to interruptions have been configured as EXTI (External Interruptor), while for normal I/O pins, only generic GPIO (General Purpose Input/Output) are used. In total, 80 out of 100 pins are used.

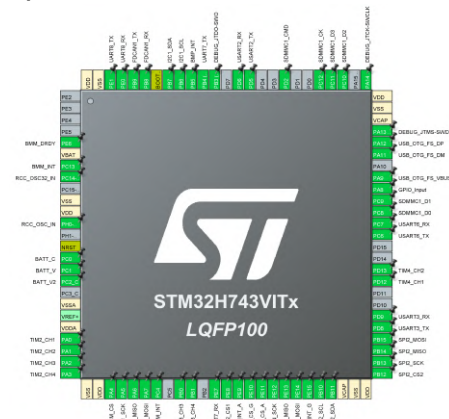


Fig. 1.4 Pin selection with STM32cubeIDE

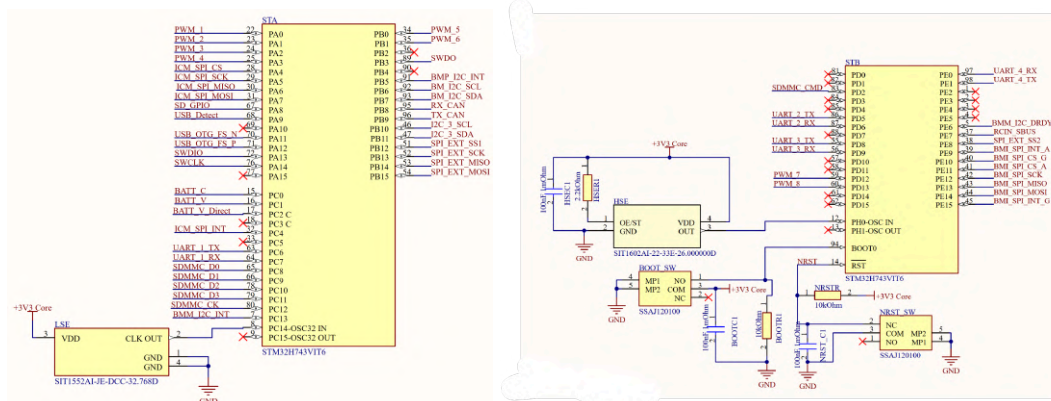


Fig. 1.5 MCU schematics

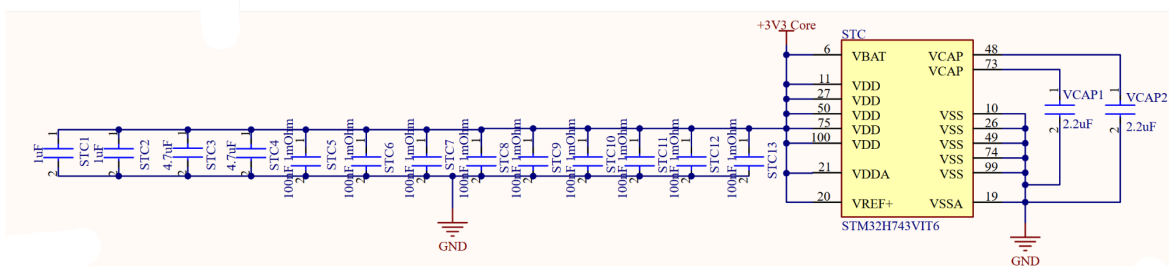


Fig. 1.6 Decoupling capacitor scheme

1.3.2. IMUs: ICM20602 and BMI088

Professional boards often carry multiple IMUs (Inertial Measurement Units). Some of them use the same type of IMU, though having different ones can provide greater reliability. As this board has to be compatible with ArduPilot, both IMUs must be compatible as well. Both sensor ICs are 6-axis, with a gyroscope and an accelerometer.

- **ICM-20602:**

This is a 6-axis motion tracking device with an LGA package, created by InvenSense. Its maximum operating voltage is up to 3.45 V, sufficient for the power source of the board, which is 3.3 V. The connection with the MCU is via a high-speed SPI bus. This is due to the need to obtain rapid and reliable real-time data regarding the drone's movement, as required by ArduPilot. TVS diodes and decoupling capacitors have been applied.

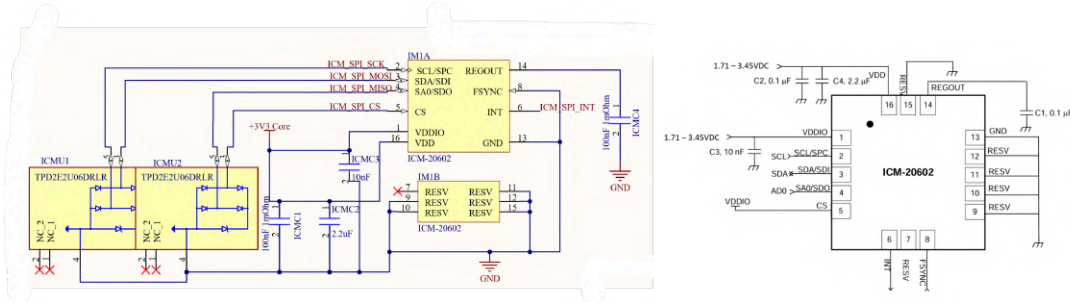


Fig. 1.7 ICM-20602 and datasheet scheme [17]

- **BMI088:**

This is a common 6-axis motion tracking device with an LGA package, created by Bosch. The operating voltage of this sensor is similar to the MCU. The connection with this IC is via high-speed SPI, on a completely different bus with respect to the ICM-20602. This is done for reliability. Additional interrupts for this IC are not used; only the minimal pins are connected. TVS diodes and decoupling capacitors have been used.

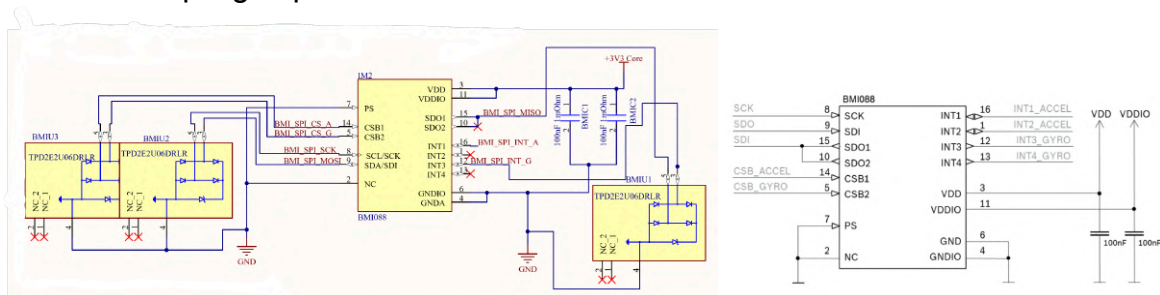
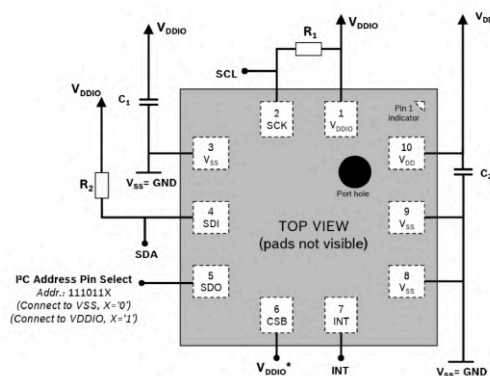


Fig. 1.8 BMI088 and datasheet scheme [18]

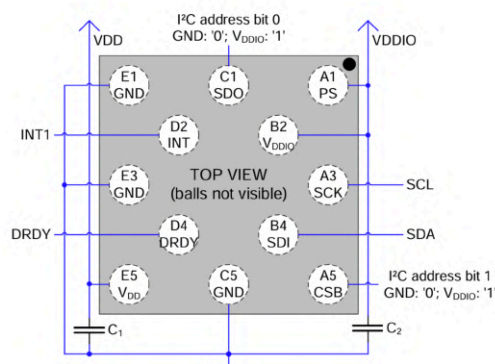
1.3.3. Barometer and Magnetometer: BMP388, BMM150

Both sensors are connected using the same I2C bus. The magnetometer and barometer are not critical components for basic stabilization, so I2C is ideal as it is a moderate speed bus. The pull-up resistors used are 2.2 kΩ, enabling communication frequencies of 100-400 kHz. Both Bosch sensors can accept up to 3.6 V, and they are ArduPilot compliant with different addresses. Furthermore, like the previous sensors, they are equipped with interrupt pins. Decoupling capacitors have been placed near the ICs.



The diagram shows the BMM150 module connected to a Raspberry Pi 3B+ via an I2C interface. The module's pins are connected as follows:

- Power:** VDD is connected to the 5V pin of the Raspberry Pi. VDDIO is connected to the 3.3V pin. GND is connected to the GND pin.
- I2C:** SDA is connected to the SDA pin of the Raspberry Pi. SCL is connected to the SCL pin.
- Interrupt:** INT is connected to the 5V pin of the Raspberry Pi.
- DRDY:** DRDY is connected to the 5V pin of the Raspberry Pi.
- CS:** CS is connected to the 5V pin of the Raspberry Pi.
- SDI:** SDI is connected to the 5V pin of the Raspberry Pi.
- PS:** PS is connected to the 5V pin of the Raspberry Pi.
- SDO:** SDO is connected to the 5V pin of the Raspberry Pi.
- DRDY:** DRDY is connected to the 5V pin of the Raspberry Pi.
- INT:** INT is connected to the 5V pin of the Raspberry Pi.
- CS:** CS is connected to the 5V pin of the Raspberry Pi.
- SDI:** SDI is connected to the 5V pin of the Raspberry Pi.
- PS:** PS is connected to the 5V pin of the Raspberry Pi.
- SDO:** SDO is connected to the 5V pin of the Raspberry Pi.



1.3.4. Differential Buses: CAN and USB-C

Common or non-professional control boards rarely carry this kind of bus. CAN bus is a highly reliable bus. The typical transceivers for this CAN as Pixhawk Standard describes are: TJA1051TK3/118 or LTC2875 (implementing the ISO 11898-2:2016). The used one is the TJA1051TK/3/IJ [21], which goes at 4.5-5.5 V. This model allows sending data at a lower voltage to the MCU using an IO power pin, up to 3.3 V.

Table 1.1: CAN

Pin	Signal	Voltage
1 (red)	VCC	+5 V
2 (blk)	CAN H	CAN High
3 (blk)	CAN L	CAN Low
4 (blk)	GND	GND

Additionally a 120 Ω termination resistor is needed. Twist cables are recommended on the peripheral side to reduce EMI (Electromagnetic Interference) to ensure the same area of interference between cables, and

reduce the same one. Additionally, decoupling capacitors have been applied as the datasheet schematic mentions.

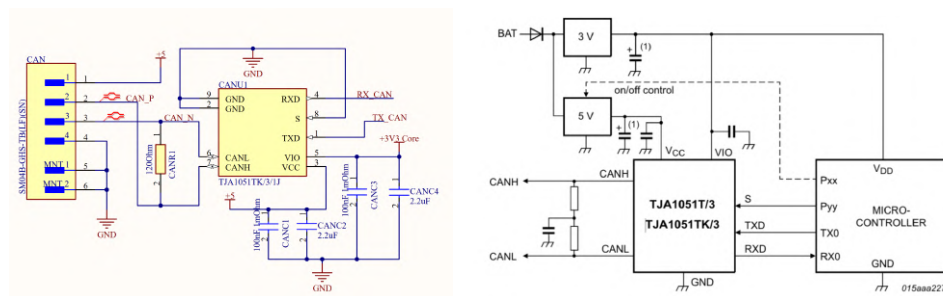


Fig. 1.11 CAN, datasheet and used schematic

- **USB bus:**

This bus is dedicated to a direct connection between the personal computer and the MCU. To function as a termination device, the USB standard requires 5.1 k Ω resistors on the CC pins.

Most drone controllers can power the PCB at 5 V from the USB; however, this PCB does not allow that. It only detects if the USB is connected. This is because the power consumption of the PCB and all peripherals can be higher than 0.5 A. While often this setup would be fine, if the user plugs a high consumption peripheral, a higher current could be triggered. If the user's computer cannot handle that, it becomes a problem.

The USB 2.0 standard limits the maximum current to 0.5 A. Despite this, for safety and preventive reasons, there is no direct power connection between the USB VBUS and the board's supply.

Additionally filtering and TVS protection for 5 V detection is provided.

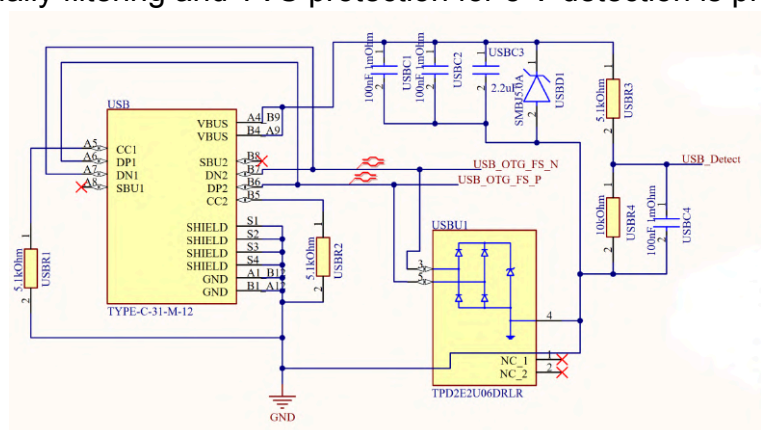


Fig. 1.12 USB-C 2.0 FS schematic

1.3.5. Debug and Peripheral Buses

All ports are within the Pixhawk standard using JST connectors.

- **Debug SWD:**

This bus is the most important one, it allows direct programming between the user's debugger and the STM32. There are 2 Pixhawk standards;

nevertheless, none of them was applied. A custom connector has been used, composed of: 3.3 V as a power line, SWDIO to program the MCU, SWCLK as a clock line, NRST to reset the MCU and additionally a SWO optional line devoted to printing debug commands.

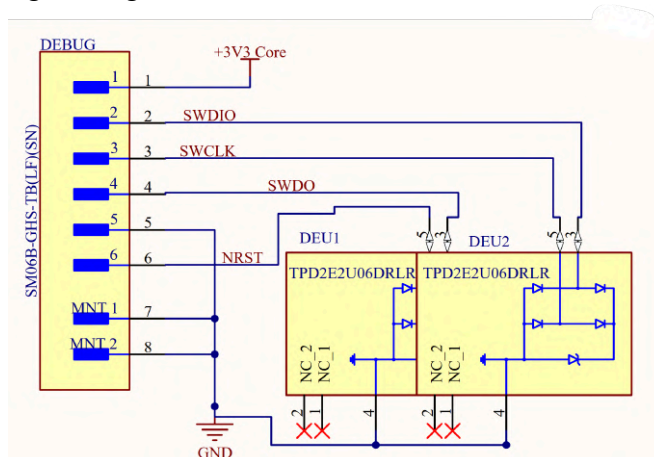


Fig. 1.13 Debug connector

- **SPI:**

The SPI bus connector follows the Pixhawk standard. A clarification regarding signal naming is necessary; the industry is transitioning from terms like MISO (Master Input Slave Output) to the modern CIPO (Controller In Peripheral Out).

When wiring, this line must be connected to the peripheral's MISO or SDO (Serial Data Output). Confusion appears when components like sensors use SDI/SDO labeling. In these cases, the connection follows the data flow: the MCU's MOSI (acting as SDO) must be connected to the peripheral's SDI (Serial Data Input). The previous convention was more simple, direct connection MOSI to MOSI and MISO to MISO.

Table 1.2: SPI

Pin	Signal	Voltage
1 (red)	VCC	+5 V
2 (blk)	SCK	+3.3 V
3 (blk)	MISO	+3.3 V
4 (blk)	MOSI	+3.3 V
5 (blk)	SS1	+3.3 V
6 (blk)	SS2	+3.3 V
7 (blk)	GND	GND

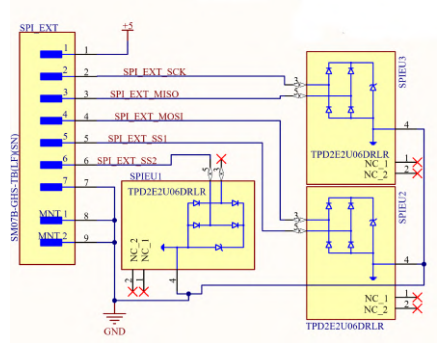


Fig. 1.14 SPI connector

- I2C:**

The design mostly has followed the Pixhawk standards except for the resistor value. The pull up resistor used is 2.2 k Ω , which handles more or less the same bus speed without having any issues.

Table 1.3: I2C

Pin	Signal	Voltage
1 (red)	VCC	+5 V
2 (blk)	I2C SCL (1.5 k Ω recommended)	+3.3 V
3 (blk)	I2C SDA (1.5 k Ω recommended)	+3.3 V
4 (blk)	GND	GND

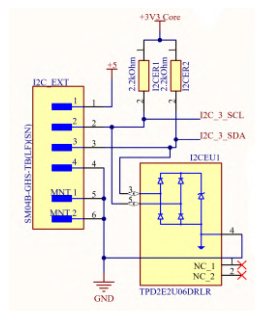


Fig. 1.15 I2C connector

- Telemetry:**

Similar to Pixhawk standards, optional dedicated hardware flow control pins are not used (RTS = Request To Send and CTS = Clear To Send). On the peripheral side, TX/RX must be crossed; this means the TX line at the MCU has to be connected to the RX line of the peripheral.

Table 1.4: Telem

Pin	Signal	Voltage
1 (red)	VCC	+5 V
2 (blk)	TX (out MCU)	+3.3 V
3 (blk)	RX (in MCU)	+3.3 V
4 (blk)	CTS (in MCU) (optional)	+3.3 V
5 (blk)	RTS (out MCU) (optional)	+3.3 V
6 (blk)	GND	GND

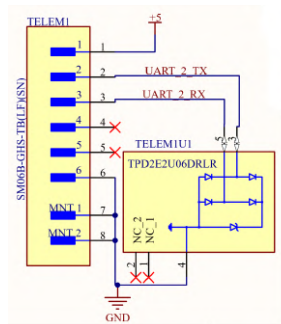


Fig. 1.16 Telemetry connector

- **Basic GPS:**

The Pixhawk standard provides multiple versions for the GPS port. The full GPS mode contains pins for a buzzer, LED and safety switch. However, this configuration is only present in a small group of devices, so it has been discarded. Only the Basic GPS connector has been applied.

Table 1.5: GPS

Pin	Signal	Voltage
1 (red)	VCC	+5 V
2 (blk)	TX (out MCU)	+3.3 V
3 (blk)	RX (in MCU)	+3.3 V
4 (blk)	External I2C SCL	+3.3 V
5 (blk)	External I2C SDA	+3.3 V
6 (blk)	GND	GND

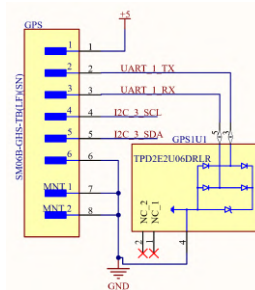


Fig. 1.17 GPS connector

- **2.54 mm Pins:**

To provide motor and servo outputs, 8 typical 2.54 mm connector pins are used. These 8 PWM outputs come from 3 different MCU timers (Timers 2, 3, and 4). Moreover, the outputs can provide not only PWM signals but also other standards.

Regarding the connection of SBUS and RCIN, 2 pins are available to receive signals via a UART-RX. A direct connection is usually complex because SBUS is an inverted logic signal. An external inverter IC would be needed in case of using an ARM Cortex-M4 (like the STM32F405) due to the lack of inversion on its UART. Despite this, the STM32H7 provides internal signal inversion within the IC package.

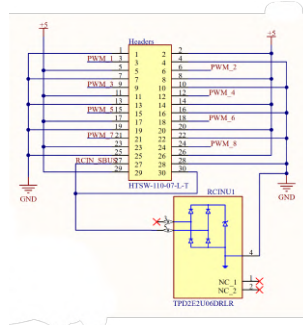


Fig. 1.18 2.54 mm connectors

1.3.6. Direct Power and Battery 5V

This PCB has been designed to be powered by 3 means: connecting a Pixhawk Power cable, connecting with a 2.54 mm pins UBEC cable or connecting directly to the battery.

- **Pixhawk Power Connector:**

This method uses the power connection of the Pixhawk standard; the lines should provide at least 2.5 A. It is important to mention that this connection provides two analog measurements: current and voltage from the battery.

These measurements were initially filtered with an RC filter, but it is crucial to understand the real source of these values. If they come from an amplifier, an RC filter works well. However, if they come from other means, such as a shunt resistor or voltage divider, the filter's resistor affects the final measurement. After testing the board, only a filtering capacitor is required, as the system measures values directly from a voltage divider and shunt resistor, sending the signal to the MCU ADCs.

Table 1.6: Power connector

Pin	Signal	Voltage
1 (red)	VCC	+5.3 V
2 (blk)	VCC	+5.3 V
3 (blk)	Current (analog)	+3.3 V
4 (blk)	Voltage (analog)	+3.3 V
5 (blk)	GND	GND
6 (blk)	GND	GND

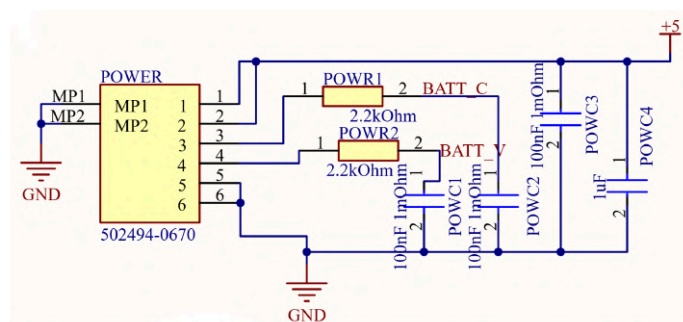


Fig. 1.19 Power Pixhawk connector

- **Battery Connection:**

This circuit has been designed to connect the battery directly to the board or by connecting an UBEC cable. The main component of this circuit is a buck converter. This was designed using Webench with an input of 6-36 V (nominal of 4S-6S) and an output of 5.5 V with a maximum current of 3 A.

The frequency used is 504 kHz with an efficiency of +90%. Regarding the capacitors, their ESR (Equivalent Series Resistance) values are close to the ideal, although actual component characteristics vary slightly.

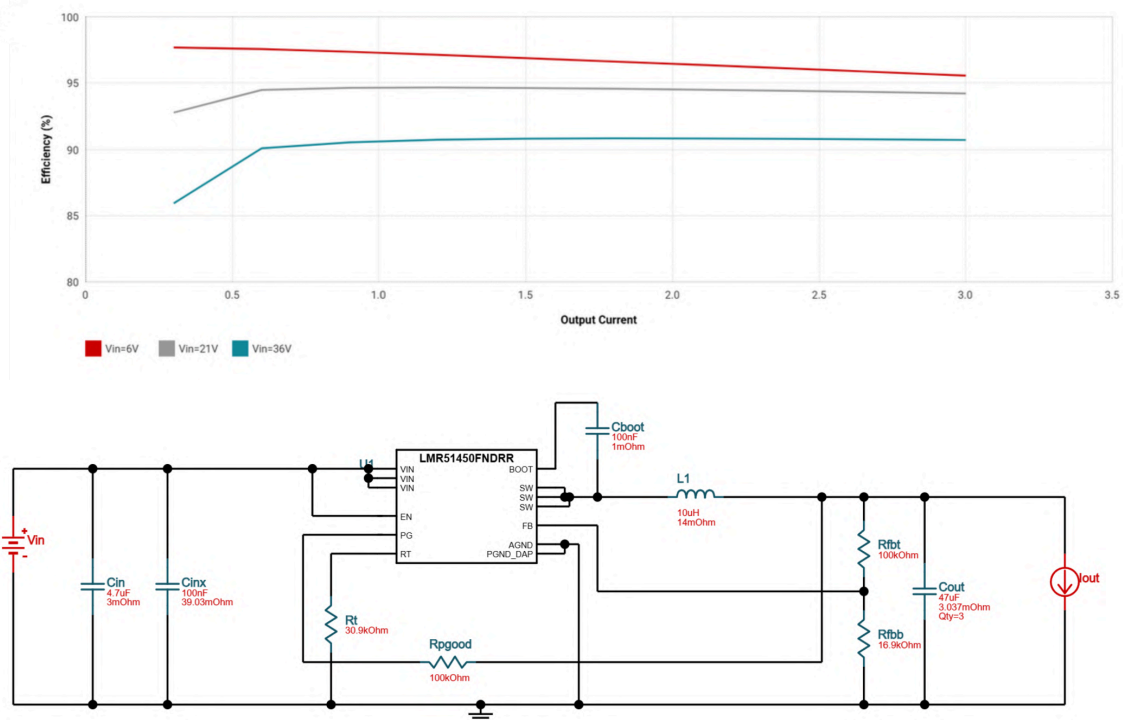


Fig. 1.20 Converter efficiency and circuit 6-36 V to 5 V [22]

Additionally a reverse protection circuit and a 3 A fuse have been applied at the input. This reverse protection circuit is composed of a P-MOSFET of 100 V, a 100 k Ω resistor and a 12 V Zener diode. While the MOSFET's orientation provides the reverse polarity protection, the Zener diode and resistor are critical for gate-source voltage clamping. Since the input voltage can reach 36 V (nominal 6S), applying this potential directly to the gate would exceed the MOSFET's maximum rating (following the datasheet a maximum of -20 V), leading to dielectric breakdown. The Zener diode limits the voltage drop across the gate to a safe -12 V, while the resistor limits the current through the Zener, ensuring the MOSFET operates safely within its specifications regardless of the high input voltage.

At the output of the converter, the design includes a TVS protector against voltage increment, a Schottky diode to protect against reverse current, and a 2.54 mm header to connect with a UBEC cable. The Schottky diode will reduce the output voltage from 5.5 V to around 5.1 V. Furthermore, it is important to use an external polymer capacitor at the battery connector to protect against noise (470 μ F with low ESR).

Moreover, a simple voltage divider was designed to detect battery presence and voltage level; however, due to ArduPilot mapping constraints, this voltage measurement on microcontroller ADC-3 was not possible to implement.

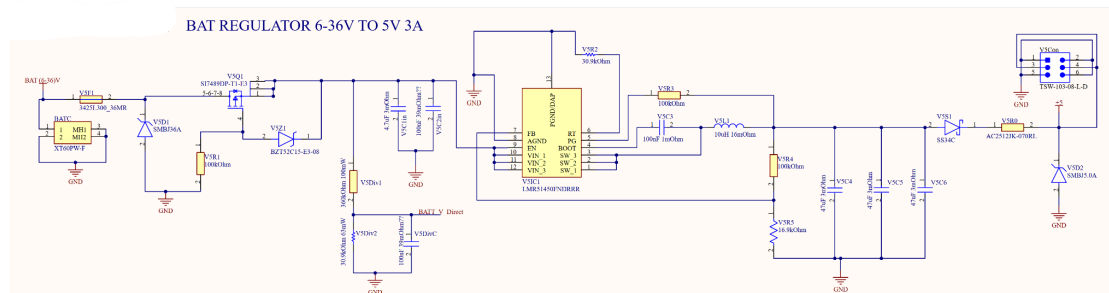


Fig. 1.21 Overall circuit diagram 6-36 V to 5 V

1.3.7. Power 3.3 V

This circuit powers the entire board, with the exception of the external peripherals and the CAN transceiver. Two identical, independent circuits have been designed (one for peripherals and the second for the board).

These converters, developed with Webench, accept an input of 3.8 V to 6 V and provide an output of 3.3 V with a maximum current of 2 A. The maximum estimated current draw is below 1.25 A. This simple buck converter operates at 1.39 MHz and achieves an efficiency of +90%. As it is a high-frequency converter, 100 nF ceramic capacitors are ideal to filter noise near the sensors and the MCU.

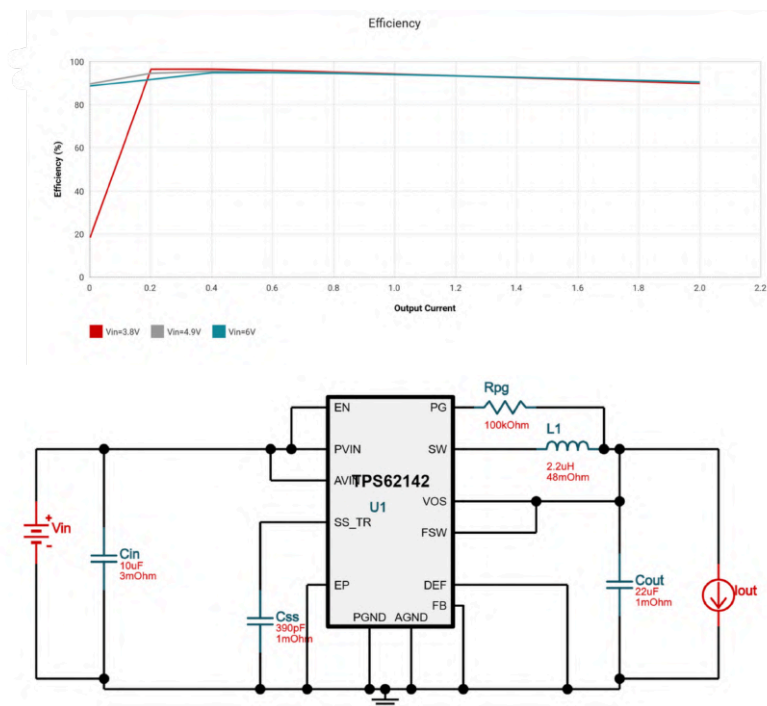


Fig. 1.22 Converter efficiency and circuit 5 V to 3V3 [23]

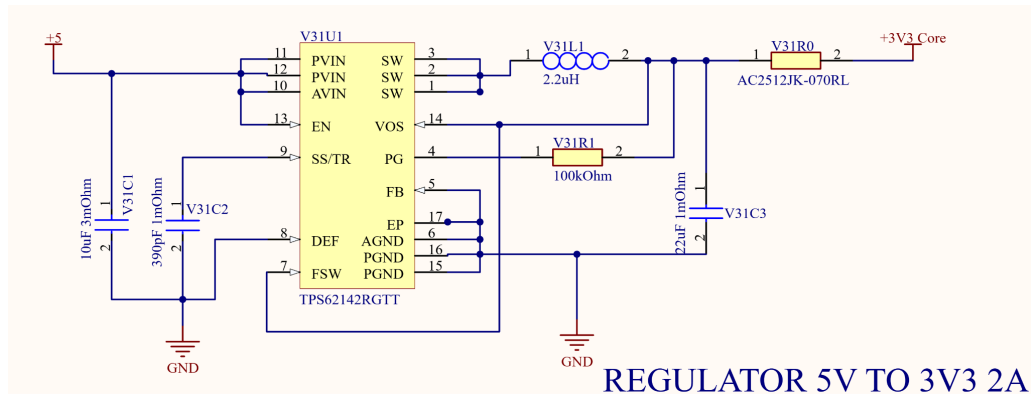


Fig. 1.23 Overall circuit diagram 5 V to 3V3

1.3.8. Storage: MicroSD

Drone boards commonly store data within Flash or RAM. While internal RAM is sufficient for the MCU's operation, external Flash storage is essential if the user wants to record flight logs.

The vast majority of microcontrollers save this data using the SPI bus. However, high-end drones avoid this due to speed limitations and DMA sharing constraints (Direct Memory Access). Better methods can be applied, such as SDMMC (Secure Digital and MultiMediaCard). This bus can transfer between 24-48 MB/s, depending on the card used. There are multiple modes; the chosen one is the 4-bit mode, which allows fast storage (storing 4 bits per clock cycle), using CMD for commands and CLK as the clock.

To create this circuit, trace length matching is necessary between lines. Additionally, pull-up resistors are required on all lines except the clock.

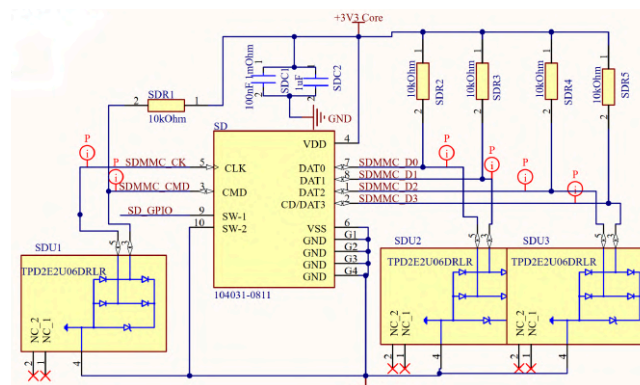


Fig. 1.24 Overall circuit diagram MicroSD

1.4. Other considerations

All resistors and capacitors selected for the design are 0402 size or larger to minimize assembly costs and potential defects.

Moreover, wherever feasible, component standardization was prioritized. Using the same component type across multiple systems such as choosing a single standard

value for capacitors or resistors significantly reduces production costs by simplifying the BOM (Bill of Materials).

Component availability was also key. Manufacturers' available quantity and lifetime cycle has to be considered, as well as potential alternatives.

Additionally, the design includes dedicated test points on important power and signal lines. For instance: un-tented vias (exposed vias) can be used as points for signal probing.

CHAPTER 2. PCB ROUTING DESIGN

This chapter provides a simple explanation of the PCB routing strategy. It begins by addressing the layer stackup and impedance constraints, followed by the routing process and concludes with the files required for manufacturing and assembly.

2.1. Manufacturing Capabilities and Layer Stackup

Manufacturing has been one of the most challenging parts of this project. The 2 biggest constraints were tolerance requirements and material availability. To produce the board, the chosen manufacturer was JLCPCB. This explanation focuses only on rigid PCBs.

2.1.1. Capabilities

The capabilities of the PCB manufacturer have a decisive role in the design process, defining the limits for DFM (Design for Manufacturability) and PCB Assembly (PCBA) feasibility. The board was designed in compliance with the manufacturer's constraints. The key manufacturing capabilities and design rules are summarized below [24]:

1. Stackup and materials:
 - Number of layers: up to 32 layers.
 - Core material: FR-4, Al, Co, Rogers or PTFE (Polytetrafluoroethylene).
 - Core dielectric constant: usually 4.5 for FR-4.
 - Maximum PCB dimensions 670x600 mm and tolerance 0.1 mm.
 - Thickness: 0.4-4.5 mm.
 - Surface finish: HASL, ENIG or OSP.
2. Routing and copper:
 - Impedance tolerance: 10%.
 - Copper layer weight: 0.5 oz, 1 oz, 2 oz, etc.
 - Track width and spacing (3.5/3.5 mil) (this depends on copper weight).
 - Track tolerance: 20%.
 - SMD pad to pad clearance: 0.15 mm.
 - BGA pad diameter: 0.25 mm.
 - Copper distance clearance: 0.2 mm.
 - Pad to track clearance: 0.1 mm.
3. Drilling and vias:
 - Hole size: 0.15-6.3 mm (5.9-248 mil).
 - Blind/buried vias.
 - Plated and non-plated slots or holes.
 - Via-to-via spacing: 0.45 mm.
 - Via to hole clearance: 0.2 mm.
 - Min NPTH/PTH annular ring distance: 0.45/0.2 mm.
4. Soldermask and silkscreen:
 - Soldebridge: 0.1 mm.
 - Soldemask expansion: 1:1.
 - Soldemask dielectric constant: 3.8.

- Silkscreen width and clearance: 6 mil/0.15 mm.
- Panelization, and many more.

2.1.2. Layer Stackup

The layer stackup comprises the main board materials and is critical because it defines aspects such as signal integrity, EMI (Electromagnetic Interference) behavior, power integrity, cost and reliability.

The main objective of this project is to develop a flight controller prototype, making cost a relevant factor. To reduce expenses, a standard controlled impedance stackup has been chosen. While this is not a high-density drone stackup (which typically consists of 6-10 layers), it is more robust and feasible for repairs. This approach provides a balance between professional performance and prototyping flexibility.

Table 2.1: Layer stackup

Layer	Material	Thickness (mil)	Thickness (mm)
L1	Outer Copper Weight1oz	1.38	0.0350
Prepreg	7628, RC 49%, 8.6 mil	8.28	0.2104
L2	Inner Copper Weight	0.60	0.0152
Core	1.1mm H/HOZ with copper	41.93	1.0650
L3	Inner Copper Weight	0.60	0.0152
Prepreg	7628, RC 49%, 8.6 mil	8.28	0.2104
L4	Outer Copper Weight1oz	1.38	0.0350

This 1.6 mm stackup is a base standard in JLCPCB [25], composed of an FR-4 core and Prepreg 7628 (glass fiber with semi-cured epoxy and a dielectric constant of 4.4). Copper layers range from 0.5 oz for inner layers and 1 oz for outer layers.

It is important to mention that the stackup selection depends on trace impedance, connection complexity and environment. To reduce manufacturing costs, all vias are through-hole; this means every vertical connection crosses all four copper layers.

Regarding the layout, having a symmetric stackup is crucial to reduce thermal stress in the Z-axis and prevent warping. The designed drone PCB uses L1 and L4 as the primary signal layers, while L2 is dedicated to grounding (as a reference plane) and L3 is used for grounding and power distribution.

2.2. Trace Impedance Calculation

Trace impedance and reference grounding is quite important for signal transmission.

The trace characteristic impedance can be defined as:

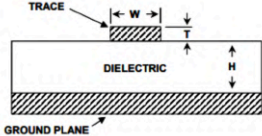
$$Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}} \quad \text{or} \quad Z_{approx} \approx \sqrt{\frac{L}{C}} \quad (2.1)$$

Where L and C stands for inductance and capacitance while R and G stands for resistance and conductance. When designing a track:

- If the track increases width, the surface area increases. This causes higher capacitance (more charge can be stored) and lower inductance. The result is a lower characteristic impedance.
- If the distance between the reference plane and the signal track increases, capacitance decreases and inductance increases. The result is a higher characteristic impedance.
- If the dielectric constant of the material is higher, capacitance increases. The result is a lower characteristic impedance.

In the design of a PCB, the most used formulas are from IPC-2141 standard, though these can contain an error of 7%. The symbol ϵ_r stands for material relative permittivity or dielectric constant. It measures how much electrical energy can contain the material. For instance: 1 for vacuum and 4.5 for FR-4.

$$Z = \frac{87}{\sqrt{\epsilon_r + 1.41}} \ln \left(\frac{5.98H}{0.8W + T} \right)$$



$$Z = \frac{60}{\sqrt{\epsilon_r}} \ln \left(\frac{1.9B}{0.8W + T} \right)$$

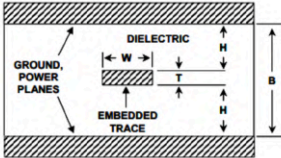


Fig. 2.1 IPC-2141 standard formulas (7% tolerance) (good for 50 Ω) [26]

Another set of formulas commonly used are the Wadell's impedance equations, these are much more accurate around 1%.

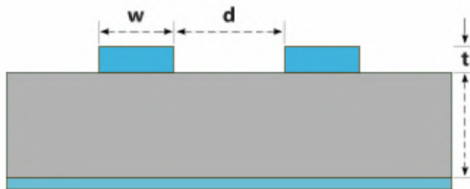
There are multiple online models and calculators available, from simple to complex ones. However, most of them do not take into account dielectric losses (the imaginary part of the dielectric constant), copper losses (skin effect and surface roughness), radiation losses, high-frequency dispersion, or fiber weave effects (statistical non-uniformity of the PCB glass fiber).

Another important aspect is via impedance and specially differential impedance. Differential pairs transmit signals with opposite polarities. Their close proximity ensures tight electromagnetic coupling, making transmission more reliable. If external

noise affects one signal, it affects the other in the same way, allowing the noise to be cancelled out. As well as a single track, a differential track can be coplanar or non-coplanar (surrounded by a ground plane). The differential impedance formula from the physical point of view can be defined like:

$$Z_{diff} = 2Z_{odd} = 2\sqrt{\frac{L_{trace} + L_{mutual}}{C_{trace} + 2C_{mutual}}} \quad (2.2)$$

If we increase the spacing between lines, $\uparrow C_{mutual}$, obtaining a results of $Z_{diff} = 2Z_0$. Same result as having a single trace. Despite this formula, the design process does not use this calculus directly as it can become quite difficult to determine these internal values. Instead, design processes use IPC estimation formulas and alternatives, depending on frequency range. An example of these equations is [27], [28], [29]:



$$Z_0 = \frac{87}{\sqrt{\epsilon_r + 1.41}} \ln \left(\frac{5.98 \times h}{0.8 \times w + t} \right)$$

$$Z_d = \frac{174}{\sqrt{\epsilon_r + 1.41}} \ln \left(\frac{5.98 \times h}{0.8 \times w + t} \right) \left(1 - 0.48 \exp \left(-0.96 \frac{d}{h} \right) \right)$$

Fig. 2.2 IPC-2141 differential formulas for microstrip (5-10% tolerance)

2.2.1. Impedances on the PCB

Regarding the designed PCB, three main impedances were calculated. While the target impedance for the main traces is 50 Ω , this was not strictly followed throughout. Since this is not a radio frequency board, trace widths were varied to optimize connectivity and thermal relief.

In order to provide CAN connectivity, CAN bus requires a 120 Ω differential impedance. While USB FS, requires a 90 Ω differential impedance. In parallel with the layer stackup [25], these dimensions were chosen to provide a reliable connection.

Table 2.2: CAN/USB bus differential bus impedance

Impedance (Ω)	Type	Signal Layer	Top Ref	Bottom Ref	Trace Width	Trace Spacing
120	Differential Pair (Non coplanar)	L1	/	L2	7.1700	14.0000
90	Differential Pair (Non coplanar)	L1	/	L2	11.2600	8.0000
90	Differential Pair (Non coplanar)	L4	L3	/	11.2600	8.0000

2.3. PCB Routing

Routing a PCB seems an easy task, though it is not. Manufacturer tolerances, number of layers, material used, component design, component spacing and reparation design can cause a lot of problems. There are many topics to expose but I will try to focus on those that are important.

The designed board can be seen in the following pictures.

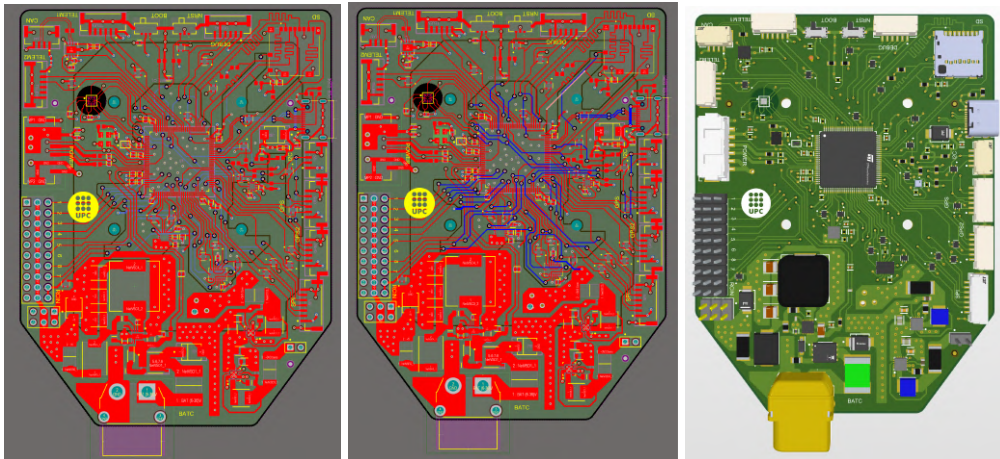


Fig. 2.3 PCB design (Left→ L1 to L4, Center→L4 to L1, Right→3D STEP)

The board outline and mounting holes (including copper clearance) were designed using AutoCAD and Altium, aiming to reduce size and repair operation. During the design many component footprints were found with errors. Datasheet revision and comparison is required prior position.

2.3.1. Power supply design and planes

A 4-layer stack-up was selected for this board design to maximize performance. The use of continuous internal planes provides a solid reference, minimizes crosstalk (traces do not cross split planes to avoid crosstalk), lowers EMI (Electromagnetic Interference), improves heat distribution and increases current handling capabilities.

- **Layer 1 (L1):** This is the principal routing layer containing most of the connections. Power converter traces were widened (+60 mil) to safely handle at least 3 A. These are visible at the bottom of the figure below.
- **Layer 2 (L2):** This is a solid copper GND plane (ground plane), acting as a low-impedance reference for the signals on L1.
- **Layer 3 (L3):** This is a mixed power/ground plane divided into three zones: an outer ring providing 5 V (powering outer Pixhawk connectors), an intermediate semi-circle distributing 3.3 V to onboard SMD devices (Surface Mount Device) and an inner GND pour to shield L4 and prevent crosstalk.
- **Layer 4 (L4):** This layer serves as an alternative routing path when L1 is congested, utilizing vias for transitions. By placing all SMD components on the top side and using L4 only for alternative routing, the design is on single-sided assembly, significantly reducing manufacturing costs.

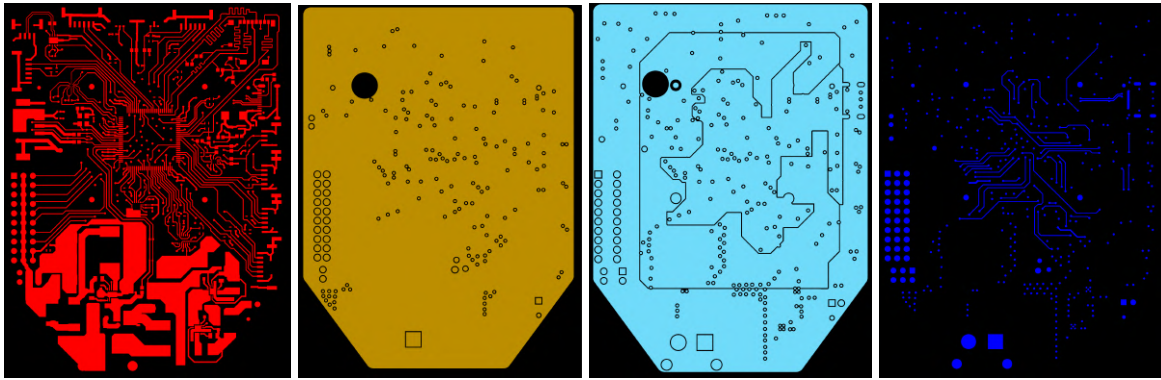


Fig. 2.4 Planes (L1→connections, L2→GND, L3→ reference, L4→ connections)

Standard 0.3 mm hole / 0.6 mm diameter vias were utilized throughout the design to minimize manufacturing costs. These through-hole vias are capable of handling approximately 1.5 A of current. Strategically placed vias are essential for thermal management, current distribution in reference planes and signal routing between layers.

To develop power traces the IPC-2152 standard has been used. This standard is key for manufacturing on power transmission lanes.

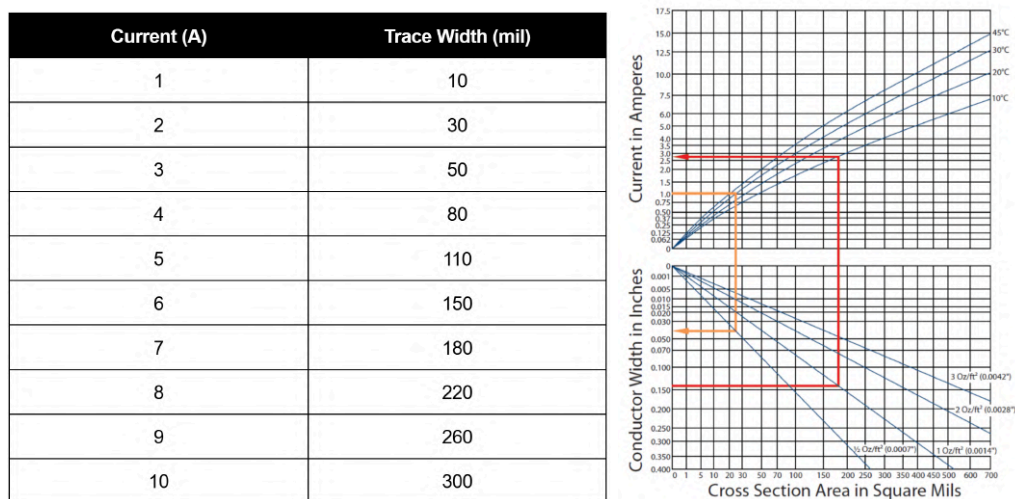


Fig. 2.5 IPC-2152 trace table and Nomograph (both related) [30]

The 5 V converter, powered directly by the battery, is shown in the figure below. This converter was designed following the datasheet's recommended layout to provide at least 3 A. It is surrounded by GND planes/thick traces to minimize noise coupling and crosstalk with adjacent circuits. To connect this 5 V power line to the L3 power plane, a group of vias was used. Specifically, 0.3/0.6 mm vias were chosen; larger vias were avoided as they could cause solder bridging, which is not recommended.

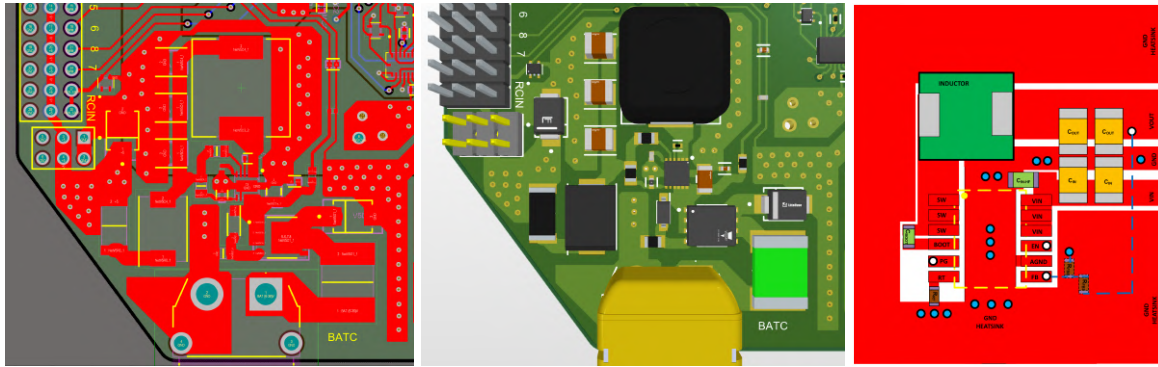


Fig. 2.6 5 V Converter (Left→ routing, Center→3D, Right→Datasheet layout)

This board contains two 3.3 V converters that provide an external power source and an internal power source for the board. In terms of design, they are quite similar to the 5 V converter. Therefore, decoupling capacitors should be placed near the output of the converter to ensure signal stability.

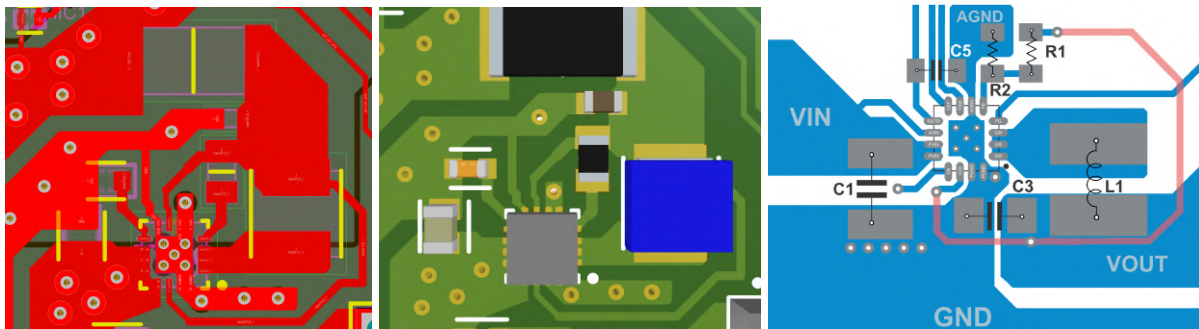


Fig. 2.7 3.3 V Converter (Left→ routing, Center→3D, Right→Datasheet layout)

2.3.2. Sensor and MCU design

The routing strategy for the sensors and the MCU is quite similar. Decoupling capacitors must be placed in close proximity to these components to minimize noise, ensuring their placement does not obstruct other connections. However, there are specific design constraints to consider:

- **Magnetometer:** To prevent magnetic interference, no copper pours were placed underneath or near the sensor. The traces are narrow (5 mil), approaching the manufacturer's minimum width limit.
- **BMI088 IMU:** The two MISO pins, which carry the same output, were merged into a single trace. For very high-frequency applications, this might cause signal integrity issues.
- **MCU and Clocks:** The STM32 datasheet provided the reference design for connecting the external oscillators (clocks), decoupling capacitors, and power delivery network.

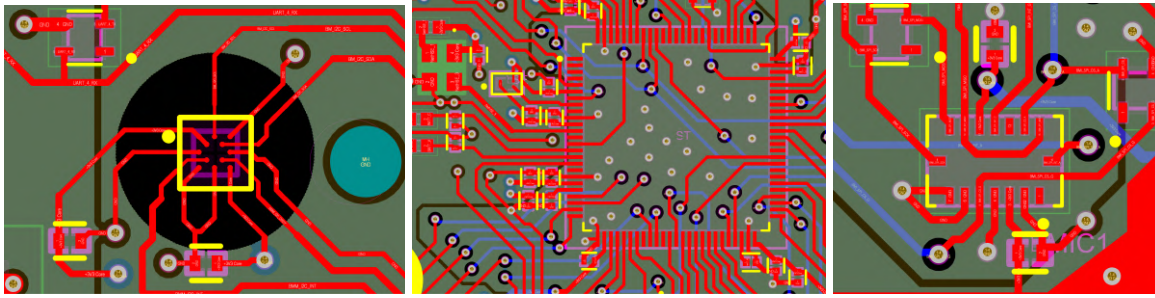


Fig. 2.8 Other connections (Left→ Compass, Center→ MCU+clocks, Right→ BMI088)

2.3.3. Connectors and special buses

The Pixhawk power connector serves as an auxiliary power source, supplying 5.3 V to the 5 V power plane while enabling real-time battery monitoring.

A critical interface on the board is the SDMMC bus, which uses parallel data lines operating at high frequencies. To ensure signal synchronization across the 4-bit bus, length tuning was applied.

An accordion pattern was selected for this purpose, although trombone or sawtooth patterns are valid alternatives. A potential drawback of length tuning is the risk of creating crosstalk and acting as an antenna.

Standard practices recommend a spacing of at least three times the trace width (3 W rule) to reduce crosstalk, especially for frequencies above 50 MHz. To reduce antenna radiation, a solid ground reference plane was used. Coplanar tracks, termination resistance and via shielding are recommended to reduce antenna effects. This design could be further optimized by refining the tuning geometry and increasing trace spacing to mitigate noise.

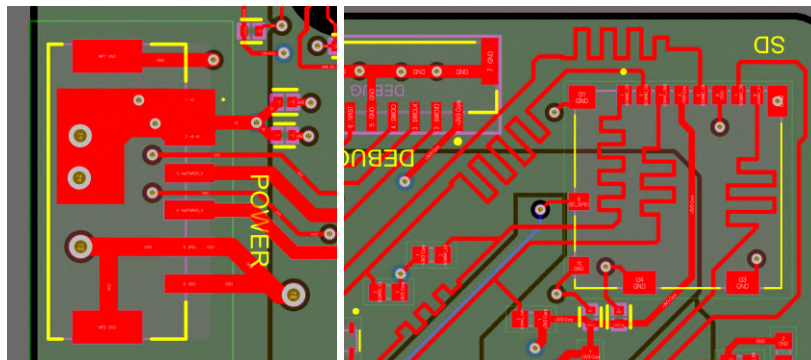


Fig. 2.9 Left→ Pixhawk power connection, Right→ SDMMC bus

The CAN and USB buses are also key components of the design. Both follow the trace width calculated for the required impedance. The USB bus is routed with 2 differential pairs, merging them into a single differential pair with vias within the manufacturer's tolerance limits.

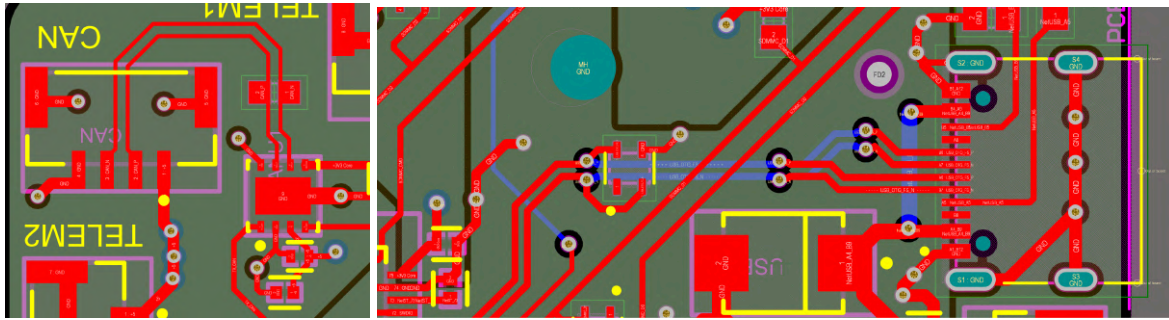


Fig. 2.10 Left→ CAN bus, Right→ USB FS bus

2.3.4. Fiducials and possible improvements

Fiducials are a key component of the design. These marks of exposed copper are essential for the machinery that places the components. They must be positioned at the edges of the board, forming a rectangle. Usually, manufacturers require them to be 1 mm in diameter with a 0.5 mm mask clearance/expansion.

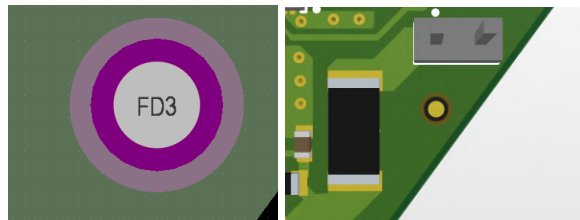


Fig. 2.11 Fiducial

The design is not perfect and has potential for improvement. Improvements may include the analog sensing traces and the power distribution network. Specifically, decoupling capacitors should be positioned strictly between the via and the IC pin to force current through the capacitor, rather than connecting the via directly to the pin. Also, the accordion traces require wider spacing to reduce crosstalk.

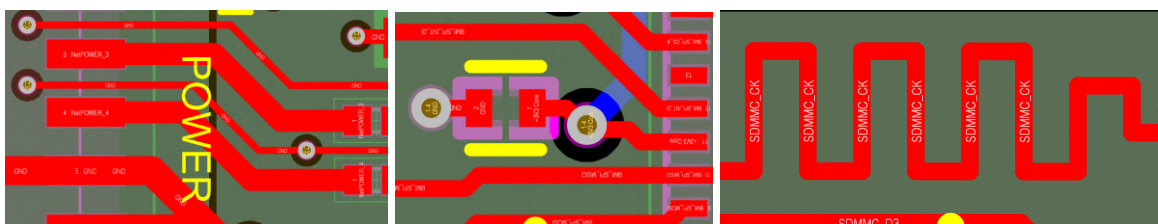


Fig. 2.12 Left→ Analog measure traces, Center→ Decoupling, Right→ Accordion

90° corners shall be avoided to maintain constant trace width. Sharp corners create impedance discontinuities that can cause signal reflections, potentially degrading the quality of transmissions. Moreover, sharp corners can contribute to EMI (Electromagnetic Interference), abrupt change in geometry concentrates the electric field, which can lead to signal radiation.

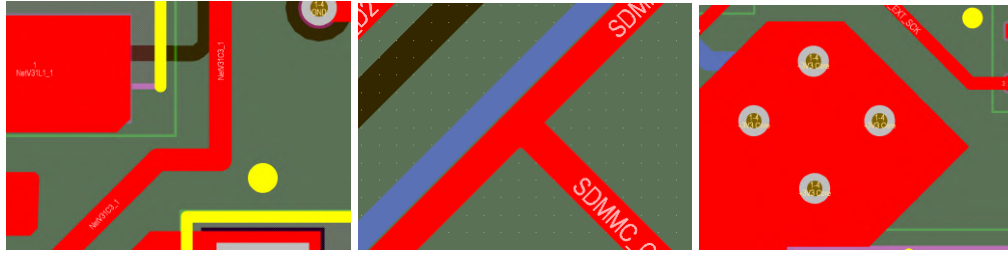


Fig. 2.13 Copper corners

2.4. Manufacturing files

There are multiple manufacturing files inside the industry. These can be divided into 2 sectors, manufacturing files for boards and manufacturing files for mounting and testing. The files used in this project have been GerberX2, PnP (modified) and BOM (modified).

2.4.1. PCB

- **Gerber files (i.e: Gerber X2):** This is a set of files containing 2D vector representations of the board for each specific layer, including soldermask, pad/via positions, and copper traces. They provide the essential geometric data and guidelines required to execute the manufacturing process.
- **OBC++:** Similar to Gerber, though its structure is like a database. Owned by Siemens but quite used in the industry nowadays.
- **IPC-2581:** Modern file format, it provides information from the board manufacturing to even mounting and testing. It is a single file that comprises everything.
- **Drilling files:** Provides position and size of PTH (Plated Through Hole) and NPTH (Non-Plated Through Hole) drills to the CNC drilling machine.
- **Others:**
 - Board outline
 - Fabrication notes, etc.

2.4.2. PCBA

- **Pick and Place (PnP or CPL):** It provides the X-Y coordinates, rotation and layer side of every component to guide the automated assembly machinery. This file has been modified by hand to fulfill the manufacturer's requirements for this project. A total of 146 components. An example would be the following table:

Table 2.3: Example of PnP file

Designator	Comment	Layer	Footprint	Mid X	Mid Y	Rotation	Description
USBR1	5.1kΩ	Top	RESC160 8X55N	94.4880 mm	87.8840 mm	90	Resistor
GPS2	SM06B-G HS-TB(L F)(SN)	Top	FP-SM06 B-GHS-T B_LF_SN -MFG	93.5590 mm	49.1490 mm	90	CONN HEADER SMD R/A 6POS 1.25MM

Crucial considerations for manufacturing: It is important to note that the position and rotation of components in the design files must strictly match the manufacturer's coordinate system; otherwise, the board will be assembled incorrectly. For this project, JLCPCB was used as the manufacturer. Their online viewer allows for component alignment and PCB design verification.

The production files must comply with the manufacturer's specifications, as manual rotations made within the online viewer will not update the original source files. It is highly recommended to select the "manufacturer file revision" option and manual review when ordering the boards to verify that all components are correctly oriented before production.

- **Bill of materials (BOM):** A comprehensive list of all components required for the assembly. It maps each reference designator to its specific MPN (Manufacturer Part Number) and package type. This file has been modified by hand to fulfill the manufacturer's requirements for this project. A total of 44 different components. In the example, the final column lists the manufacturer part numbers.

Table 2.4: Example of BOM file

Comment	Description	Designator	Footprint	LibRef	Quantity	LSCS part
BMP388,	P-Sensor SPI I2C 1, 25 Bar SMD	BAR	PQFN50P20 0X200X80-1 0N	BMP388	1	C779278
100nF 1mΩ,	CAP CER 0.1UF 16V X7R 0402	BMIC1, BMIC2, ...	FP-GRM15- MFG	CMP-210 0-03641-2	30	C71629

- **Others:**
 - Solder paste file: File used to create stencil, specific solder paste layer of a Gerber file.
 - Assembly drawing: Usually a PDF file containing PCB/PCBA drawing, often used on quality inspection.
 - Test and inspection files: There are multiple files to test multiple things. From AOI (Automatic Optical Inspection) to ICT (In-Circuit Test).
 - 3D STEP file, etc.

CHAPTER 3. SOFTWARE PORTING

This chapter provides an overview of how to port ArduPilot firmware to custom STM32 hardware, as well as the required testing to evaluate its firmware functionality.

3.1. Creation of files

The board is designed to run ArduPilot software. To install this software, both a bootloader and the firmware must be compiled. A bootloader (sometimes called a bootstrap) is a critical program that acts as a bridge between the hardware and the operating system or main application. It functions similarly to a car's starter motor: it initializes the system before the main software takes control. Common examples in personal computing include BIOS and UEFI.

In the STM32 architecture, the bootloader's role extends to flashing new firmware onto the chip. There are 2 types: the manufactured bootloader (located in a ROM like memory, non-erasable and created by STMicroelectronics) and IAP bootloader (In-Application Programming, custom program placed at the beginning of the flash memory, its job is to handle firmware updates before handing over control to the main program).

Building this firmware requires a specific foundation. ArduPilot runs on top of ChibiOS [31], a system that handles the RTOS and HAL functions necessary for the hardware.

As the process of integrating ChibiOS and creating firmware for custom boards can be complex, I will describe it in a simplified guide covering all the necessary steps. This explanation assumes the use of a Windows computer. Check this for more information: [link](#).

First of all, to install Linux, open "Powershell" in admin mode. Install wsl:

```
wsl --install
```

Reboot the computer after the installation. Once rebooted, install ubuntu with your custom user and password:

```
wsl --install -d Ubuntu
```

Then run Ubuntu as admin. It is mandatory to update and install git:

```
sudo apt update && sudo apt install git -y
```

To start with ArduPilot, clone the repository. It will appear in the home directory of the Linux environment.

```
git clone https://github.com/ArduPilot/ardupilot.git
```

Once with ArduPilot, the best option is to update and check if it is usable. To do so, try these commands:

```
cd ardupilot
git submodule update --init --recursive
Tools/environment_install/install-prereqs-ubuntu.sh -y
. ~/.profile
```

To test and check an actual commercial grade drone board code:

```
./waf configure --board MatekH743
./waf copter
```

If at the end of the command line shows completed or something similar it is possible to build a firmware for a microcontroller.

Knowing that the environment works, the next step is to start creating the custom firmware and bootloader. Open the following folder with commands or manually:

```
cd ardupilot/libraries/AP_HAL_ChibiOS/hwdef
```

Inside this directory, it is mandatory to create a new folder, in my case I've named "**CustomH7**". Add the following files into the folder: the hwdef.dat (firmware file), hwdef-bl.dat (bootloader file) and default.parm (optional file for firmware). The explanation about the creation of these files can be found at the end of this chapter.

To check if they are recognized (hwdef-bl and hwdef mainly):

```
ls -la
```

Return to the ArduPilot root directory to build the firmware and the bootloader. The process involves three main steps: building the bootloader, building the firmware and compiling the both. Finally, verify the output files to ensure the build was successful.

```
cd ~/ardupilot
./Tools/scripts/build_bootloaders.py CustomH7
./waf configure --board CustomH7
./waf copter
```

In case of troubleshooting with errors, the following commands are to delete previous data created.

```
./waf distclean  
rm -rf build/
```

All these command lines are devoted to creating arducopter firmware. If the final user only wants to create rover or plane firmwares, it is only necessary to substitute the command:

```
./waf copter or ./waf copter --board CustomH7  
Other possible firmwares to test:  
• ./waf plane  
• ./waf antennatracker  
• ./waf sub  
• ./waf blimp  
• ./waf rover
```

3.2. Uploading

It is time to upload the bootloader and the firmware inside the board. This can be done in multiple ways, via SWD or DFU. SWD is preferable, it is required to download STM32CubeProgrammer or STM32CubeIDE, though I don't really recommend this last one.

All these programmes can be found in “ardupilot/build/CustomH7/bin/”. Here we can find the 3 main files (bootloader, firmware and bootloader + firmware), these can be in the form of .hex, .bin or .apj. Hexadecimal or binary files are for uploaders/debuggers like the STM32CubeProgrammer [32], while .apj file is used to upload the code in Mission planner.

It is possible to upload firstly the bootloader and then the arducopter firmware, though there are faster ways. To keep it simple, inside “ardupilot/build/CustomH7/bin/”, there is a file containing both programs named arducopter_with_bl.hex.

The first step is to open STM32CubeProgrammer and connect the microcontroller via SWD using a debugger (i.e: ST-LINK V3). To ensure a stable connection, it is recommended to set the BOOT0 pin high and perform a power cycle.

Always keep the PCB powered via an external source like a battery. USB power can be unstable and the debugger device often cannot supply current for these boards.

Activating BOOT0 is mandatory to prevent connection failures caused by continuous software resets. This pin forces the board to bypass the Flash and boot from the system memory (manufacturer's internal bootloader).

Upon opening STM32CubeProgrammer, select the interface (typically ST-LINK) and click connect. Once the connection is established, it is recommended to perform a full chip erase to clear previous data. This is done by clicking the button with the eraser icon. These steps can be seen in the following picture.

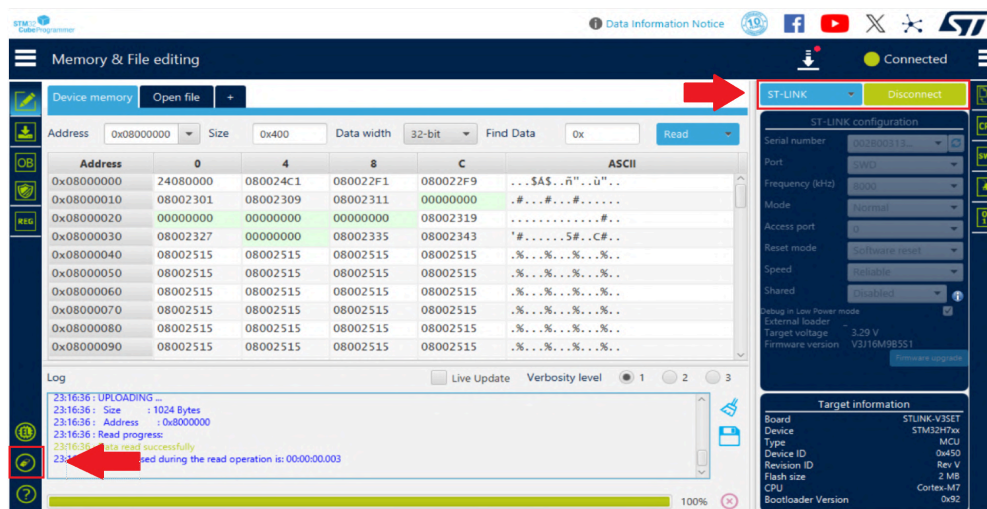


Fig. 3.1 STM32CubeProgrammer connection and memory delete

After connecting, open the arducopter_with_bl.hex file. Verify that the address is set to 0x08000000 (base address of the Flash memory for STM32H743). Then, click download to flash the data into the MCU. Note: the initial bootloader and firmware address is 0x08000000 while considering only firmware is 0x08020000.

Finally, disconnect the debugger, remove the BOOT0 connection, and perform a power cycle to reboot the board into normal operation.

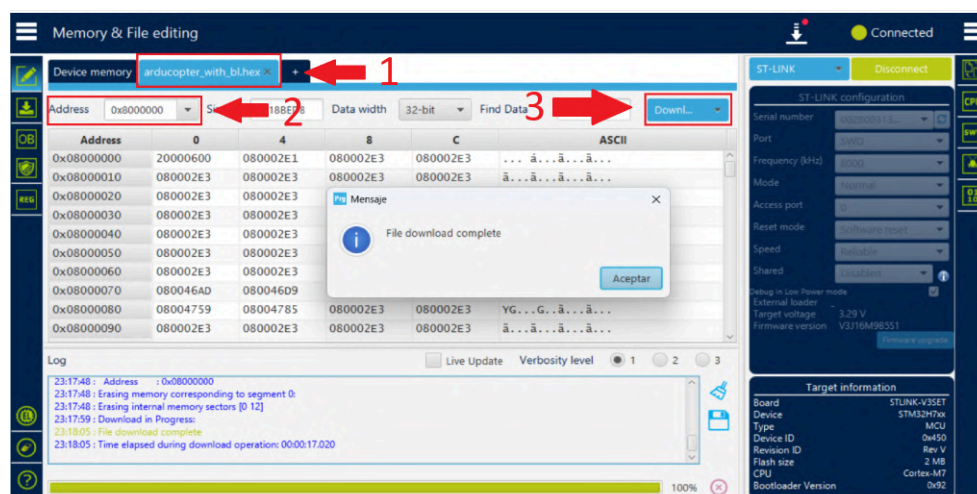


Fig. 3.2 Uploading steps (in step 2, address should be: 0x08000000)

Right now, the board has all necessary programs loaded. If the user uses a microSD card, ensure it is FAT32 formatted. Without a card, the board might not work properly.

To verify that it works, connect the board via USB (it is recommended to power on/off the device a few cycles, sometimes it does not work at first connection). Open device manager; you should see a new device listed under COM/LPT ports. Open Mission Planner, select the new COM port and connect at a baud rate of 115200. In the message section, errors and information will appear.

3.3. Simple firmware testing

To check that the board works well, a minimum of tests shall be applied. The minimum required setup connected to the boards to perform these tests is: a compass or GNSS receiver, a radio receiver and motor or actuator. These are the minimum set of test and their possible solution:

- The board boots correctly and connects to the GCS via USB. If not, check power distribution, clock configuration and hardware design.
- Firmware loads and parameters can be read and written from the GCS. Otherwise check flash configuration and bootloader settings.
- If the HUD shows messages such as “failed to init xyz”, the corresponding peripheral has not been configured correctly in hwdef.dat or in parameters.
- RC input, GNSS receiver and compass are detected and provide valid data. If not, verify hardware, protocol selection and configuration.
- Verify IMU detection and orientation. IMU orientation can be tested by enabling/disabling each IMU using the INSx_USE parameters.
- Check that the board can arm correctly. If arming fails, check hardware and configuration, maybe force it via parameters.
- Verify current and voltage measurements. If values are incorrect, check analog scaling in hwdef.dat and parameters.
- Test motor and servo outputs. If outputs do not respond, check pin assignment, PWM configuration and hardware design.
- Verify that all designed communication interfaces like UART, SPI, I2C, CAN, etc; are functional.
- Other important tests can be: barometer test, failsafe test, logging test (microSD test), reboot and power-cycle tests, DMA test or thermal stability test.

3.4. hwdef.dat, hwdef-bl.dat and default.parm

These hwdef.dat and hwdef-bl.dat files are the most important files to port a microcontroller into ArduPilot. These files should not contain any error in typing and in order, because minimal changes won't allow ChibiOS to create the firmware and the bootloader, or even they won't work once uploaded inside the MCU, most probably blocking the communication with the SWD bus and with any other available bus.

The hardware and software of the board have to be implemented according to ArduPilot python scripts. These rules explain what components can be used and

what functions the microcontroller pins can take, the documentation can be found in the following [link](#).

Regarding the software files, a good example is the [fmuv3](#).

3.4.1. hwdef.dat

This file allows the creation of the firmware. The first step is to define the board nucleus and its memory in KB in binary.

```
# Core
MCU STM32H7xx STM32H743xx

FLASH_SIZE_KB 2048
FLASH_RESERVE_START_KB 128
```

Followed by the clock configuration, the system timer and the board information. The clock frequency is 24 MHz, both clocks in bypass configuration. We want a stable system frequency of 400 MHz using the PLL (Phase-Locked Loops) and the general system timer 5. Board information ID/USB_product stands as 1001 for prototyping, USB_Vendor is optional though a random vendor can be used for testing. Lastly, the name of the board and the product.

```
# Clock
OSCILLATOR_HZ 24000000
STM32_HSE_BYPASS
STM32_LSE_BYPASS
STM32_SYSCLK 400000000
STM32_ST_USE_TIMER 5

# Board names
APJ_BOARD_ID 1001
USB_VENDOR 0x2DAE
USB_PRODUCT 0x1001
USB_STRING_MANUFACTURER "CustomH7"
USB_STRING_PRODUCT "FlightController"
```

Defining the buses pins:

```
# I2C buses
PB6 I2C1_SCL I2C1
PB7 I2C1_SDA I2C1
PB10 I2C2_SCL I2C2
PB11 I2C2_SDA I2C2

# ICM SPI bus
PA5 SPI1_SCK SPI1
PA6 SPI1_MISO SPI1
PA7 SPI1_MOSI SPI1
PA4 ICM_CS CS

# External SPI bus
PB13 SPI2_SCK SPI2
```

```

PB14 SPI2_MISO SPI2
PB15 SPI2_MOSI SPI2
PE8 EXT_CS1 CS
PB12 EXT_CS2 CS

# BMI SPI bus
PE12 SPI4_SCK SPI4
PE13 SPI4_MISO SPI4
PE14 SPI4_MOSI SPI4
PE11 BMI_ACCEL_CS CS
PE10 BMI_GYRO_CS CS

# USB FS bus
PA11 OTG_FS_DM OTG1
PA12 OTG_FS_DP OTG1
PA9 VBUS INPUT

# Sensor data ready pins
PE9 BMI_ACCEL_DRDY INPUT
PE15 BMI_GYRO_DRDY INPUT
PC4 ICM_DRDY INPUT

# Serial pins (NODMA = No DMA)
PC6 USART6_TX USART6
PC7 USART6_RX USART6
PD5 USART2_TX USART2
PD6 USART2_RX USART2
PD8 USART3_TX USART3
PD9 USART3_RX USART3
PE0 UART8_RX UART8
PE1 UART8_TX UART8
PE7 UART7_RX UART7 NODMA

# CAN bus pins
PB8 CAN1_RX CAN1
PB9 CAN1_TX CAN1

```

Definition of buses, battery measurement and motor pins.

```

# Order of buses: (0, 1, 2, 3, 4, 5, ...)
I2C_ORDER I2C1 I2C2
SERIAL_ORDER OTG1 USART6 USART2 USART3 UART7 UART8

# PWM with GPIO capability (always starting from 50), (TIM=timers)
PA0 TIM2_CH1 TIM2 PWM(1) GPIO(50)
PA1 TIM2_CH2 TIM2 PWM(2) GPIO(51)
PA2 TIM2_CH3 TIM2 PWM(3) GPIO(52)
PA3 TIM2_CH4 TIM2 PWM(4) GPIO(53)
PB0 TIM3_CH3 TIM3 PWM(5) GPIO(54)
PB1 TIM3_CH4 TIM3 PWM(6) GPIO(55)
PD12 TIM4_CH1 TIM4 PWM(7) GPIO(56)
PD13 TIM4_CH2 TIM4 PWM(8) GPIO(57)

# Battery measurement ADCs with scale factor
PC0 BATT_CURRENT_SENS ADC1 SCALE(1)
PC1 BATT_VOLTAGE_SENS ADC1 SCALE(1)

```

```
# Sensor address and velocity of bus
I2CDEV baro I2C1 0x76 400000
I2CDEV compass I2C1 0x10 400000

# Sensor bus speed definition range, SPI mode and number of connected devices
SPIDEV icm20602 SPI1 DEVID1 ICM_CS MODE3 2*MHZ 8*MHZ
SPIDEV bmi088_g SPI4 DEVID1 BMI_GYRO_CS MODE3 8*MHZ 8*MHZ
SPIDEV bmi088_a SPI4 DEVID2 BMI_ACCEL_CS MODE3 8*MHZ 8*MHZ
SPIDEV spi_ext SPI2 DEVID1 EXT_CS1 MODE0 2*MHZ 8*MHZ
```

Sensor drivers and rotation, MicroSD configuration, storage and DMA priority. True rotation can be verified with Mission Planner by monitoring the HUD (Heads-Up Display) in the Flight Data window, ensuring that the artificial horizon responds correctly to the physical board movements or by using the tuning window.

```
# Sensor drivers, address and rotation
BARO BMP388 I2C:0:0x76
COMPASS BMM150 I2C:0:0x10 false ROTATION_NONE

#Sensor drivers and rotation
IMU InvenSense SPI:icm20602 ROTATION_YAW_90
IMU BMI088 SPI:bmi088_a SPI:bmi088_g ROTATION_PITCH_180

# MicroSD card pins definition and SD mode detection
PC8 SDMMC1_D0 SDMMC1
PC9 SDMMC1_D1 SDMMC1
PC10 SDMMC1_D2 SDMMC1
PC11 SDMMC1_D3 SDMMC1
PC12 SDMMC1_CK SDMMC1
PD2 SDMMC1_CMD SDMMC1
PA8 SD_DETECT INPUT PULLUP

# System PID page storage (usually page 14, away from the main code) and size to store values
STORAGE_FLASH_PAGE 14
define HAL_STORAGE_SIZE 32768

# Bus DMA priority (IMU sensors first on the queue)
DMA_PRIORITY SPI1* SPI4* CAN1* SDMMC* USART6*
```

3.4.2. hwdef-bl.dat

This file allows the creation of the bootloader. Really similar to hwdef.dat but much simpler.

```
# Core
MCU STM32H7xx STM32H743xx

# Clock configuration (frequency and mode)
OSCILLATOR_HZ 24000000
STM32_HSE_BYPASS
STM32_LSE_BYPASS

# Board names
APJ_BOARD_ID 1001
USB_VENDOR 0x2DAE
```

```

USB_PRODUCT 0x1001
USB_STRING_MANUFACTURER "CustomH7"
USB_STRING_BOOTLOADER "CustomH7 Bootloader"

```

Regarding the memory it is mandatory to use this strict configuration:

```

# Mandatory configuration for Bootloader STM32H743
FLASH_SIZE_KB 2048
FLASH_RESERVE_START_KB 0
FLASH_BOOTLOADER_LOAD_KB 128
APP_START_OFFSET 0x20000

# Bus pins
PA11 OTG_FS_DM OTG1
PA12 OTG_FS_DP OTG1
PA9 VBUS INPUT

PC6 USART6_TX USART6
PC7 USART6_RX USART6

# Chip select (often applied in hwdef-bl.dat)
PA4 ICM_CS CS
PE11 BMI_ACCEL_CS CS
PE10 BMI_GYRO_CS CS
PE8 EXT_CS1 CS
PB12 EXT_CS2 CS

```

Configuration of bus order and SDMMC. It is mandatory to use the FATFS library to write and read the MicroSD. Moreover SD is enabled to modify and upload firmware.

```

# Bus serial order of the bootloader
SERIAL_ORDER OTG1 USART6

# Standard configuration drivers for SDMMC (MicroSD)
define CH_CFG_USE_MEMCORE 1
define CH_CFG_USE_HEAP 1
define CH_CFG_USE_SEMAPHORES 0
define CH_CFG_USE_MUTEXES 1
define CH_CFG_USE_DYNAMIC 1
define CH_CFG_USE_WAITEXIT 1
define CH_CFG_USE_REGISTRY 1

# SDMMC pins (MicroSD)
PC8 SDMMC1_D0 SDMMC1
PC9 SDMMC1_D1 SDMMC1
PC10 SDMMC1_D2 SDMMC1
PC11 SDMMC1_D3 SDMMC1
PC12 SDMMC1_CK SDMMC1
PD2 SDMMC1_CMD SDMMC1
PA8 SD_DETECT INPUT PULLUP

# Allow the use of FATFS for the MicroSD
define HAL_OS_FATFS_IO 1

# Allow the user to upload the firmware via SD
define AP_BOOTLOADER_FLASH_FROM_SD_ENABLED 1

```

3.4.3. default.parm

This file modifies the parameters inside ArduPilot, they can be changed manually using Mission Planner. That's because it is an optional file.

```
# Serial 1 (following the order in hwdef, it is UART 6)
# Protocol 5 (stands for GNSS receiver) and velocity of communication
SERIAL1_PROTOCOL 5
SERIAL1_BAUD 115

# Serial 2 (following the order in hwdef, it is USART 2)
# Protocol 1 (stands for telemetry) and velocity of communication
SERIAL2_PROTOCOL 1
SERIAL2_BAUD 57

# Serial 3 (following the order in hwdef, it is USART 3)
# Protocol 5 (stands for GNSS receiver) and velocity of communication
SERIAL3_PROTOCOL 5
SERIAL3_BAUD 115

# Serial 4 (following the order in hwdef, it is UART 7)
# Protocol 23 (stands for RCIN)
SERIAL4_PROTOCOL 23
SERIAL4_OPTIONS 0

# Serial 5 (following the order in hwdef, it is UART 8)
# Protocol 1 (stands for telemetry) and velocity of communication
SERIAL5_PROTOCOL 1
SERIAL5_BAUD 57

# DroneCAN and CAN driver
CAN_D1_PROTOCOL 1
CAN_P1_DRIVER 1

# Disable safety pin (safety pin usually is required in some special boards)
# Delay 3 ms to wait sensor response
BRD_SAFETYENABLE 0
BRD_BOOT_DELAY 3000

# Battery configuration for HEX Cube Power Brick
BATT_MONITOR 4
BATT_VOLT_MULT 10.1
BATT_AMP_PERVLT 17.0

# IMU and compass orientation
AHRS_ORIENTATION 0
COMPASS_ORIENT 0
```

CHAPTER 4. SPACE APPLICATIONS AND REQUERIMENTS

This chapter investigates potential aerospace applications for drones and presents a preliminary analysis of the designed board, focusing on the microcontroller's radiation tolerance and its resilience against SEUs (Single Event Upset).

It is recommended to consult ANNEX-2, where Western industrial space standards have been overviewed, as well as the space environment for drones.

4.1. Space Applications

In outer space, there are few sites where a UAV (Unmanned Aerial Vehicle) can be flown easily. Being an aircraft without a human pilot, a UAV requires a significant amount of atmosphere to operate. The term “drone” can describe an unmanned vehicle used in different environments such as ocean, air, terrain, or space. Despite this versatility, only surface aerial vehicles are studied in this project.

Some of the solid-surface environments where aerial drones can operate are listed below. While other bodies exist, these are the primary candidates where a UAV, with specific modifications, could be deployed.

There are multiple possible applications. As mentioned previously, it is mandatory to fly the vehicle in an atmosphere environment to consider it as a UAV. Some of them can be:

- **Applications inside the ISS:**

The drone can be used as a supporting entity inside the ISS to monitor data, evaluate the environment, and assist with daily tasks.

This application was demonstrated by the Int-Ball (Internal Ball Camera) drone, created by JAXA. The experiment's objective was to evaluate the perspective inside the ISS and reduce the astronauts' workload, establishing a direct connection to ground stations.

Other possibilities include environmental monitoring and hazard mitigation, such as detecting fires or damage from space debris impacts and acting upon them. The biggest constraint in this scenario is the thermal dissipation of the drone's electronics.

- **Applications in planet atmospheres:**

- **Mars:** The first drone flying on this planet was the Ingenuity. Its mission was to scan the environment, map it, and evaluate the best route for the Perseverance rover.

The biggest constraints found were the low air density, pressure, and extreme temperature changes during day-night cycles (thermal cycling).

Currently, NASA is developing new drones to operate inside the Mars atmosphere.

There are other applications for the use of drones, such as: transport of goods, collection of samples in difficult access zones, non-static meteorological stations, and acting as transmitters of information between stations.

- **Venus and Titan:** In the case of Venus, the biggest drawbacks are the extreme pressure, temperature and corrosion. While applications similar to Mars were considered, these constraints prevent surface operations. VAMP (Venus Atmospheric Maneuverable Platform) is a concept drone designed to fly at an altitude of 50-65 km (a layer with more favorable conditions) for data acquisition, such as mapping, radiation analysis, and atmospheric study.

In the case of Titan, a drone mission is much more feasible. Scheduled for 2028, NASA plans to send a drone named Dragonfly to search for organic components. The main constraint on Titan is the extremely low ambient temperature. Similar to previous examples, multiple purposes are sought: mapping, geologic exploration, sample extraction and analysis.

4.2. Spenvis Radiation Simulations and PCB Changes

To obtain space environment data, the SPENVIS program [33] developed by ESA was used. Due to scope constraints, only Earth and Mars evaluations were considered. All simulations have been performed using an estimated mission duration of 2 years. As the solar activity period varies with time (11.2 year cycle), all analyses have been calculated starting from January 1, 2026, to align with high solar activity.

The main COTS (Commercial-off-the-shelf) components analyzed are:

- Primary component STM32H743VIT6 processor .
- Secondary components:
 - Drone sensors: ICM20602, BMI088, BMP388 and BMM150.

The TID (Total Ionizing Dose) evaluation was performed using the Shieldose-2 model, considering a silicon target and aluminium shielding.

Defining the input parameters for long/short-term SEU has been a complex task and it is a mere approximation. Due to absence of specific radiation test data for the STM32H743VIT6. Since the STM32H7 relies on a 40 nm CMOS process, the sensitive volume approximated to $0.2 \mu\text{m} \times 0.2 \mu\text{m} \times 0.2 \mu\text{m}$, this approximation is coherent with current 40 nm processors (typically x5, in other words: $40\text{nm} \times 5 = 200 \text{ nm} = 0.2 \mu\text{m}$) [34].

The parameters chosen for the Weibull function are:

- $1 \text{ MeV} \cdot \text{cm}^2 / \text{mg}$: As it is not a hardened device, this low threshold assumes high sensitivity to ionizing particles.

- $5 \cdot 10^{-8} \text{ cm}^2/\text{bit}$: This saturation cross-section corresponds to the typical physical area of an SRAM cell in 40 nm technology.
- $W = 80$ and $S = 1$: Generic Weibull curve.

Regarding proton induced upset rates, the model used has been the Bendel function with standard values $A = 12 \text{ MeV}$ and $B = 7.09 \text{ MeV}$, standard values for generic silicon with 40 nm.

4.2.1. LEO (Low Earth Orbit) Radiation

One of the worst LEOs is the SSO (Sun-Synchronous Orbit). This orbit is affected by the SAA (South-Atlantic Anomaly), the fluence of protons and electrons from north and south poles by the magnetic field and the general outer space radiation from SPEs (Solar Particle Events) and GCRs (Galactic Cosmic Rays). The chosen orbital parameters are the 700 km height (semi-major axis 7078 km), 0° of eccentricity, 98° of inclination. Additional parameters are not considered important for these simulations. This was chosen to replicate the worst case scenario that the ISS orbit would experience.

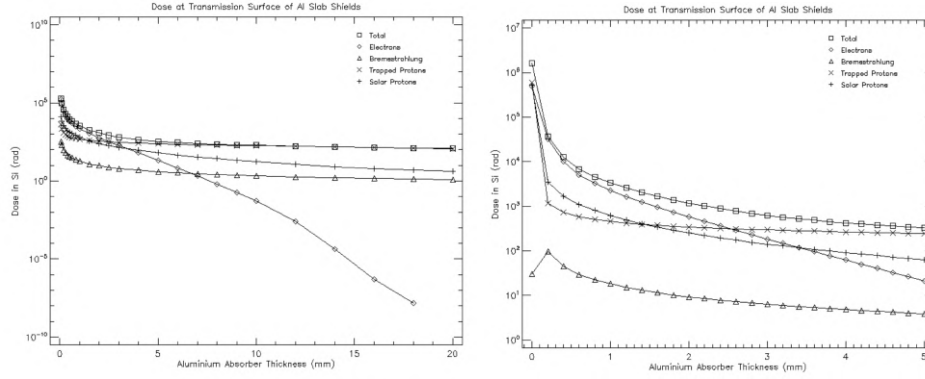


Fig. 4.1 LEO Shieldose-2 results (same plot)

The maximum TID (Total Ionizing Dose) demonstrated and documented for STM32H753 is 47 krad, (version H743 is practically identical, [35]). To withstand this radiation level during 2 years of mission and a RDM (Radiation Design Margin) of 2, the maximum radiation environment should not exceed 23.5 krad.

A minimum of 0.5 mm of aluminum shielding satisfies this requirement. However, for structural and safety reasons, thicknesses of 2 mm and 4 mm have been implemented, obtaining a total dose of less than 5 krad for the complete mission.

For the calculation of SEU (Single Event Upsets): AP-8 (protons), SAPHIRE-SPEs, ISO-15390-GCR and CREME-96 models are considered. The STM32H743 contains 1MB of RAM (main focus as it is volatile and sensitive). This 1MB contains 1048576 bytes of data, in other words, 8388608 bits. The calculation of the SEUs is using the following formula:

$$SEU_{device} = SEU_{spenvis} \cdot \text{Number of bits} \quad (4.1)$$

Long-term SEU simulations consider the total mission life with average radiation values. For 2 mm of aluminum is 910 errors, while for 4 mm is 246 errors (along 2 years).

Table 4.1-2: Long-term SEU for 2-4 mm of Al

Total mission single event upsets			Total mission single event upsets		
Device	Effect	(bit ⁻¹)	Device	Effect	(bit ⁻¹)
STM32H7	Direct ionization	1.0685E-04	STM32H7	Direct ionization	2.7863E-05
	Proton induced ionization	1.6128E-06		Proton induced ionization	1.5037E-06
	Total	1.0846E-04		Total	2.9367E-05

Short-term SEU represents the worst-case simulation scenario, where a high amount of radiation interacts with the device over a short period. Considering the data in the table below and the microcontroller's memory size, a 2 mm shield would create 1653 errors per day, while using a thickness of 4 mm reduces this to around 451 SEUs per day.

Table 4.3: Short-term SEU for 2 mm of Al

Segment averaged and total SEU rates							
		Mission total			Mission segment 1		
Device	Effect	(bit ⁻¹)	(bit ⁻¹ s ⁻¹)	(bit ⁻¹ day ⁻¹)	(bit ⁻¹)	(bit ⁻¹ s ⁻¹)	(bit ⁻¹ day ⁻¹)
DEFAULT	Direct ionization	1.4375E-01	2.2791E-09	1.9692E-04	1.4375E-01	2.2791E-09	1.9692E-04
	Proton induced ionization	1.5344E-04	2.4328E-12	2.1019E-07	1.5344E-04	2.4328E-12	2.1019E-07
	Total	1.4390E-01	2.2816E-09	1.9713E-04	1.4390E-01	2.2816E-09	1.9713E-04

Table 4.4: Short-term SEU for 4 mm of Al

Segment averaged and total SEU rates							
		Mission total			Mission segment 1		
Device	Effect	(bit ⁻¹)	(bit ⁻¹ s ⁻¹)	(bit ⁻¹ day ⁻¹)	(bit ⁻¹)	(bit ⁻¹ s ⁻¹)	(bit ⁻¹ day ⁻¹)
DEFAULT	Direct ionization	3.9146E-02	6.2066E-10	5.3625E-05	3.9146E-02	6.2066E-10	5.3625E-05
	Proton induced ionization	1.1056E-04	1.7529E-12	1.5145E-07	1.1056E-04	1.7529E-12	1.5145E-07
	Total	3.9257E-02	6.2241E-10	5.3776E-05	3.9257E-02	6.2241E-10	5.3776E-05

This discrepancy between both values long and short terms is coherent, it exposes that during the mission the errors will be reduced but when a dangerous SPE occurs, the number of SEUs can be increased drastically. A short event can be more dangerous than the whole microcontroller life in the entire mission. This discrepancy requires protection strategies like watchdogs, ECC and redundancy.

To avoid these problems STM32H743 contains ECC (Error Correction Code), CRC (Cyclic Redundancy Check), IWDG (Independent Watchdog), WWDG (Window Watchdog) and MPU (Memory Protection Unit). Despite these commercial grade methods, it seems that the STM32 can resist these conditions with possible reboots but real test data and further research is necessary to confirm these aspects with possible degradation over time.

Moreover as a curiosity it is possible to plot radiation flux vs LET (Linear Energy Transfer) in short and long term SEU in Spenvis. When higher LET, less fluence we obtain, which is coherent.

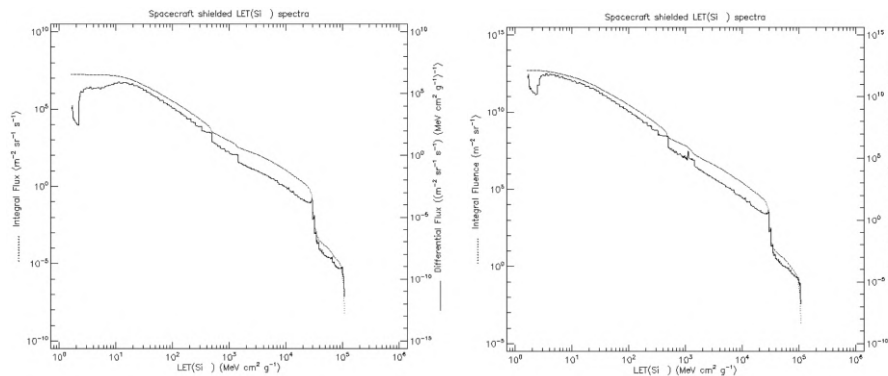


Fig. 4.2 Fluence vs LET with 4 mm Al (Left→Short-term) (Right→ Long-term)

4.2.2. Mars Radiation

Since the Martian surface cannot be simulated with SPENVIS, the closest circular orbit considered is at an altitude of 100 km, with an inclination of 98°. In this environment the atmosphere and magnetic attenuation is minimum compared to the Earth. AP-8 and AE-8 models are not required because protons and electrons are not trapped as in Van-Allen belts (they don't exist), reducing these sources of radiation. The models that have been considered are SAPHIRE for SPEs, ISO-15390 for GCR and CREME-96 to simulate.

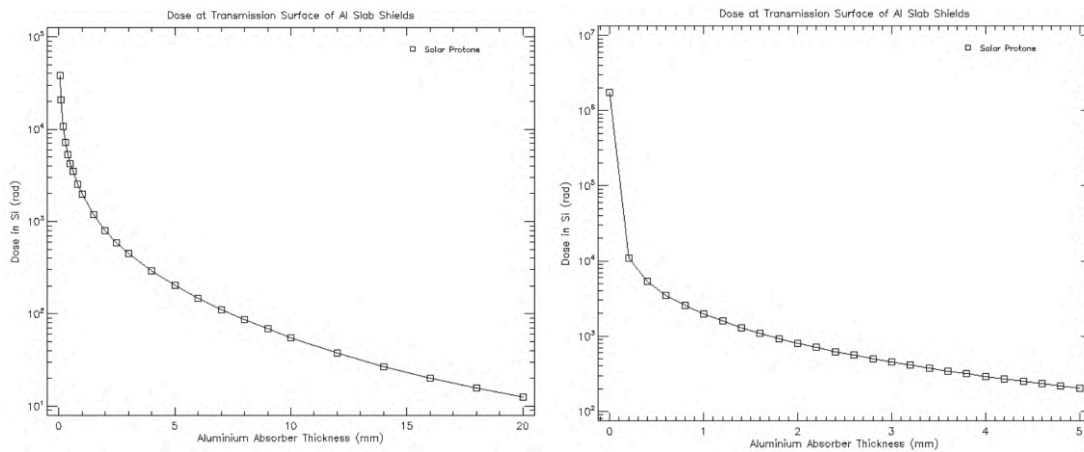


Fig. 4.3 Mars Shieldose-2 results (same plot)

Using Shieldose-2, the minimum thickness of aluminum should be 0.02 mm for 23.5 krad. For structural and safety reasons 2 and 4 mm of aluminium have been considered, obtaining less than 5 krad.

In case of long-term SEU simulation, the number of errors is 2858 for 2 mm and 761 for 4 mm of aluminum.

Table 4.5-6: Long-term SEU for 2-4 mm of Al

Total mission single event upsets			Total mission single event upsets		
Device	Effect	(bit ⁻¹)	Device	Effect	(bit ⁻¹)
STM32H7	Direct ionization	3.4007E-04	STM32H7	Direct ionization	9.0241E-05
	Proton induced ionization	5.8561E-07		Proton induced ionization	4.5158E-07
	Total	3.4065E-04		Total	9.0692E-05

In case of short-term SEU simulation, the number of errors per day is 3235 for 2 mm and 614 for 4 mm of aluminum.

Table 4.7: Short-term SEU for 2 mm of Al

Segment averaged and total SEU rates							
		Mission total			Mission segment 1		
Device	Effect	(bit ⁻¹)	(bit ⁻¹ s ⁻¹)	(bit ⁻¹ day ⁻¹)	(bit ⁻¹)	(bit ⁻¹ s ⁻¹)	(bit ⁻¹ day ⁻¹)
DEFAULT	Direct ionization	2.8117E-01	4.4579E-09	3.8516E-04	2.8117E-01	4.4579E-09	3.8516E-04
	Proton induced ionization	3.5394E-04	5.6117E-12	4.8485E-07	3.5394E-04	5.6117E-12	4.8485E-07
	Total	2.8152E-01	4.4635E-09	3.8565E-04	2.8152E-01	4.4635E-09	3.8565E-04

Table 4.8: Short-term SEU for 4 mm of Al

Segment averaged and total SEU rates							
		Mission total			Mission segment 1		
Device	Effect	(bit ⁻¹)	(bit ⁻¹ s ⁻¹)	(bit ⁻¹ day ⁻¹)	(bit ⁻¹)	(bit ⁻¹ s ⁻¹)	(bit ⁻¹ day ⁻¹)
DEFAULT	Direct ionization	5.3157E-02	8.4280E-10	7.2818E-05	5.3157E-02	8.4280E-10	7.2818E-05
	Proton induced ionization	2.4879E-04	3.9445E-12	3.4080E-07	2.4879E-04	3.9445E-12	3.4080E-07
	Total	5.3406E-02	8.4675E-10	7.3159E-05	5.3406E-02	8.4675E-10	7.3159E-05

It is coherent to encounter more TID (Total Ionizing Radiation) at LEO orbit than on Mars, due to the high flux of trapped protons-electrons in the Van Allen belts and the SAA (South Atlantic Anomaly). On the other hand, we have more SEUs on Mars, due to the absence of geomagnetic shielding, exposing the device to the unattenuated spectrum of GCR (high-LET ions), which are the primary drivers of SEU in deep space.

In conclusion, the protection strategy must be adapted to the orbit: while physical shielding is the critical countermeasure for LEO to mitigate TID, it is less effective against high-energy particles in the Martian environment. Therefore, Mars missions rely mainly on active mitigation (redundancy, ECC mechanisms) to maintain stability against frequent errors.

4.2.3. Conclusion of PCB, COTS and general changes

- **Regarding COTS (Commercial-off-the-shelf) components:** A rigorous qualification and selection process is mandatory. Since components usually contain distinct internal structures ranging from pure silicon dies to mechanical vibrating elements. These shall operate well below their rated limits to reduce stress and extend operational life (derating). To mitigate outgassing risks, a thermal vacuum bake-out campaign is mandatory prior to integration.

Redundancy is a must in this type of missions; heterogeneous redundancy, LCL (Latch-up Current Limiters), and software redundancy are common techniques. All components shall work inside the operating temperature and pressures and always placed near mounting points with staking (epoxy) to withstand accelerations and vibrations. In LEO, the primary challenge is heat dissipation due to the vacuum, while Mars requires active heating to prevent component freezing.

Typical drone components like the BMM150 (magnetometer) can measure the magnetic field in LEO conditions, its resolution is capable of doing so. In the case of the BMP388 (barometer) it won't work in LEO and on Mars. It would be able to evaluate the environment only if it is inside the ISS or during launch phase. Filling critical structures with SF6 gas can suppress arcing (and could be monitored by the BMP388), it adds structural complexity. Therefore, solid potting or conformal coating is selected as the primary mitigation strategy against arcing and corona effects.

- **Regarding PCB design and materials:** A high-Tg material such as a special FR-4 and epoxy substrate is selected to ensure mechanical stability. To optimize thermal dissipation and structural integrity, the layout should incorporate thermal vias, teardrop connections, continuous ground planes (inner and outer), and maximize copper pours.

To mitigate tin whiskers it is mandatory to use SnPb solder. For surface finish, ENIG (Electroless Nickel Immersion Gold) is preferred over HASL (Hot Air Solder Leveling) due to its superior resistance to corrosion and atomic oxygen, due to gold being non-reactive.

To protect against ESD (Electrostatic Discharge), TVS space grade diodes are considered. Spark gaps are not effective in void conditions, and for low pressure atmospheres they can produce short circuits as their basis is on Paschen law. To avoid arcing and coronas effect, copper connection shall be rounded and never perpendicular or sharp edges. Lastly, a conformal coating is required to prevent electrical arcing, protect against conductive debris, and mitigate thermal stress.

- **Note:** Sharp edges are prohibited in electronics because they concentrate the electric field, causing uncontrolled corona discharges and arcing. However, on the spacecraft exterior, this effect is useful. Tungsten needles are installed to intentionally bleed off static charge via field emission, preventing high-voltage accumulation that could damage the solar panels.

CHAPTER 5. TESTS, ISSUES, COST AND ENVIRONMENT

This chapter details from a broad perspective, the performed tests with the designed flight controller, the resolved and remaining issues, the cost of the prototype's production, as well as its environmental and social impacts.

5.1. Tests and issues

The developed board was validated using a custom 3D printed case integrated into a Hexsoon EDU 450 drone frame, equipped with a Here4 (GNSS and magnetometer), radio receiver, and telemetry module. All peripherals were connected via standard Pixhawk connectors. 3 flight tests were conducted in Loiter mode: an anchored safety test, a free flight within a geofence, and an autonomous 4 waypoint mission. **The board demonstrated excellent stability and full functionality in all tests.**

Additionally, the direct power connection was tested using a 24 V power supply with no issues. However, specific requirements were identified: a 24 MHz HSE (High Speed External oscillator) is mandatory for firmware compatibility, and ESD protection improvement on specific buses. During testing, continuous handling caused an electrostatic discharge that damaged the oscillator, requiring its replacement. Moreover, the onboard magnetometer could not be calibrated due to environmental noise, and MicroSD communication remained inactive, likely due to a configuration conflict in hwdef.dat or hardware crosstalk.



Fig. 5.1 Left→ First test with anchor, Right→ Second test within a geofence

For more information check ANNEX-3: Testing, Project Issues, Comparison and Cost.

5.2. Hardware budget

The cost of manufacturing the boards is shown in the following table:

Table 5.1: Prototype cost

Object:	Cost with taxes:
JLCPCB production This contains: - 5 boards (2 mounted) - 1 stencil board	344.04 €: - 227.29 € material - 116.75 € tariffs
3 extra clocks to substitute	27 €
3D printed case material and screws	2.5 € approx
Total approx:	373.54 €

This is only a rough approximation. Unit costs are higher during prototyping but decrease significantly when manufactured at scale.

While the prototype unit cost is 186.77 €, mass production would drop the price to 65-85€ approx compared with direct competitors, which typically oscillate between 90 € and 160 €. For instance: MATEK H743 90-120€, Holybro Kakute H7 v2 120 €, and the Holybro Pixhawk 6C Mini 150 €. Specialized industrial variants like the mRo Control Zero 280 € or CUAV Nora+ 220 € represent a higher price.

5.3. Environmental and social impact

All components used, as well as the PCB with a Lead-Free HASL finish, are RoHS compliant. The main source of energy used during this project has been electricity. The electrical consumption and the Co2 produced by this consumption, supposing 0.246 kgCO₂/kWh (data from 2019) and 0.15 €/kWh approx:

Table 5.2: Co2 production and Cost

Material	Consumption (approx)	Hours of consumption (approx)	Cost approx (0.15 €/kWh)	Co2 produced approx (0.246 kg/kWh)
Laptop	100 W	+600 h	9	14.76
		TOTAL:	9 €	14.76 kg

In terms of social impact, this project addresses SDG 9 (Sustainable Development Goals) (Industry, Innovation, and Infrastructure) and SDG 4 (Quality Education), [36]. By exposing the design process of a complex PCB, it promotes an open-hardware philosophy, demonstrating that industrial-grade control systems (STM32H7-based) can be achieved at a fraction of the cost of alternatives. Consequently, the development of cost-effective flight controllers facilitates the use of drones for humanitarian purposes globally, such as localization of missing persons or delivering supplies.

CONCLUSIONS

This TFM covers several topics, many of them from a broad perspective. The main focus was the development of control hardware for drones. All the objectives have been successfully completed:

1. Develop a functional drone control PCB.
2. Create the drivers to manage the drone control PCB with ArduPilot.
3. Evaluate the current design and expose space requirements.

To achieve these goals, standardization played an essential role in both hardware and software development. Following industry standards and manufacturer recommendations is vital to prevent errors during the manufacturing process and to ensure compatibility between complex systems.

Although the main objectives were met, some technical challenges remained. Specifically, the magnetometer and MicroSD functions could not be fully established on the board. Future versions will be necessary to implement these features, along with more robust ESD protection and system redundancy to increase reliability. Furthermore, moving toward a multi-layer design and using double-sided mounting would significantly reduce the board's area, though this involves a tradeoff with higher manufacturing costs. Other technical improvements, such as optimizing DMA sharing and refining hwdef files, are also mandatory for a final product.

In conclusion, this TFM, which was developed during 5 months, demonstrates that it is feasible to create a custom drone control PCB and port it to the ArduPilot firmware. Additionally, the project included an evaluation of the board's component potential for space missions and possible applications. This is an open-source work that stands between a hobbyist and a professional level. Since it is an experimental design, its performance cannot be fully guaranteed, but it serves as a solid base for future developments.

All documents created available can be found in the following link:
https://github.com/AlexBoa45/TFM_Custom_FC_Ardupilot.git

ANNEX- 1. BASIC PCB ANALYSIS

This annex provides a simple overview of how commercial drone PCBs are built. It first explains the main components of a drone control PCB and then presents 4 examples of actual hobby and commercial PCBs.

A-1.1. Main components of a drone control PCB

The principal objective of this TFM is to create a simple PCB to control a multirotor drone; however, many PCBs are designed to add the FPV goggle system with the control system. This characteristic will not be explained in this TFM as well as the development communication system, because it compromises much more theory and workload making a very extensive thesis. It will only focus on the control basis.

There are several components that can be part of a drone control system, some of these are:

A-1.1.1. Computational Unit

The core of any drone is the computational unit. Though there are many kinds of computers, drones only use three types of dedicated computers. These are:

A-1.1.1.1. MCU: Microcontrollers

These computational units (Microcontrollers) are the most common in drones today. They allow for real-time flight control, manage multiple sensors, offer low power consumption and low cost. While they handle lightweight tasks easily, they present specific constraints: limited processing capacity, low memory, and the inability to execute complex Operating Systems.

Some of these microcontroller are:

- **STM32 family:** Created by STMicroelectronics, it is a family of microcontrollers of 32 bits based on ARM Cortex-M structure. These chips are the main option in the development of drone control PCB, because of many reasons such as: cheap, good consumption, they can work at high frequencies, good integration of peripherals, usage of multiple interruptors, DMA handling and general reliable support in industry. Though it can be a bit difficult to program in comparison with ESP32 and ATmega.
- **ESP32 family:** Created by Espressif Systems, it is a family of microcontrollers of 32 bits based on Xtensa LX6/LX7 and RISC-V. These chips are a pretty used alternative to STM32 controllers. Its main difference is the usage in communications systems mainly. They are very cheap and supported by a large community.
- **Atmega family:** Created by Microchip, it is a family of microcontrollers of 8 bits based on AVR architecture. Very simple, cheap microcontrollers, used on

multiple boards like Arduino Uno. The main clock goes at 8-20 Mhz. From a development perspective, basic drones can be created using these controllers, specifically through bare-metal programming. However, for high-performance or complex UAVs, the use of these basic microcontrollers is not enough.

- **NXP Kinetis and i.MX RT family:** Created by NXP and Freescale (same company since 2015), it is a family of microcontrollers of 32 bits based on ARM Cortex-M7. Some people say i.MX RT chips are a “crossover MCU”, this is because these chips tend to be similar to MPU (Microprocessor Unit) (up 600 Mhz). Used on industrial and high control applications. Indeed, NXP is an official partner of PX4 products.
- **TI MSP430 or Tiva-C family:** Created by Texas Instruments, it is a family of microcontrollers of 16-32 bits based on MSP430 and ARM Cortex-M4F architecture. MSP430 chips are being used on ultra low power consumption and sensor treatment while Tiva-C (known as Stellaris before), used on drone payload applications.
- **Renesas RX or RA family:** Created by Renesas, it is a family of microcontrollers of 32 bits based on ARM Cortex-M23 architecture. These MCUs are used in automotive, aerospace and industrial applications that carry high safety features such as ISO 26262.

A-1.1.1.2. MPU: Microprocessors

Despite being used mainly on higher levels of data processing, it is a considerable alternative especially when the drone system is quite complex. They can execute tasks without many difficulties: use of AI, drone vision, mission logic, etc.

In comparison with MCUs and FPGAs, these microprocessors are not often used as the control thread, instead they are used as a payload or control computers with OS.

Some of these microprocessor are:

- **ARM Cortex-A72** (Raspberry Pi 4)
- **ARM Cortex-A SoCs**

A-1.1.1.3. FPGA: Field-Programmable Gate Array

Though MCUs are quite dedicated for drones, FPGAs can even be a better option but with a cost.

These chips/computers are built for computing efficiency, allowing extreme parallelism, low latency, and flexibility. But with the constraints of being more complex to programme, higher valuable cost, and variable power consumption.

Some of these FPGAs are:

- **Xilinx Zynq:** Used in computing vision, as well as inside of some space satellites.
- **Intel Cyclone:** Used in communication systems.
- **Lattice ICE40:** Used in small drones.

A-1.1.2. Sensors

Sensors are key in order to develop a drone. There are multiple types of sensors but only a few of them are essentials. The most important one is the Inertial Measurement Unit.

A-1.1.2.1. IMU: Inertial Measurement Unit

Without this sensor a drone wouldn't be possible in general aspects. The vast majority of chips named IMU contain more than 1 sensor. These are:

- **Inertial Measurement Unit:** Measures the acceleration of the drone. Using this acceleration, it is possible to estimate the position. As a general concept, the most basic acceleration measurement structure is a spring that allows electrical pass. If an acceleration is created the spring circuit will let pass more or less current.

This sensor acquires drift over time, many people use gravity to filter the obtained data.

- **Gyroscopic Measurement Unit:** Measures the rotation (rad/s). Using this velocity of rotation, it is possible to estimate the angle created by the drone. As the inertial measurement, the gyroscopic measurement can drift over time too.

Many MEMS (Micro-Electronic Mechanical System) gyroscopes are based on a vibrating piezoelectric structure (often made from artificial quartz). When the drone leans or rotates, the Coriolis effect causes a shift in the vibration pattern. This shift is detected electrically and used to measure angular velocity.

- **Magnetometer** (Optional in some IC).

Sometimes the datasheet of the IMU chip names it: 6-axis IMU, that means inertial + gyros, while 9-axis IMU means inertial + gyro + magnetic. For instance chips like: MPU 6050, ICM-20948, etc.

A-1.1.2.2. Others sensors

There are many sensors, they can be inside the control PCB or added as an external peripheral, some of them are:

- **Barometer:** A barometer measures the atmospheric pressure. As the drone changes altitude, the sensor detects variations in pressure, which can be used to estimate the change in height.
- **Magnetometer:** Measures the magnetic field of the earth to determine the direction of the drone. When it is near an electrical station it may not work properly.
- **Laser/Ultrasonic rangefinder:** Sensor that uses laser or ultrasounds, to measure a certain distance.
- **GNSS receptor:** Global Navigation Satellite System receptor, this sensor provides the position data respect the globe. Usually in NMEA format (National Marine Electronics Association). If the sensor is covered and does not have clear sky sight, it may not work.
- **Optical Flow:** Sensor used to track ground movement to determine drone position. Using previous and current ground images a movement pattern can be determined.

A-1.1.3. Memory

There are multiple types of memory. I will only focus on those which are devoted to drone systems. There are 2 types of memory, internal (inside the main MCU or sensor ICs) and external.

- **Static RAM:** Volatile memory used in MCU to store variables such as register, cache, etc.
- **Dynamic RAM:** Volatile memory used on high performance MCU-MPU. Most of the time it is used for video storage, navigation, SLAM, etc.
- **Flash memory:** Main non-volatile memory found in MCU such as STM32, ESP32 and so on. It saves all the firmware, code and data.
- **EEPROM (Electrically Erasable Programmable ROM):** Smaller than the Flash memory. It is used to store multiple variables such as PID values, offsets, etc.
- **SD Card and eMMC:** Non-volatile memory with high capacity. Mainly they are used to store camera images and telemetry logs.

A-1.1.4. Basic buses

To establish communication between chips like MCUs and MPUs, various bus standards are currently utilized. A fundamental distinction must be explained between the physical interface (wires, cables or traces) and the electrical protocol (how bits are encoded and transmitted). This analysis focuses on data transmission across both domains:

A-1.1.4.1. I2C

When we talk about I2C, the physical and electrical mediums are defined. Invented in 1992 by Philips Corporation. Its data transmission can handle 100 kbits/s or 400 kbits/s, structured as a master-slave, 2 threads (SDA for data transmission and SCL as a master clock).

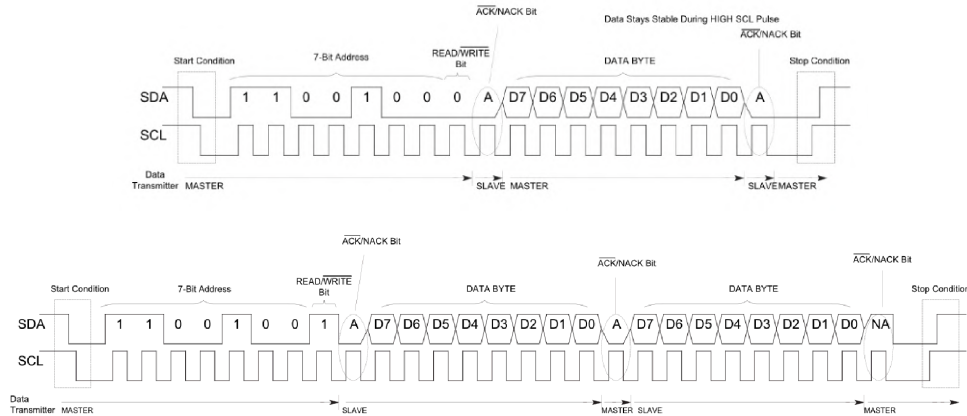


Fig. A-1.1 I2C bit structure (up → sending, down → receiving) [37]

As a curiosity, this bus needs 2 pull-ups resistors to be functional. One in each cable. It is a typical bus used on simple drone PCB boards.

A-1.1.4.2. UART/USART

This bus is specified in the electrical manner only. The physical manner of this bus can be defined by multiple serial ways as the RS-232 standard, TTL serial and many more.

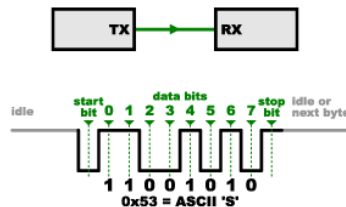


Fig. A-1.2 Simple serial structure

On the electrical aspect, UART stands for Universal Asynchronous Receiver Transmitter, while USART adds the difference of being Synchronous/Asynchronous. The key point is that USART can use an external serial SCK clock line, in order to develop an optional synchronous serial transmission; similar to SPI. Providing higher transmission speeds.

Start Bit (1 bit)	Data Frame (5 to 9 Data Bits)	Parity Bits (0 to 1 bit)	Stop Bits (1 to 2 bits)
------------------------	------------------------------------	-------------------------------	------------------------------

Fig. A-1.3 UART frame

A-1.1.4.3. SPI

The SPI bus is defined by its physical and electrical aspects. Invented in 1982 by Motorola, it is designed for short distances, operating in full duplex with high data transmission rates (1 MHz to +100 MHz).

Being a synchronized protocol, it consists of 4 lines: a clock determining when to read (controlled by the master), a MOSI line (Master Out Slave In) that sends master data, a MISO line (Master In Slave Out) that sends slave data, and finally SS (Slave Selector, also known as CS: Chip Select).

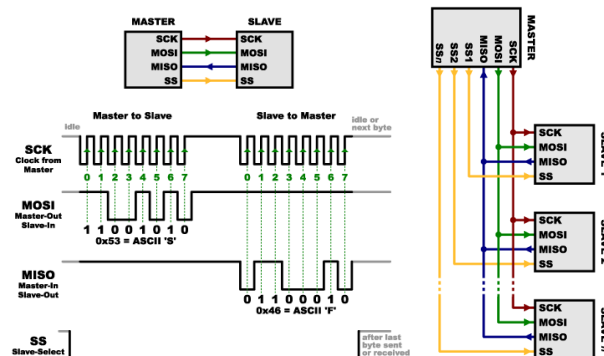


Fig. A-1.4 SPI structure [38], [39]

A-1.1.4.4. USB: Universal Serial Bus

Developed in 1996 by multiple entities such as Intel, IBM, Microsoft, etc. This bus is quite different in comparison with SPI and I2C, much more tricky.

- **Physical layer:**

Created as a serial bus (USB), it is designed with a master-slave architecture. The Master (Host) is composed of two components: First, the Host Controller, which is in charge of detecting new devices (enumeration), managing data, power, and bus activity. Second, the Root Hub, which is in charge of the physical USB hierarchy on the host system.

USB has a 7 bit address. If more hardware ports are required, external hubs can be added, with a maximum of 7 tiers/layers.

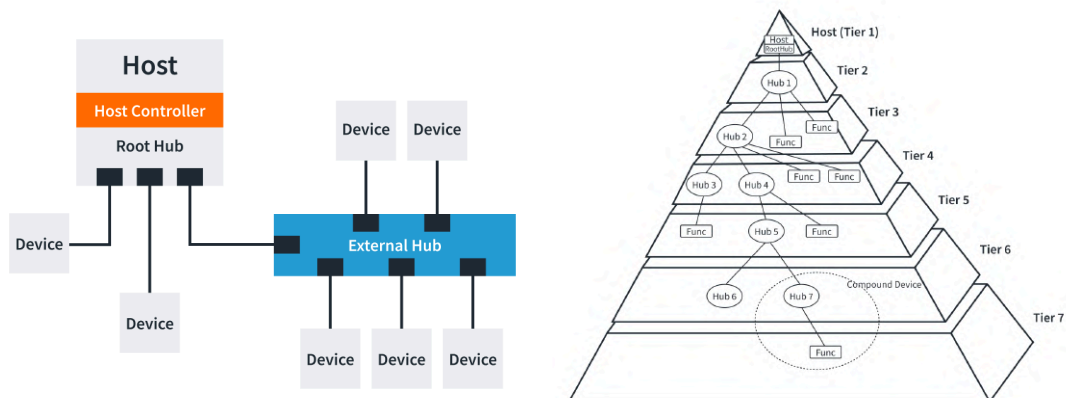


Fig. A-1.5 USB master-slave and tiers [40]

There are many types of USB connector. USB OTG (On-The-Go) is the specification that allows small devices to act as a host. Quite important in the development of a PCB.

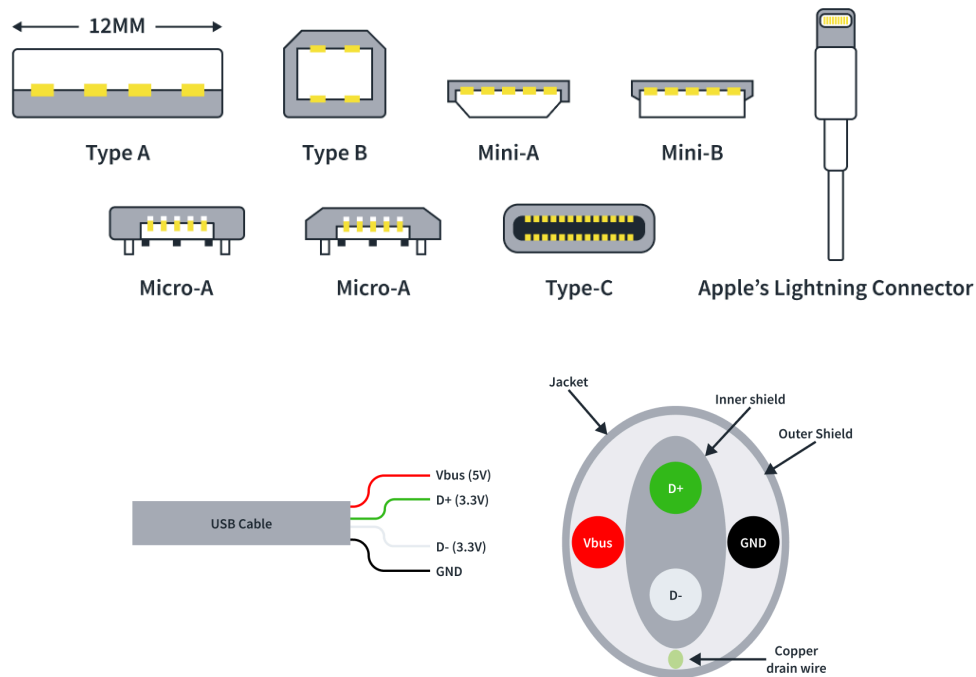


Fig. A-1.6 Types of USBs and cable structure [40]

The USB cable is composed of a differential line twisted 90 Ω connection at 3.3 V, a Vbus line at 4.4 - 5.25 V and a ground cable. What's more, the whole cable is coaxial in order to reduce the noise.

There are different kinds of transmission speeds, depending on the length of the cable. This is because the longer the cable the more susceptible it is to noise. For instance a cable of 5 m can reach speeds of 480 Mb/s or 480 Mhz.

- **Electrical layer:**

The signals that are sent by the USB are coded in NRZI-S (Non-Return-To-Zero Inverted Space). This binary transmission uses a stuffing change of bit on every N bits sent in order to maintain the synchronization.

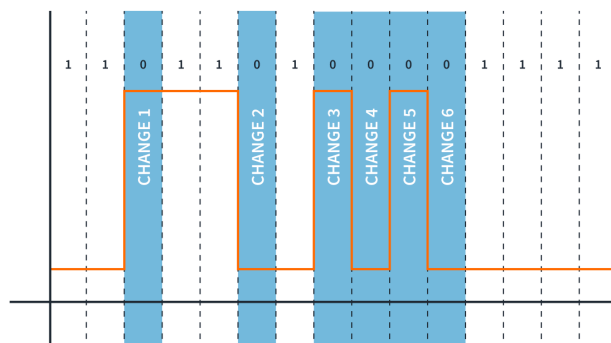


Fig. A-1.7 NRZI-S coding [40]

Moreover, depending on the electrical state of the differential pair (for instance: D+ High and D- Low), the bus signals condition the beginning of each message, such as the SOP (Start of Packet) or speed identification.

The communication, specifically in USB 2.0, is half-duplex and transmits data in an LSB (Least Significant Bit) first format.

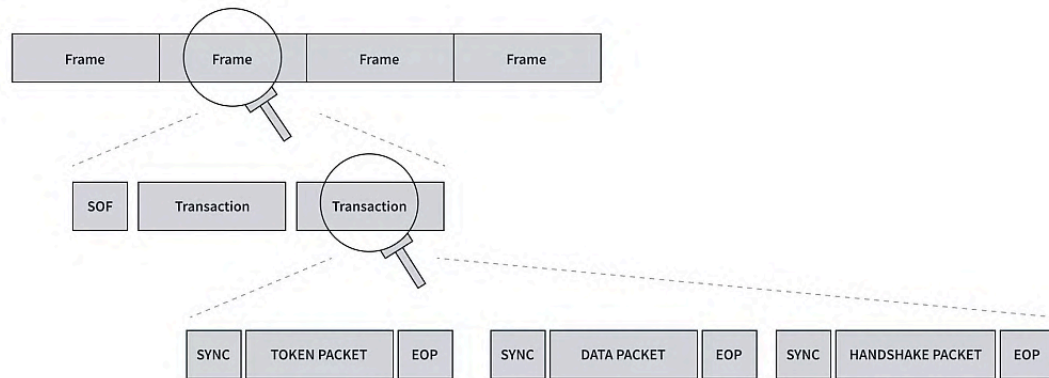


Fig. A-1.8 USB 2.0 transmission frames [40]

The typical USB 2.0 Frame starts with an SOF (Start Of Frame) packet, followed by the transactions. Each transaction is composed of three stages: a Token Packet (specifying the type of transaction), a Data Packet, and lastly a Handshake Packet.

Each individual packet is framed by synchronization bits at the start and EOP (End Of Packet) bits at the end. Additionally, all packets support CRC (Cyclic Redundancy Check) to ensure data integrity.

A-1.1.4.5. CAN: Controller Area Network

This bus configuration started in 1983 at Robert Bosch GmbH, then in 1987 the bus was introduced by Intel and Philips. Applied mainly in vehicles and instrumental platforms, this half-duplex bus has been one of the most important buses in the world.

Standardized in 1993, it has different parts: ISO 11898-1 data link layer, ISO 11898-2 physical layer for high speed, ISO 11898-3 physical layer for low speed, error tolerance, etc.

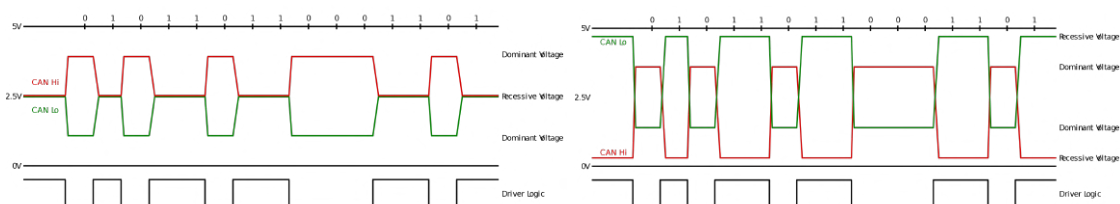


Fig. A-1.9 CAN transmission (left → high speed, right→ low speed) [41]

In order to avoid half-duplex problems, CAN bus uses CSMA/CR (Carrier Sense Multiple Access with Collision Resolution), allowing a successful transmission. All the connections are done via a 120 Ω bus differential pair.

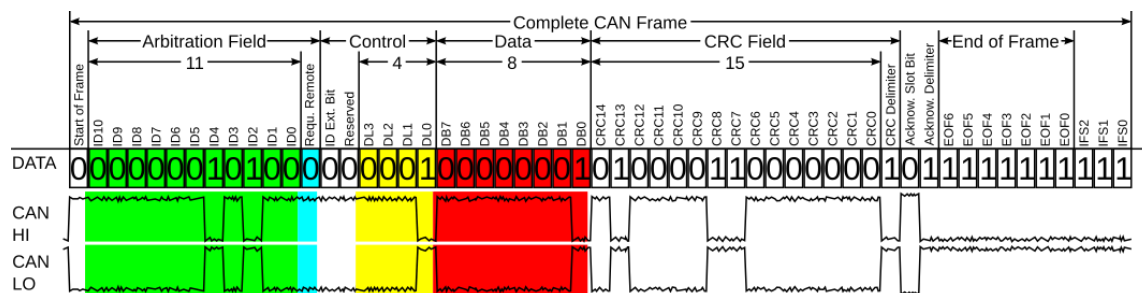


Fig. A-1.10 CAN frame [41]

A typical CAN frame is composed of:

- 1 bit, start frame
- 11 bits, identification
- 1 bit, stuff synchronism
- 1 bit, frame type (remote frame or data frame)
- 1 bit, identifier extension
- 1 bit, reserved bit
- 4 bits, data length
- 0-8 bytes, data
- 16 bits, cyclic redundancy check
- 2 bits, acknowledgement
- 7 bits, end of frame
- 3 bits, spacing

A-1.1.5. Power Management

Power management circuits provide the energy source for the main PCB, as well as for other circuits such as the ESCs (Electronic Speed Controllers). If a fault occurs in these circuits, it is possible to destroy the complete board. In this power management section, all circuits operate on DC (Direct Current). AC is not used in the vast majority of drone electronic circuits.

A-1.1.5.1. Linear regulator (LDOs)

Linear regulators reduce the voltage by dissipating the extra energy as heat. A very simple example can be a dropdown from 5 V to 3.3 V using resistors in series (voltage divider).

Most of these IC (Internal Circuit) regulators have thermal shutdown in case of overheating and current limiting circuits. It is common to see zener diodes in these kinds of circuits, allowing a maintenance of a constant voltage.

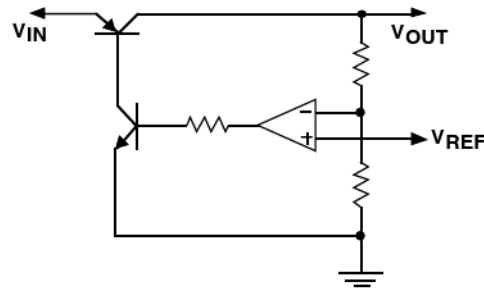


Fig. A-1.11 Simple LDO (Linear Drop-Out) Regulator

These kinds of regulators are clean, stable and create low-noise, however; they are very inefficient and produce a lot of heat.

A-1.1.5.2. Switching regulator

The concept of these switching regulators is based on the rapid transfer of energy packets. By modulating the connection to the power source, the circuit can increment or decrease the output voltage. Mainly, it is composed of a series of MOSFETs used as switches, inductors for energy storage, and capacitors for filtering and stability.

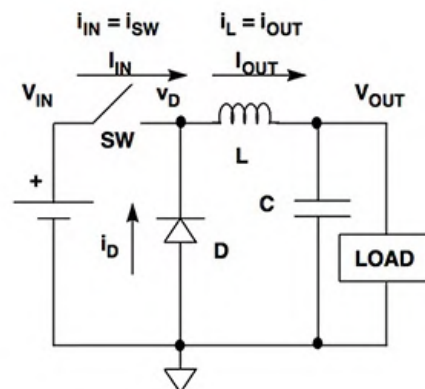


Fig. A-1.11 Very Basic Buck Converter (Switching regulator) [42]

It is very efficient, around 85-95%, being able to supply high currents; however the switching step creates noise. So an important step in applying this kind of circuit is the shielding process, in order to reduce EMI (ElectroMagnetic Interference).

A-1.1.5.3. BEC: Battery Elimination Circuit

BEC (Battery Eliminator Circuit) functions as a linear or switching regulator designed to power electronics directly from the main battery. Most of the time, this BEC is integrated into the ESC (Electronic Speed Controller) circuitry. However, professional drones incorporate multiple independent BECs to ensure power stability and distribute thermal loads.

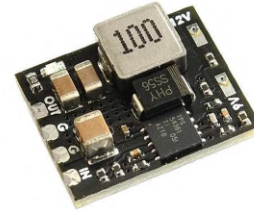
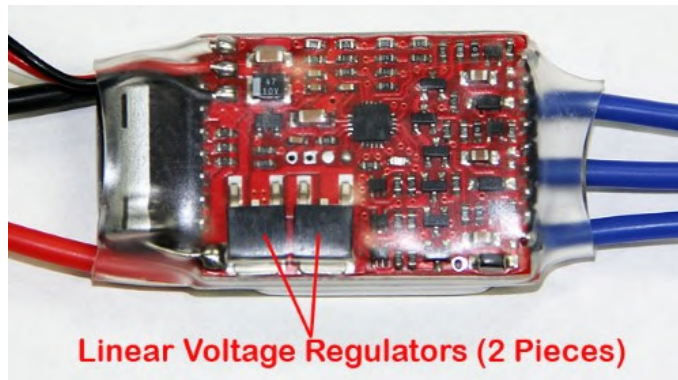


Fig. A-1.12 Left → ESC with 2 LDO and a small MCU, Right → Small BEC circuit [43]

The name BEC comes from the elimination of the dedicated battery used in legacy receivers.

A-1.1.5.4. Filtering

This section explains different methods in order to have a stable voltage/current source, from filtering frequencies to even avoiding current drops.

- **Decoupling and Bypass Capacitors:**

Capacitors are key in PCB design. Decoupling capacitors their main role is to provide a local charge to IC during current spikes and secondarily have a DC stability filtering AC to ground. While Bypass capacitors, their main role is to short high-frequency noise to ground, suppressing any AC signal inside the main DC source.

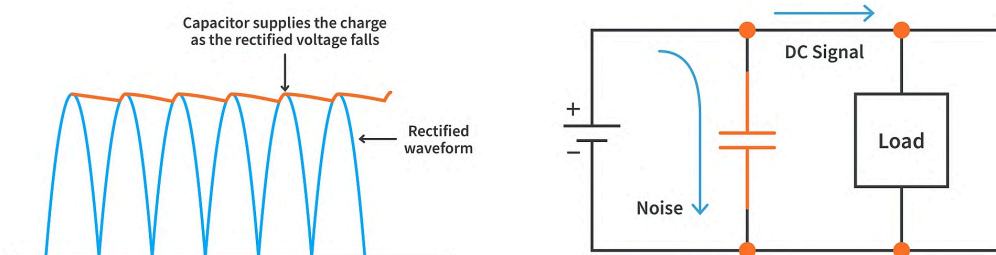


Fig. A-1.13 Left→ Capacitor rectification to drops, Right→ AC filtering from DC source, [44]

As an advice, it is important to use different capacitors in parallel to filter multiple frequencies. In case of having a high current battery circuit a large electrolytic or polymer capacitor can be useful. Ceramic capacitors are more durable and can resist heat in a better way.

Typical capacitors:

1. High frequencies: 0.01-0.1 μF ceramic capacitors
2. Medium frequencies: 1-10 μF ceramic capacitors
3. Low frequencies: +100 μF electrolytic or polymer capacitors for bulk decoupling (devoted for battery management)



Fig. A-1.14 Types of capacitors

- **Ferrite Beads:**

Ferrite is a material used in many applications, such as chip construction, transformers, magnets, memories, etc. In PCB design, it assumes the role of a frequency selective filter.

Its main function is blocking AC noise by acting as a high impedance path for high frequencies and low impedance for low frequencies (DC). Its usage is focused on power lines and sensitive ICs. For drone PCBs, typical values are 60-600 Ω at 100 MHz. Some components, known as ferrite chokes, contain a wrapped coil to provide a stronger inductive effect at lower frequencies. Warning: Never place this component at the output of a regulator, as it may cause instability and ringing.

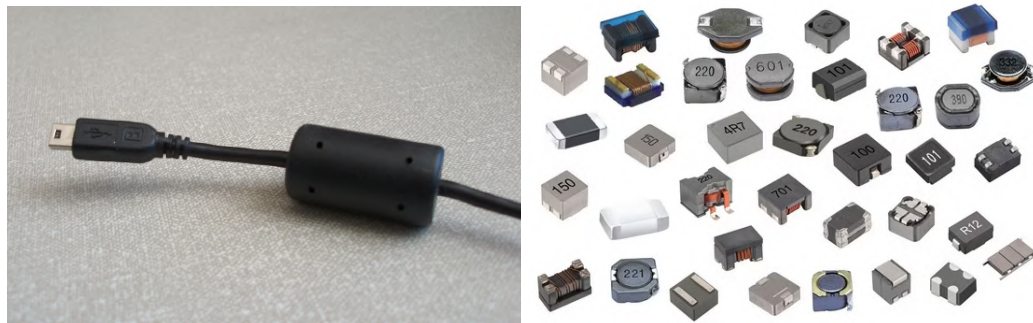


Fig. A-1.15 Left→ Ferrite around a cable, Right → PCB ferrite beads

As a curiosity, some old power cables have the ferrite necking around a cable, preventing high-frequency noise from escaping the device.

- **Inductors:**

Inductors are used to smooth the current in power regulators (by storing energy) and to filter noise, acting as a crucial part of the filter stage. While large coils are not common on drone PCBs, small surface-mount inductors with values between 2 μH to 50 μH are frequently found in the power section. In fact, the vast majority consist of a ferrite core (shielded construction) to improve performance and reduce size.

- **LC Filters: Inductance-Capacitance Filters**

There are multiple types of filters, such as BPF (Band Pass Filter) or HPF (High Pass Filter) used for RF/Radio Control purposes. However, for DC filtering, the LPF (Low Pass Filter) is the standard.

Leaving aside complex mathematical responses like Butterworth or Chebyshev, the most typical simple passive filters in a power supply line are:

1. L-type Filter
2. π -type Filer
3. T-type Filter

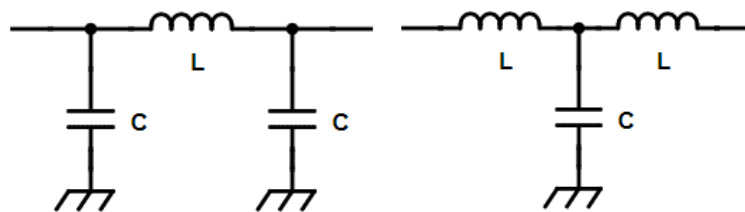


Fig. A-1.16 Left $\rightarrow$ π -type, Right $\rightarrow$ T-type [45]

The impedance of the capacitor and inductor is frequency dependent. The typical cutoff frequency is 5-20 kHz. Typical values are 10-100 μ F and 10-50 μ H. Bigger component values can produce a slow reaction in the system, causing a voltage drop during current spikes.

It is important to note that in case of a switching regulator the cut off frequency has to be lower than the switching frequency, providing a clean DC power.

- **Diodes:**

There are multiple diodes that are used in PCBs, some of them:

- **Schottky Diodes**

It is used on power supply rails and DC-DC converters. It is an improved version of the typical diode with a voltage drop of 0.2-0.4 V. The main usage of this diode is that it can block backward current, allowing the current to go in a certain direction only. What's more, it can handle fast switching frequencies, good in case of a switching regulator. Reverse leaking current in this diode can be dangerous in some circuits. It is important to select a model with the minimum possible leakage to avoid circuit instability.

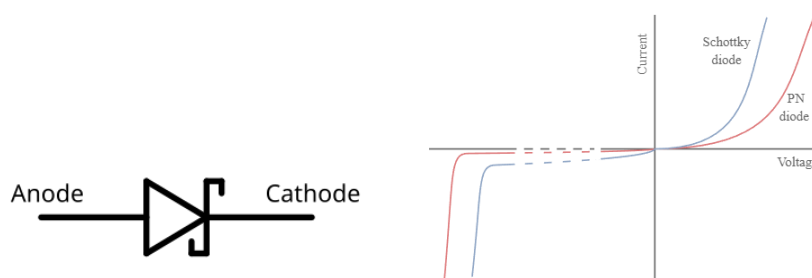


Fig. A-1.17 Schottky Diode [46]

- **TVS (Transient Voltage Suppression) Diodes**

It is used on battery lines and sensitive sensors. Permits a protection against current spikes such as those generated by motor transients, ESD (ElectroStatic Discharge).

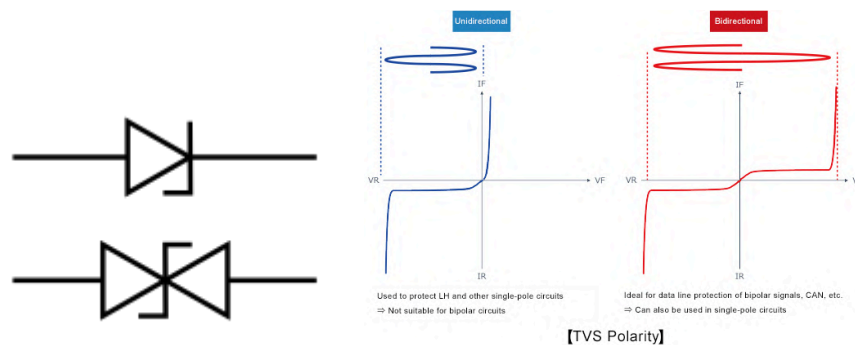


Fig. A-1.18 TVS (Transient Voltage Suppression) Diodes [47]

- **Flyback or Freewheeling Diodes**

They are not used on control PCB; instead, they are used in ESC (Electronic Speed Controller), servos, inductances and high current draining circuits. This diode provides a safe path for current continuity when voltage spikes are created, like when a motor-MOSFET circuit is switched-off for instance.

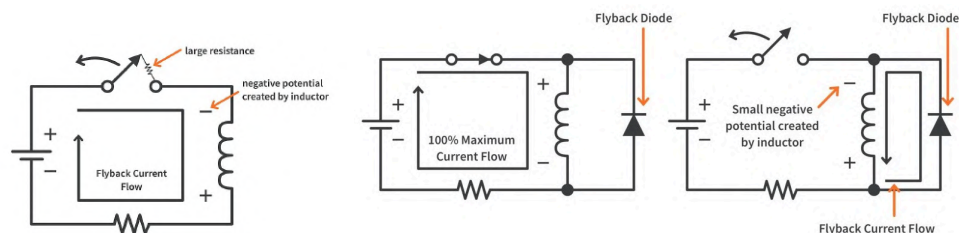


Fig. A-1.19 Schematic of how Flyback diode works [48]

Indeed, it is a simple diode with a special function worth mentioning. Often a Schottky diode, as it is a fast response diode.

- **Zener Diodes**

Zener Diodes are used on MCU (Microcontroller) protection pins and voltage reference circuits. They clamp the voltage to a stable level at the point of connection, protecting the system from high voltage levels. Additionally, they are characterized by a specific Zener voltage. This allows them to conduct current in reverse when a specific voltage threshold is reached, stabilizing the line.

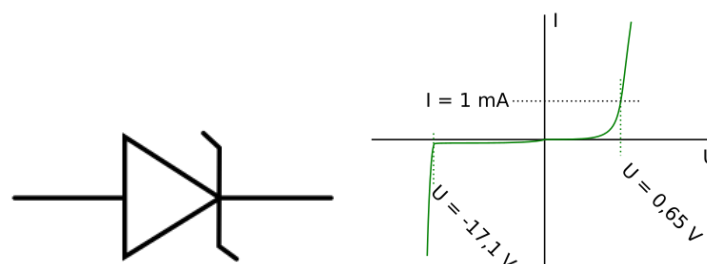


Fig. A-1.20 Zener Diode [49]

- **Rectifier Diodes**

This diode is mainly used for converting AC to DC current. In fact it is a normal diode, but it is worth mentioning by its importance. Commonly seen as a diode bridge.

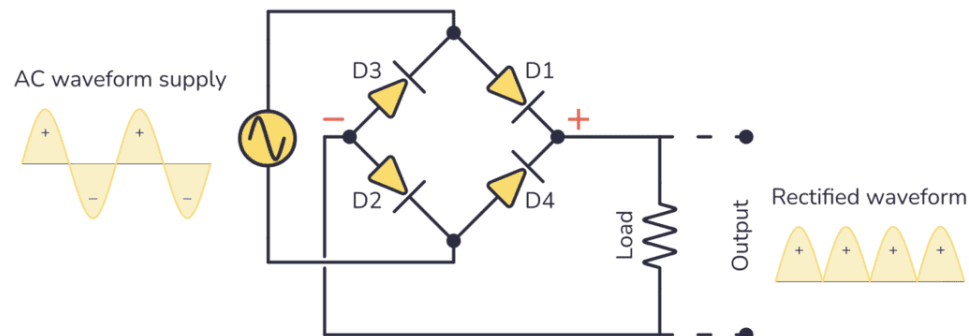


Fig. A-1.21 Basic schematic of a diode bridge [50]

- **Fusibles and transistors:**

I will not detail every variation of these components due to their vast diversity, especially regarding transistors. However, when combined with the previously mentioned parts, they ensure a reliable power supply for the main circuits. Fuses open the circuit if too much current is applied. Transistors, acting as switches (gates), allow power control. When combined with diodes or configured as active gates, they provide highly efficient reverse polarity protection.

A-1.1.6. Other EMI (ElectroMagnetic Interference) considerations

This section exposes additional considerations that have not been shown in the filtering section, such as grounding and power distribution networks.

- Shielding with GND on sensitive IC.
- Not place high current traces/lines next to sensitive signals.
- High-speed signals may be delivered in differential pairs.
- Some small drones carry supercapacitors, though it is strange to find them.
- In the development of a PCB, consider at least a layer of reference ground.

A-1.1.7. SWD and DFU

The vast majority of drones are flown with an onboard MCU. Some MCUs are programmed and debugged with 2 methods, these are:

A-1.1.7.1. SWD: Serial Wire Debug

The main purpose of the SWD is to programme and debug a MCU board. SWD uses a 2 pin interface composed of SWDIO as input/output pin and SWCLK as Serial Wire Clock. It is an alternative to JTAG (Joint Test Action Group) interface designed to program ARM Cortex-M MCUs.

Its main advantages are real-time debugging, reduced number of pins and full control of the MCU. Though, it may require additional special hardware to establish the connection between the MCU board and the user's personal computer program.



Fig. A-1.22 ST-Link V2 SWD

A-1.1.7.2. DFU: Device Firmware Upgrade

DFU is used to upload firmware to the MCU board, without any debugger. Establishing a direct connection between the MCU and the personal computer via USB standard. DFU can upload new firmware in binary to the device. During this step many boards require special bootloaders. The main tools used on this method are the STM32CubeProgrammer (for STM32) and dfu-utils (open source files).

A-1.1.8. Clocks

Without a clock it would be impossible to make a computational unit work. It is an important component of the PCB, and there are several types, a few of them are:

- **Crystal resonator (sometimes named oscillators):**
Piezoelectric crystal (most of the time, quartz), that vibrates when a voltage is applied, being possible to create a basic stable clock signal. Generally these crystal oscillator circuits require external adjustment capacitors and shielding to prevent EMI (Electromagnetic Interference).

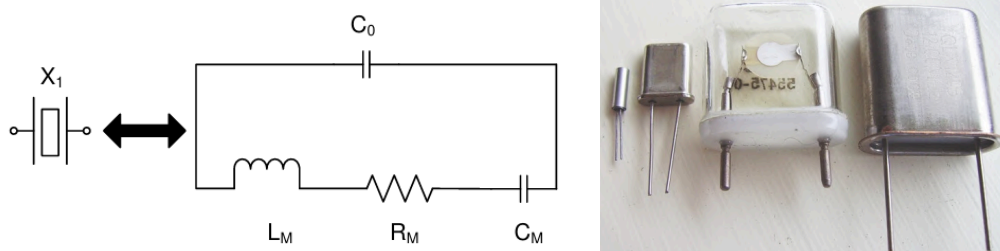


Fig. A-1.23 Symbol crystal oscillator, electrical model and real examples

MCU boards usually carry an external crystal oscillator because it is more stable and higher frequencies can be reached.

There is a similar alternative named ceramic resonators, which relies on piezoelectric effect. It has more resistance against impact with a lower cost though it is less accurate in comparison with a crystal resonator.

- **MEMS Oscillator:**

These vibrating elements are incorporated into the main MEMS (Micro Electro Mechanical Systems) architecture. They function similarly to crystal oscillators relying on resonance. An example is the IMU (Inertial Measurement Unit), where the gyroscope uses this internal vibration to detect angular velocity.

- **Oscillator ICs (Active/Programmable oscillator):**

Used by cameras, RF transceivers, some GNSS devices and accurate circuits. Oscillators ICs are packaged devices with all necessary components such as the resonator, drivers, and conditioning circuits. It provides a clean accurate stable clock frequency. There are different types such as XO (simple Crystal Oscillator), VCXO (Voltage Controlled Crystal Oscillator), TCXO (Temperature Compensated Crystal Oscillator), OCXO (Oven Controlled Crystal Oscillator), etc. It can be differentiated from the crystal resonator, because it may have at least more than 3 pins. These oscillators seek precision and low jitter conditions.

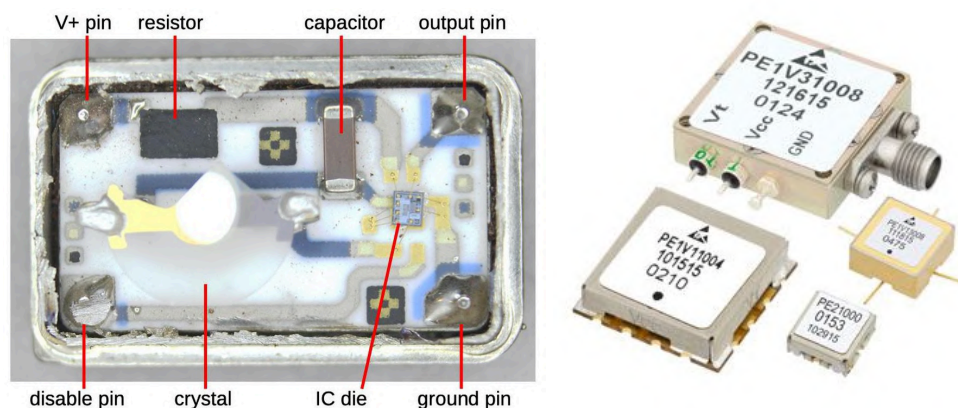


Fig. A-1.24 Left→ Example of an oscillator IC, Right→ Example of TCXO, [51], [52]

- **RC Oscillator:**

Known as “cheap and dirty” clocks, it is based on a resistor-capacitor circuit where the capacitor charges and discharges creating a clock frequency. This oscillator most of the time is located inside MCU chips (like STM32). It is quite easy to create, but the main disadvantage is that it has poor accuracy with large drift. Also, it cannot reach at max a few MHz.

- **GPSDO: GPS Disciplined Oscillator:**

A GPSDO is not a single component, but a timing technique used in high-end navigation like RTK drones. It consists of a local oscillator (typically a TCXO or OCXO) that is continuously corrected by the timing signals from GNSS satellites. The main advantage is that it is extremely accurate (sometimes with atomic clocks as in the image), though it requires a high amount of power.



Fig. A-1.25 PCIe Modular GPSDOs with atomic clock [53]

- **SSO: Spread-Spectrum Oscillator:**
It is a specialized oscillator that spreads its central frequency signal to reduce the EMI (Electromagnetic Interference) peaks. Typical usage frequencies are 25-200 MHz. These clocks are used on digital interfaces near sensitive devices, such as drone peripherals, high-speed cameras, and USB 2.0/3.0 controllers. They suffer from higher jitter compared to standard oscillators.
- **SAW Resonator: Surface Acoustic Wave:**
A specialized resonator designed specifically for RF (Radio Frequency) circuits. It uses IDT (Interdigital Transducers) to generate acoustic waves that propagate along the surface of the piezoelectric substrate. This allows it to generate very high and precise frequencies (typically 300 MHz to GHz), making it essential for drone telemetry modules and remote controllers, [54].
- **Curiosity:** Sometimes when seeing a RF (Radio Frequency) PCB, we can find frequency synthesizers; these devices often use XO or TCXO clocks to generate different RF frequencies using a method called PLL (Phase Locked Loop). For instance the RF SoCs like an nRF Nordic Semiconductors chip.

A-1.1.9. Special considerations on control protocols used on drones

There are a large variety of communication protocols, here are some of the most important ones, [55]:

- **ESC protocols:**
 - **PWM: Pulse Width Modulation:**
This protocol of communication is one of the most used ones in terms of robotics. Basically, it is a repeated square pulse sent to the ESC (Electronic Speed Controller). Depending on the duty cycle of this pulse (width of the pulse), the ESC will know if the user wants full throttle (2000 μ s) or motor off (1000 μ s). The main constraint is the high latency and susceptible to EMI (Electromagnetic Interference) and jitter problems.

- **OneShot125 or OneShot42:**
Similar as with PWM, but with a pulse range of 125-250 μ s with OneShot125 or 42-84 μ s with OneShot42. However, this protocol was replaced by DShot.
- **DSHOT (150, 300, 600, 1200):**
This protocol as in comparison with PWM, DShot is digital. It sends a 16 bit packet (11 bit throttle values, 1 bit telemetry request and 4 bits for CRC). Variant 150 sends 150 kbit/s, 300 sends 300 kbit/s, and so on. ESC can send back additional data (bidirectional). This protocol is the most used protocol on high-end drones, because no calibration is required.
- **Drone RX-Telemetry bus-protocols:**
 - **PPM: Pulse Position Modulation:**
Basically compresses multiple PWM channels into one line of data. Encoded by timing between pulses, the maximum number of channels are 8. Very simple with high latency.
 - **SBUS: Serial BUS:**
One of the most important buses in drones nowadays. Developed by Futaba it can transmit digitally max 16 channels in serial. The typical velocity goes at 100 kbaud in UART inverted format. The normal latency is at 14 ms. The sent frame is composed of: 0x0F, the channels data (22 bytes with 11 bits resolution), then a byte of flags and lastly 0x00. The actualization frequency is typically 120 Hz.
 - **IBUS:**
Quite similar to SBUS but not inverted. Communication via 1 UART cable. Maximum 16 channels with an overall frequency of actualization 300-500 Hz (15-20 ms latency). The message frame is: 0x20, 16 channels (every channel 2 bytes of 2000-100 values), and lastly a 1-2 bytes checksum. Depending on the version, IBUS can be bidirectional providing telemetry; however, CRSF or ELRS are better options.
 - **CRSF: Crossfire Serial Protocol:**
This protocol is quite important, developed by TBS (Team Black Sheep) is a serial bidirectional UART RC protocol (420 kbaud). The actualization frequency is in the order of 150-500 Hz with 3-5 ms latency. A part of being able to send 16 channels (each channel 11 bits coded on 172-1811 values, total 22 bytes), it can send telemetry such as SNR (Signal Noise Ratio), voltage, GPS, etc. The typical packet structure is: address, length payload, type of message, payload and lastly CRC.
 - **ExpressLRS (ELRS):**
The most used protocol in FPV drones, it is an open-source protocol. It uses CRSF to transport data to FC (Flight Controller). Though the RF

transmission is quite different compared to CRSF, it uses modulation LoRa or FLRC with a special packet type.

- **PWM: Pulse Width Modulation:**
Used on ESC (Electronic Speed Controller), some drones use this modulation to send RX data to the main drone controller.
- **DroneCAN:**
Introduced as UAVCAN or DroneCAN used on PX4/ArduPilot peripherals. The physical layer is CAN bus, however; other layers are much more complex. The publication of the messages is multicast/broadcast with unicast answers. The payload goes in DSDL (Data Structure Description Language) format.
- **MSP: Multiwii Serial Protocol:**
Serial binary protocol used by Betaflight, INAV and others. Simple, UART connection between GCS (Ground Control Station) and FC (Flight Controller) to send telemetry data. There are 2 versions:
 - **MSP V1:**
Simple version composed by a packet: header, size, command, payload and lastly a checksum. Simple messages with the protocol of request/response.
 - **MSP V2:**
Much more complex system, devoted for complex payloads. The packet format goes: header, identifier, type, flags, message id, payload size, payload, and lastly checksum.
- **MAVLINK:**
Communication standard telemetry UDP/TCP between GCS and autopilot, devoted to mission protocol. There are 2 versions, MAVLINK V1 (classic), and MAVLINK V2 (modernized). Quite complex, the maximum payload is 255 bytes.

A-1.2. Example 1: Acci FPV Flight Controller

This open-source project was developed by user “jesuslg123” with the objective of creating an FPV (First Person View) drone running on Betaflight.

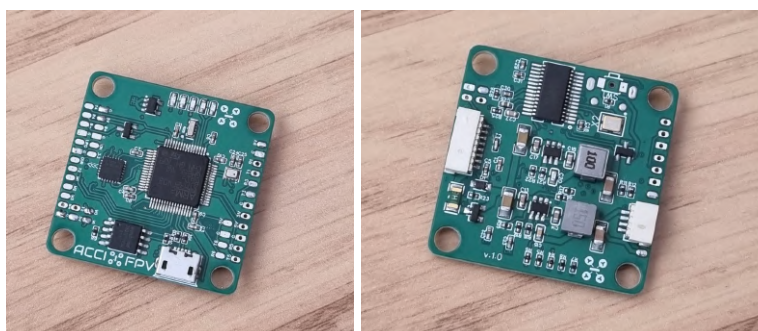


Fig. A-1.26 Acci FPV Flight Controller [56]

A-1.2.1. MCU system

This flight controller has been designed to support 2 microcontroller ICs, specifically: the STM32F405 and the AT32F435.

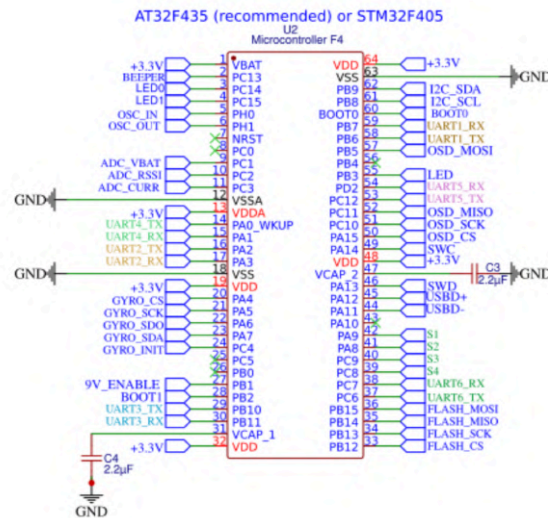


Fig. A-1.27 STM32F405 connections

The connections to this MCU are quite simple:

- 6 UART connections.
- 3 SPI connections to IMU, Flash memory and OSD (On-Screen Display).
- 1 I2C connection.
- Several LEDs and 1 beeper.
- 3 ADCs for :battery status (VBAT), FPV-RSSI connection and current (CURR).
- 6 3.3 V power supply connections.
- 1 USB bus.
- 1 SWD connection.
- 2 BOOT pins to turn the MCU to programming mode.
- 1 Oscillator connection and 4 PWM connections (S1, S2, S3, S4).

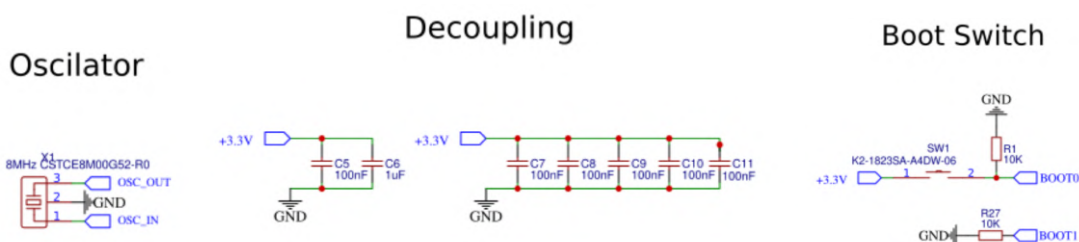


Fig. A-1.28 Oscillator, decoupling and boot connections

Other important data worth mentioning:

- It contains a simple 8 MHz oscillator directly connected without conditioning
- As a decoupling strategy it contains 6 capacitors to match the 6 3.3 V entries, plus a 1 μ F capacitor.

A-1.2.2. Sensor and actuators

Two main sensors are used, the IMU MPU6000 with 6-DoF (Degrees of Freedom), connected via SPI bus and a BMP280 barometer connected via I2C.

The connection to the power supply board is established via a JST connector. This connector carries: 4 PWM signals + the battery power source + a current measurement + UART receiver connector.

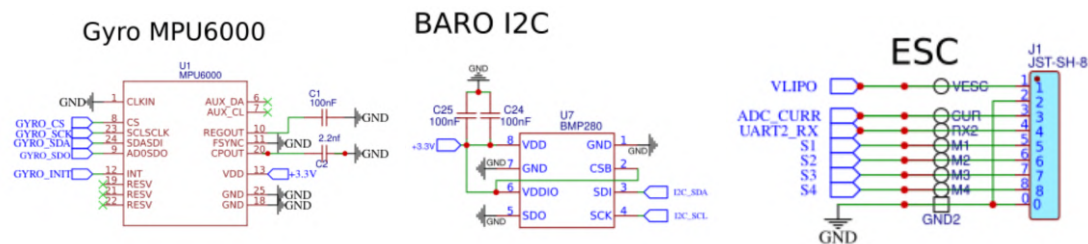


Fig. A-1.29 MPU 6000, BMP 280 and ESC connector

A-1.2.3. Communication buses

The board design manages signals across multiple buses. The I2C lines are connected with 4.7 k Ω pull-up resistors, while the USB data lines utilize 22 Ω series resistors for impedance matching. A critical design consideration is the SBUS input; since STM32 F4 microcontrollers lack internal signal inversion, an external hardware inverter is implemented to ensure compatibility with standard serial receivers running on Betaflight.

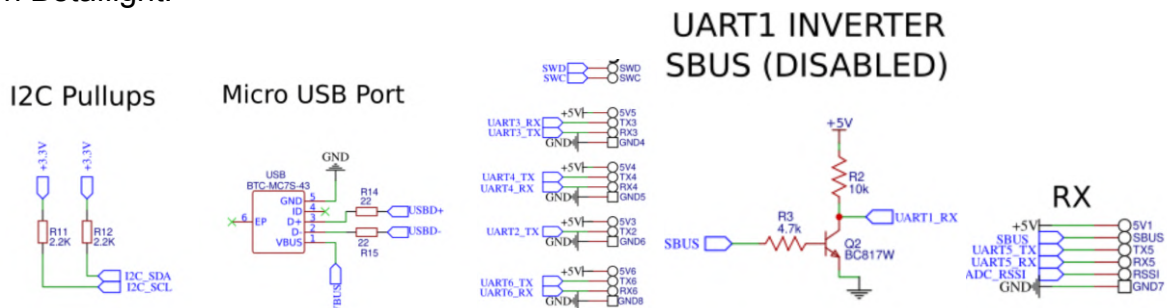


Fig. A-1.31 I2C bus, USB, UART/SWD, SBUS and radio

A-1.2.4. Power regulation system

FPV drones require higher voltages to support OSD (On-Screen Display), reaching up to 10 V.

To protect the board against reverse polarity and overcurrent, the design uses a protection circuit consisting of a P-Channel MOSFET, a diode, and a fuse.

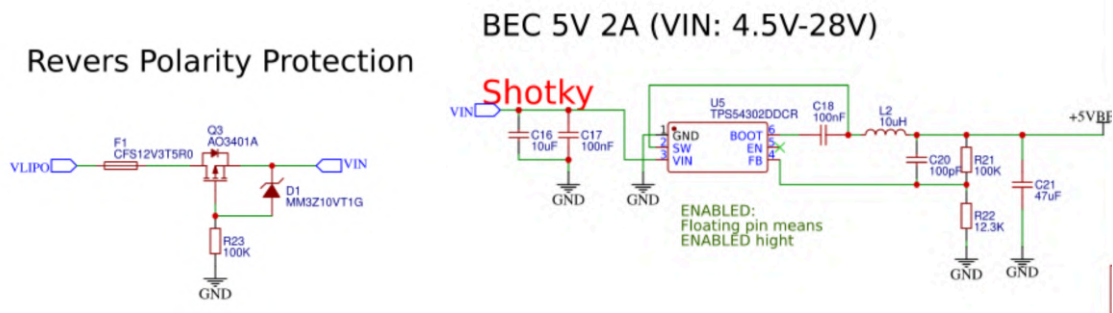


Fig. A-1.32 Reverse polarity protection and BEC at 5V

This BEC regulator circuit uses a TPS54302 chip (synchronous buck converter). Depending on the conditioning, the output can be different.

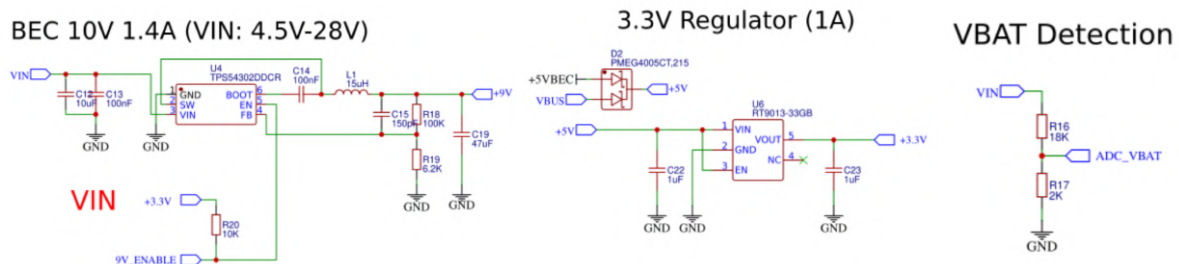


Fig. A-1.33 BEC at 10V, VBUS-3V3 regulator and VBAT detection

To regulate from 5 V to 3.3 V and measure the battery voltage, a voltage divider is applied.

A-1.2.5. Other systems: FPV and storage

Apart from the memory in the main MCU, this PCB has a direct SPI communication with a Flash IC memory (16 MB capacity approx). Additionally this board uses an AT7456E chip that provides drone data to the FPV google.

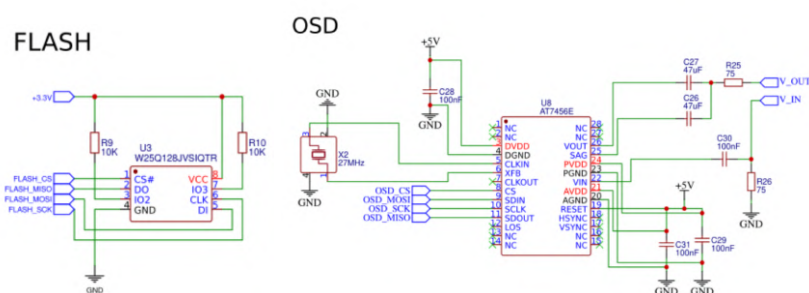


Fig. A-1.34 Flash memory and OSD

A-1.3. Example 2: PULPY_F405_FC

Designed by “Nayel”, this open-source flight controller contains the standard STM32F405RGT6 microcontroller and is fully compatible with INAV. As this board carries a similar architecture compared with the first example, sections detailing shared circuitry will be omitted to avoid repetition.

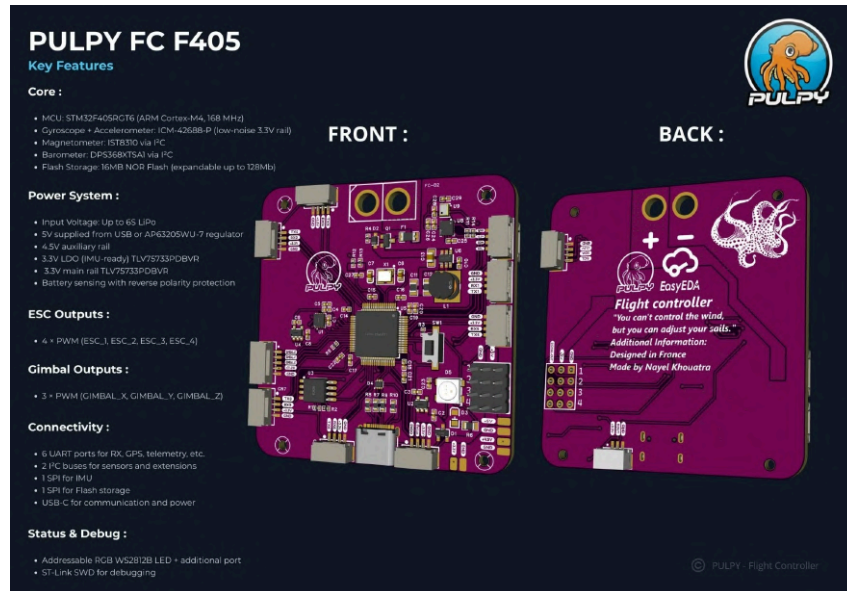
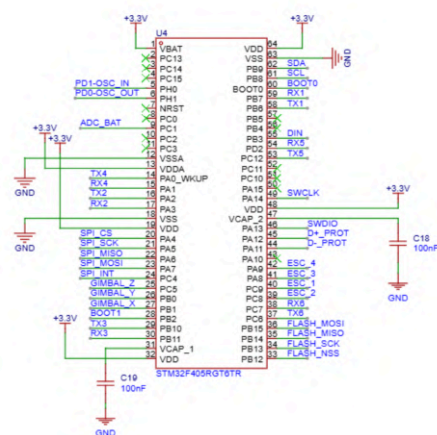


Fig. A-1.35 PULPY FC F405 [57]

This flight controller manages an IMU, magnetometer, barometer, flash memory, 4 PWM signals for ESC (Electronic Speed Controller), 3 PWM for gimbal control and connectivity via UART to multiple devices such as radio, GNSS, etc. The board can be powered by a 6S battery.

A-1.3.1. MCU system



- 1 ADC for battery measurement.
- 1 DIN connection (Data IN) for LED string.
- 6 3.3 V power supply connections for the MCU.
- 1 USB bus.
- 1 SWD connection.
- 2 BOOT pins.
- 1 Oscillator connection.
- 4 PWM connections for ESC and 3 PWM for gimbal.

Decoupling and bootloader connections are similar to the previous example. The only changes are in the oscillator configuration, which uses a simple quartz crystal, and the battery voltage sensor, which adds a low-pass filter for better signal stability.

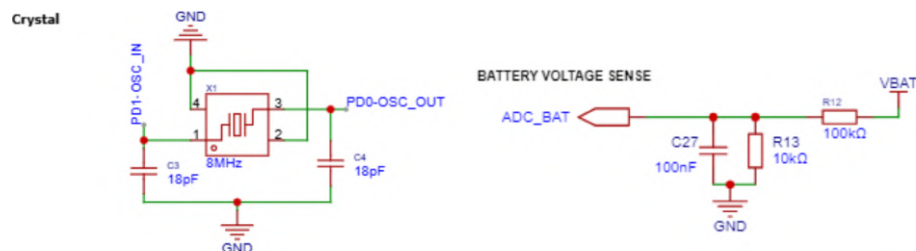


Fig. A-1.37 Oscillator and a battery voltage detector

A-1.3.2. Sensor and actuators

This board contains 3 main sensors: an IMU via SPI, and a Magnetometer + Barometer via I2C. All them with their corresponding conditioning circuitry.

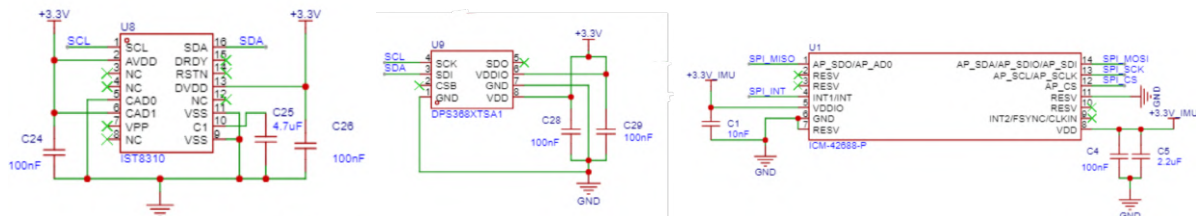


Fig. A-1.38 Magnetometer (ST8310), Barometer (DPS368XTS) and IMU (ICM-42688)

A-1.3.3. Communication buses

Connectivity options include multiple I2C and UART ports, along with a dedicated SWD interface for firmware flashing and debugging.

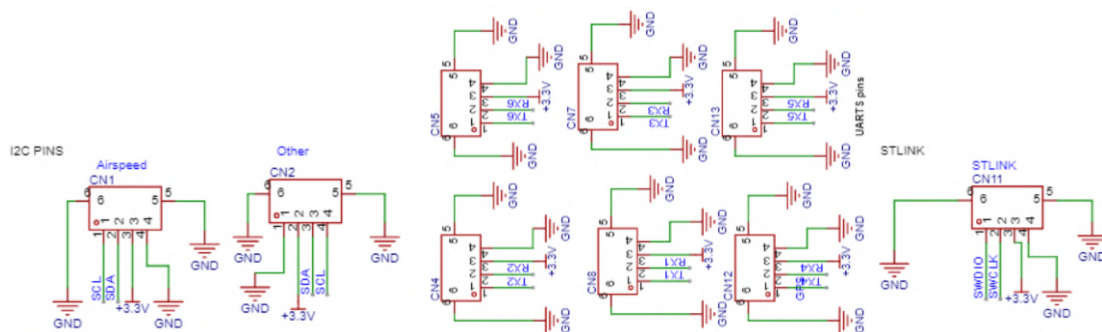


Fig. A-1.39 I2C, UART and SWD

The USB bus uses 22 Ω resistors to match the required 90 Ω differential line impedance. Furthermore, the design utilizes an ST USBLC6-2P6 for ESD (Electrostatic Discharge) protection, a low capacitance IC chosen specifically to support full-speed USB 2.0 data rates.

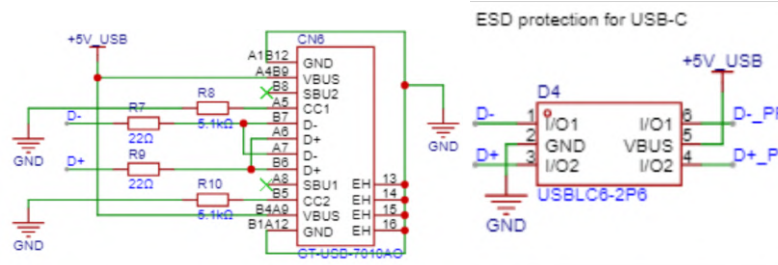


Fig. A-1.40 USB bus and conditioning

A-1.3.4. Power regulation system

Similar to the previous design, this circuit uses an AP63205WU-7 buck converter for the 5 V. A TLV75733PVB LDO then generates 3.3 V, taking input from either the onboard 5 V or USB VBUS via a diode.

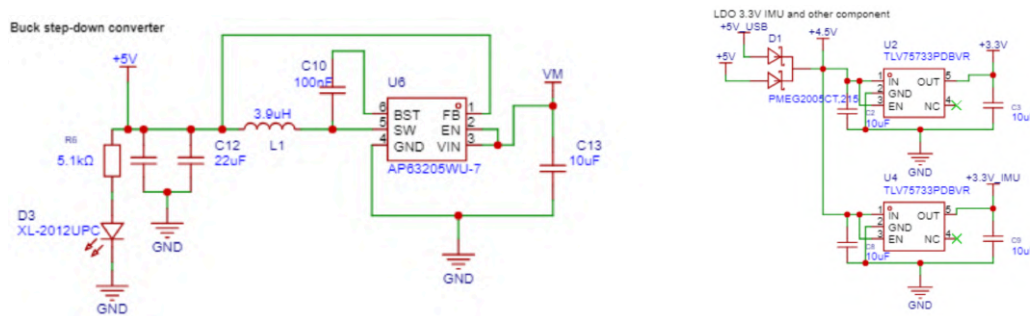


Fig. A-1.41 Battery Buck converter and linear converter from 5 V to 3V3

A-1.3.5. Other systems: Flash and LEDs

Additionally this board applies a Flash IC with SPI and a dedicated circuitry for LEDs strings.

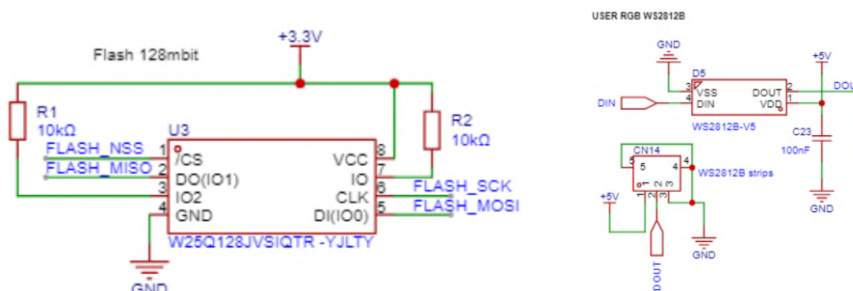


Fig. A-1.42 FLASH SPI and RGB string chip

A-1.4. Example 3: NxtPX4v2

Obtaining schematics of simple commercial grade PCBs is often challenging due to proprietary restrictions. However, the NxtPX4v2 is an exception: it is a modern, partial open source flight controller 2023-2024 designed by the HKUST UAV Group and manufactured by MicoAir Technologies.

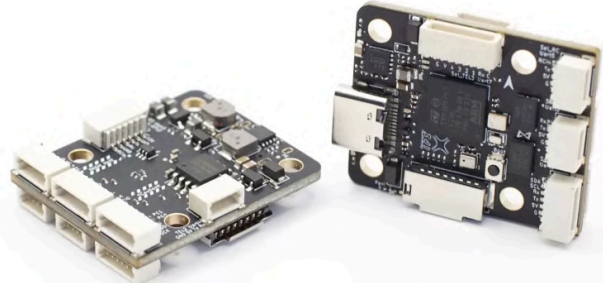


Fig. A-1.43 NxtPX4v2 [58], [59]

Compatible with the Pixhawk firmware architecture, this board offers a professional grade solution for autonomous flight.

The key upgrades over previous examples are: a more powerful MCU and more peripherals: dual IMUs, barometer, dual BECs (5 V/12 V), 8 PWM channels, and 7 UARTs. It also contains MicroSD, USB-C, and dedicated VTX/Camera ports, all within a 6.5 g weight.

Basic connection schemas will be skipped in this explanation, as they were shown in previous examples.

A-1.4.1. MCU system

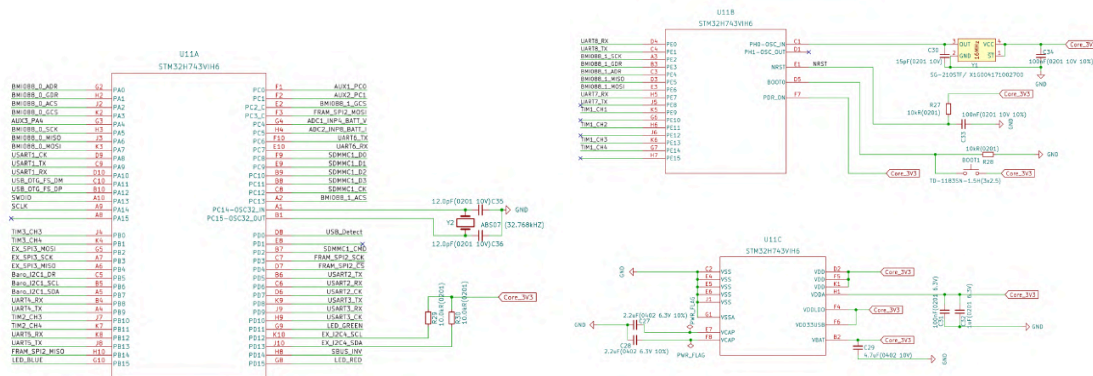


Fig. A-1.44 STM32H743VIH6

The connections are several:

- This MCU has 2 oscillators:
 - HSE (High Speed External oscillator): A 16 MHz SPXO (Simple Package Crystal Oscillator) configured in bypass mode.
 - LSE (Low Speed External oscillator) at 32.768 kHz.
- 7 UART/s connections.
- 3 LEDs connections.
- 8 PWM outputs.
- 1 SPI bus connection for FRAM and 2 SPI bus connections for the IMU.

- 1 I2C for the barometer and 1 I2C external connection.
- 1 SBUS inverted connection (unified with UART 5).
- 4 pin connections for MicroSD.
- Others: 1 SWD, 1 USB connection, 1 ADC for voltage measurement, 1 ADC for current measurement, and 3 auxiliary pins.

A-1.4.2. Sensors

2 IMU BMI088 sensors are used on different SPI buses. Additionally a I2C SPL06 barometer is used to measure the altitude.

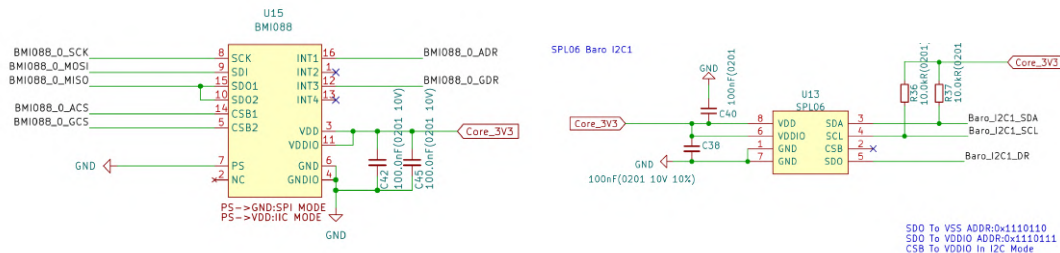


Fig. A-1.45 IMU BMI088 and Barometer SPL06

A-1.4.3. Communication buses

The connectors don't follow current Pixhawk standards; I2C contains 10 kΩ pull-up resistors. The USB-C connector has a simple configuration of TVS-Schottky diodes with decoupling capacitors (100 μF typical and 1 μF residual).

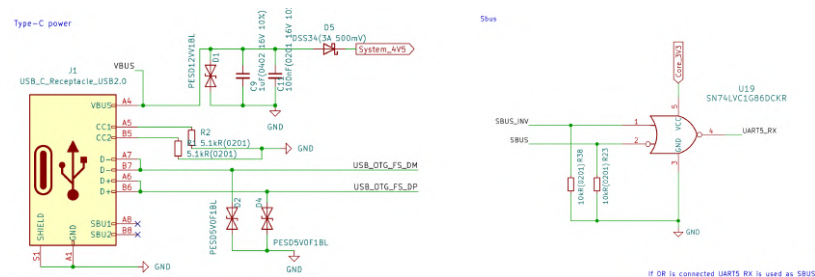


Fig. A-1.46 USB-C connector and SBUS converter

SBUS inverted with an optional selectable XOR gate in order to allow the user to use the inverted or a non-inverted signal through UART5.

A-1.4.4. Power regulation system

Regarding the battery input filtration and voltage sensing circuit, the design is perfect. However, I recommend adding a fuse for overcurrent protection and a ferrite bead for high-frequency noise suppression. Additionally, a low-ESR (Equivalent Series Resistance) bulk capacitor at the battery input is essential to manage voltage spikes.

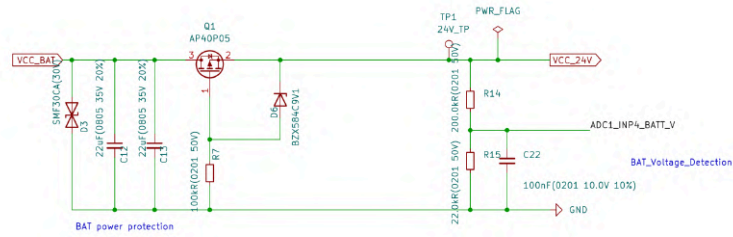


Fig. A-1.47 Main voltage filter

As for power regulation, the board uses 4 different buck converters: 9-10 V rail for the VTX, a 5 V rail for peripherals, and 2 separate 3.3 V converters for the MCU and sensors. This dual 3.3 V architecture is a perfect design choice, as it allows sufficient current capacity and provides noise isolation between the noisy digital components and the sensitive sensor circuitry.

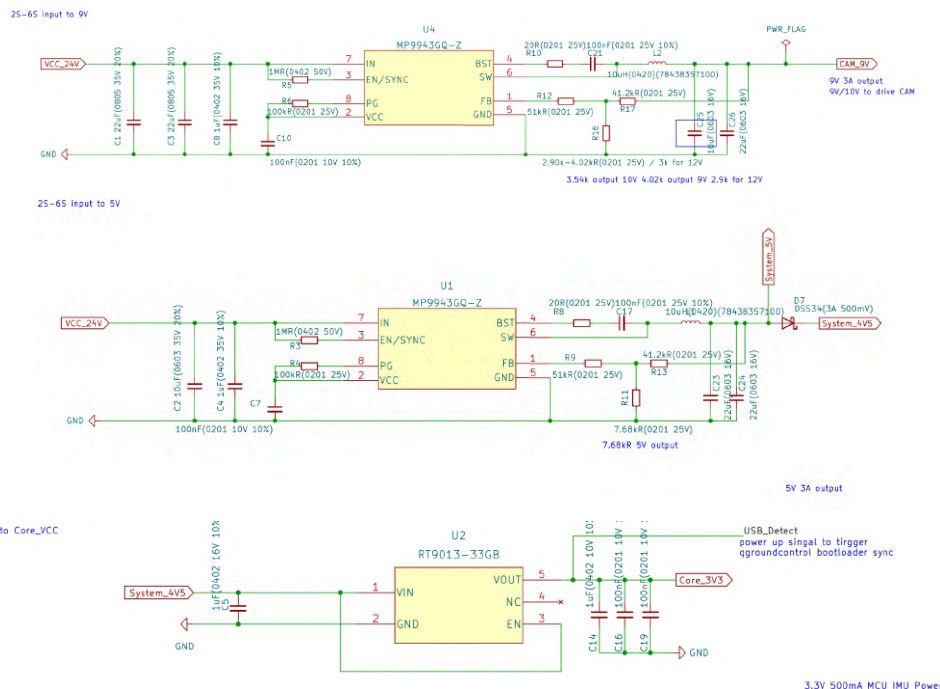


Fig. A-1.48 9-10 V Regulator, 4-5 V Regulator and 3V3 Regulator (Dual)

A-1.4.5. Storage

FRAM is connected via SPI to save rapidly critical data. MicroSD connection is established via SDMMC, which offers faster transfer speeds than SPI.

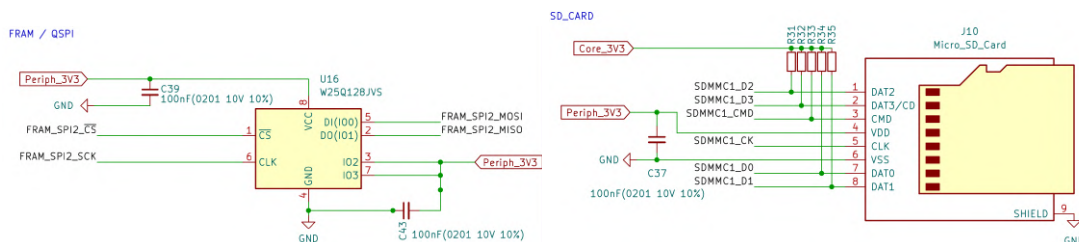


Fig. A-1.49 FRAM for MCU and MicroSD connector

A-1.4. Example 4: FC F4BY H743

This flight controller was developed by the F4BY russian research team. While several designs from this group are currently available on the market, this specific model is not. Developed in 2023, it was designed specifically for high performance applications. This product does not follow any standard.

The provided schematics are incomplete; some component names and values are omitted or incorrect.

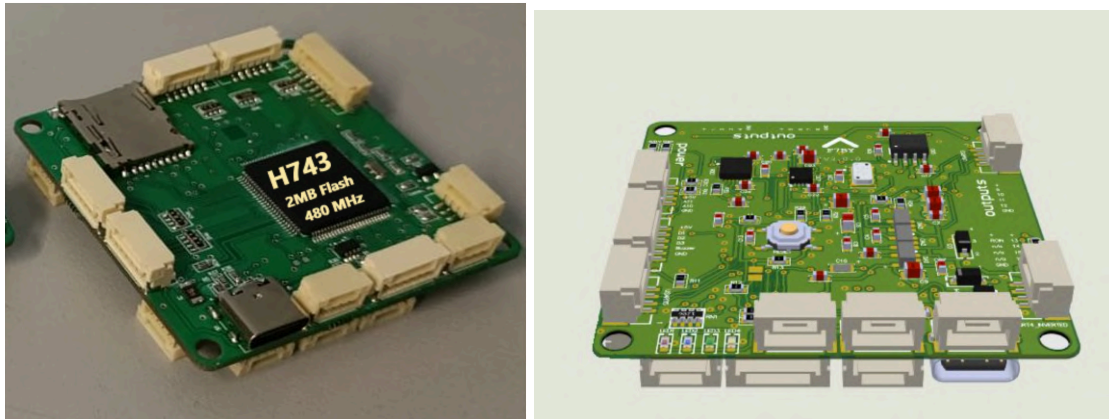


Fig. A-1.50 FC F4BY H743 [60], [61]

This board offers: 12 ESC-Servo connections, 1 USB, 5 USART, 1 Analog, 1 SPI, 1 Camera feedback connector, 1 I2C, 1 CAN, 1 MicroSD connection and 3 sensors (barometer, IMU and magnetometer).

A-1.5.1. MCU system

Similar to previous designs, this board adds 4 LEDs and a buzzer. A key difference is the utilization of 2 camera interface pins to implement an external oscillator circuit.

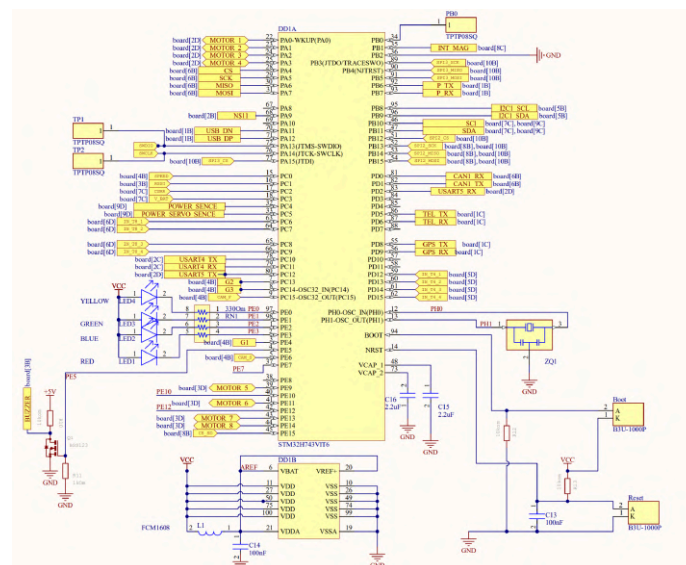


Fig. A-1.51 STM32H743VIT6

A-1.5.2. Sensors

Three sensors: an ICM-42652 IMU via SPI, QMC5883P magnetometer and a MS5611 barometer via I2C.

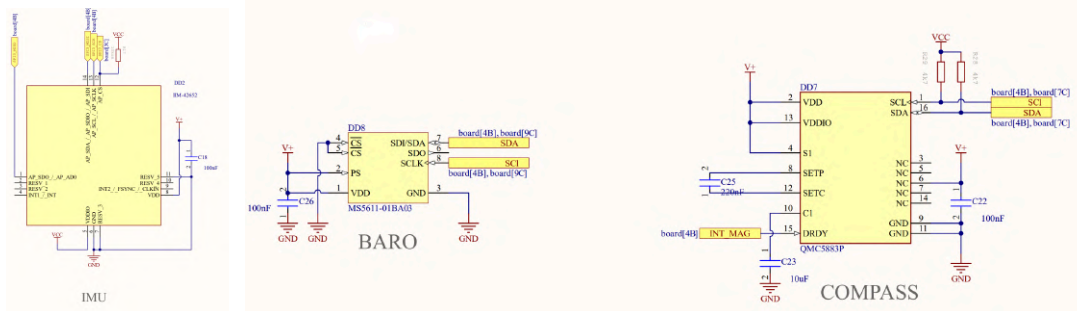


Fig. A-1.52 IMU, Barometer and Compass

A-1.5.3. Communication buses

The USB VBUS input is protected by a Schottky diode and a resettable polyfuse to prevent back feeding and overcurrent.

Regarding the CAN bus, it uses the transceiver MAX3051 IC; this SMD is in charge of converting typical CAN signals coming from CAN+ and CAN-, to serial signals for the MCU.

The motor lines include series 100 Ω resistors. These dampen signal reflections and form a RC filter with the trace capacitance.

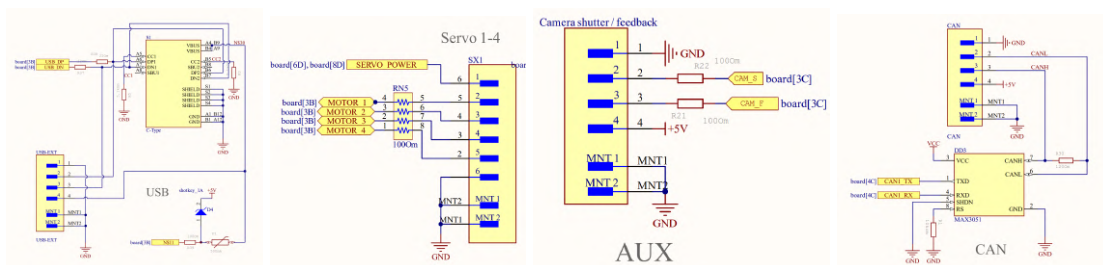


Fig. A-1.53 USB-C, PWM-Servo, Camera and CAN bus connectors

A-1.5.4. Power regulation system

The power stage utilizes a simple design, consisting of ESD protection (TVS diode), signal filtration with capacitors, and 3 buck converters. It lacks advance circuit protection such as fuse, reverse current protection, etc.

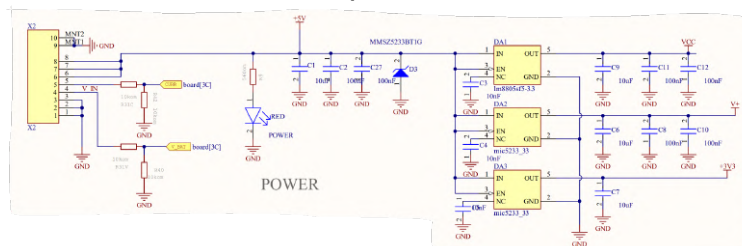


Fig. A-1.54 Main voltage regulator circuit

A-1.5.5. Storage

The board applies 2 storage methods: a MicroSD and a FRAM chip. Both peripherals are connected via the SPI bus.

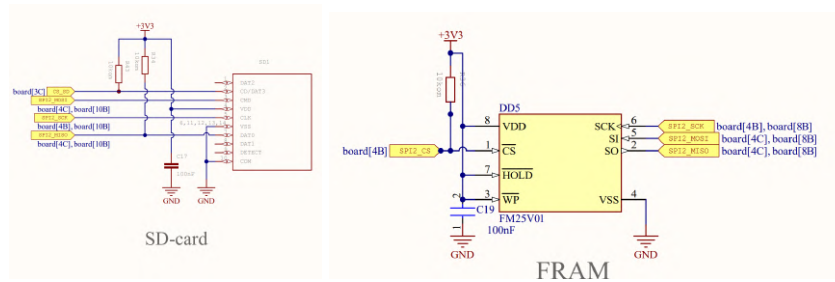


Fig. A-1.55 SD-Card connection and FRAM

ANNEX-2. SPACE ENVIRONMENT AND ELECTRONICS OVERVIEW

This annex begins by describing the space environment, then reviews the most widely used electronics manufacturing standards in the aviation, military, and space sectors.

A-2.1. Space Environment

There are not many sites in space where a drone can fly, atmosphere is mandatory in order to operate an UAV, some examples can be:

A-2.1.1. ISS: International Space Station

Similar environment compared to Earth. Temperature around 20-24°C, pressure 1 atm. Air composed mainly by N₂ 78% and O₂ 21%. As it is approximately 360 km in altitude, it is protected and affected by Van Allen belts. Despite this, 2 main constraints a drone can approach.

The first one, the need of air flow for cooling purposes as there is microgravity and no pressure difference.

The second one, the radiation from cosmic rays, radiation from the south atlantic anomaly and general space radiation. For humans the radiation received is around 0.5-1 mSv per day (72 mSv per 6 months), while the total ionizing dose for an object can be <20 rads per year. For electronics the main problem comes from the south atlantic anomaly and cosmic rays as <20 rads is relatively low.

A-2.1.2. MARS

Mars has a very thin atmosphere of 11.1 km height. Composed by Co₂ 95%, N₂ 2.7%, Ar 1.6% and other components with a gravity of 3.7 m/s². Temperature can range from -153°C to 20°C, while the common value is -63°C with a pressure of 600-1155 Pa. Mars' atmosphere varies in temperature and circulation every Martian year, though it is predictable (it has axial tilt). The only precipitation that can be found on this planet is Co₂ snowfall (water-ice rarely). Moreover katabatic winds/jumps are 3 times stronger than on Earth.

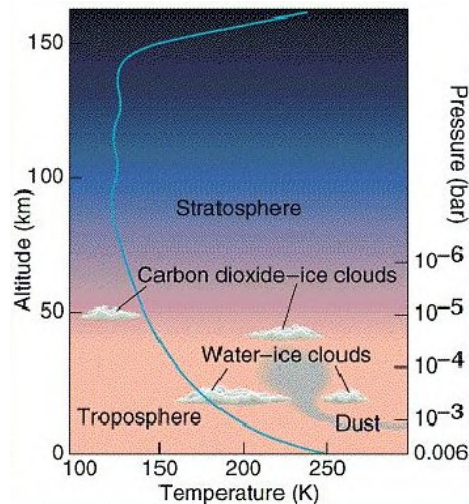


Fig. A-2.1 Mars atmosphere diagram [62]

Its atmosphere has low density 0.02 kg/m^3 , it is easy to produce dust storms of fine particles. Martian regolith can interact with surfaces eroding them, for instance by saltation effect. This regolith is being electrically charged by solar radiation and friction.

Regarding ionizing radiation, Mars has practically no magnetosphere that leads to a high radiation level (only crustal and induced magnetosphere), however the physical layer of atmosphere provides only 20-30 rads per year on ground. When a solar event takes place, hardware can be damaged or even by a SEE (single event effect). A drone can fly in the atmosphere of Mars (like Ingenuity), but the main problems are the air density, electronic protection (temperature and radiation) and regolith erosion.

A-2.1.3. VENUS and TITAN

- **Venus:**

The atmosphere of Venus is around 70-80 km. Its composition is CO_2 96.5%, N_2 3.5% with other chemicals. Much denser than on Earth, the surface temperature is around 740 K and 93 bar (92 atm or 9.3 MPa). Gravity is 8.9 m/s^2 . Venusian clouds contain large fractions of sulfuric acid, hydrochloric acid, bromine, and iodine; they cover mostly all Venusian surfaces. Lightning and sprites are possible but still under debate by the scientific community. This atmosphere suffers from the phenomenon of super-rotation (the troposphere particularly rotates faster than the planet itself).

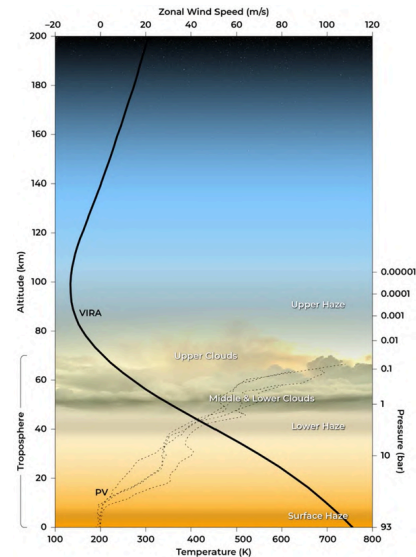


Fig. A-2.2 Venus atmosphere diagram [63]

As Mars, Venus lacks a magnetosphere. Despite this, the ionosphere separates the atmosphere from outer space and solar wind, providing an ionized surface that creates an induced magnetosphere. Due to high temperatures, the evaporating water vapor is blown away to outer space through the magnetotail. At the planet surface, radiation can be really low due this induced magnetosphere with the capability of even absorbing UV radiation. The main problems in Venus are corrosion, pressure, temperature, IR and visible radiation. If a drone has the plan to fly at high altitudes, solar wind and cosmic rays radiation shall be taken into account.

- **Titan:**

Titan composed of N₂ 93%, CH₄ 5% and Ar, is the largest moon of Saturn. Its gravity is 1.352 m/s² with a pressure of 1.5 bar (1.45 atm or 146.7 kPa) and a temperature of 93.7 K. The atmosphere is really big, the troposphere is 40 km, the thermosphere/mesosphere around 600 km and the exosphere up to 975-1200 km. Clouds and lakes can be formed by methane and ethane in Titan.

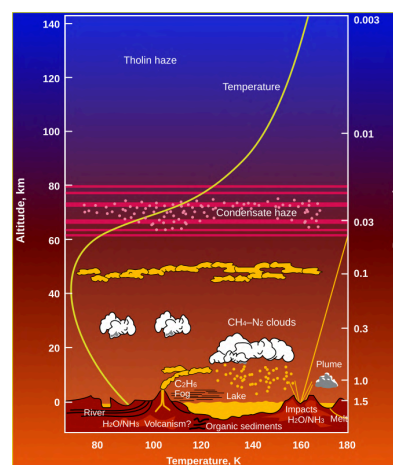


Fig. A-2.3 Titan atmosphere diagram [64]

Titan magnetic field is negligible, the only magnetic fields that act on the surface of Titan are the remnants of Saturn. The radiation is low at surface level with low levels of solar and cosmic radiation. If the magnetosphere of Saturn is altered due a corona mass ejection impact or any other cause, Titan would be affected. The main problems with electronics hardware are composition and temperature and somehow could be radiation too (when Saturn magnetic field is not present, a rare condition).

A-2.2. Standards Introduction

There are many standards devoted to space electronics applications, some of them are important to mention. All the standards mentioned are with the intention of focusing on electronics and the space environment, [65].

- **United States Military Standards (MIL-STD or MIL-SPEC or MilSpecs):**
Implemented during the cold war (1950-1980), they are used to help to standardize the objectives by the United States Department of Defense (DoD). These standards particularly comprise plenty of topics of defense-related objectives, though this project only focuses on those that comprise electronics. These standards can be defined as:
 - MIL-HDBK: Defense Handbook
 - MIL-SPEC: Defense Specification
 - MIL-STD: Defense Standard
 - MIL-PRF: Performance Specification
 - MIL-DTL: Detail Specification

The need for proper performance, maintainability, MRO (Maintenance Repair and Overhaul), and logistical usefulness was detected during WWII between British-French and American equipment. These faults were the cost of missions and human lives during this period. MIL-STD established mandatory rules for compliance between manufacturers and operators seeking absolute reliability and interoperability.

In 1994 the initiative by William J. Perry cancelled many specifications and standards due the increasing cost by the unnecessary restrictions and contractor. This Perry Memorandum opened a path for standardization. As the commercial standards were rapidly growing in the market (COTS or Commercial-off-the-shelf), the United States government was and is open to adopt them. This does not mean that MIL-STD is not used anymore, it is a good standard though IPC is being more used in some sectors such as PCB design.

- **IPC (Institute of Printed Circuits or Global Electronics Association):**
Principal industry standard; however, other standards follow this one such as: JEDEC (semiconductors), IEC/ISO (electric security, electromagnetic), MIL-STD and UL (certifications in security and materials). This standard was born in 1957 due the lack of interoperability of designers, manufacturers and assemblers. In comparison to MIL-STD, these standards were neutral based on performance, not on the procedure. Favoring the innovation in the sector

and lowering the prices. This standard was globally known when DoD (Department of Defense) adopted them. This global trade association is accredited by the ANSI (American National Standards Institute).

As a curiosity to the lecturer, these standards are not open source and they are very rigorous in the standardization worldwide. Despite this, some old open versions can be found on the internet. IPC standards only comprises PCB fabrication, assembly and packaging, no electronic component, IC or chipset is involved.

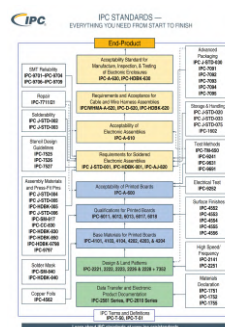
- ECSS (European Cooperation for Space Standardization):**
 Created by ESA and other national space agencies like CNES, DLR, etc. Established in 1993, this organization is in charge of developing, and maintaining a coherent single set of user-friendly standards for use in all European space activities. It replaced the old Procedures, Specifications and Standards (PSS) system, with 139 ECSS active standards. This standard is not a copy of IPC, is a completely parallel standard dedicated to the space industry.

A-2.3. IPC Standards

There are a lot of IPC standards and moreover they are not open source, so this section exposes old versions (not older than 2003) of the PCB design standard as well as open source information. IPC divides electronics products in 3 categories:

- Class 1: General electronic product**
- Class 2: Dedicated service-electronic products**
 Like class 1 but with more stringent standards.
- Class 3: High-performance electronic products**
 Used on aeronautical-military-space products. Products where high performance or performance on demand is critical and they shall provide uninterrupted service even in harsh environments.

There are multiple standards to review, but only some of them are key to mention. For the creation of PCBs IPC-2221 with their sub-specifications (for instance IPC-2222) and qualification/performance on rigid PCBs IPC-6012. For the assembly J-STD-001 (version GS for space), acceptability on assembly IPC-A-610 (very famous standard with versions GS and JS for space). For coating IPC-CC-830, and lastly requirement and acceptability for cables and supports IPC/WHMA-A-620.



For space and military products IPC addendum IPC-6012ES and the generic IPC-2221 are quite important, [66], [67]. Some of the practices inside these standards for class 3 are, [68], [69], [70], [71]:

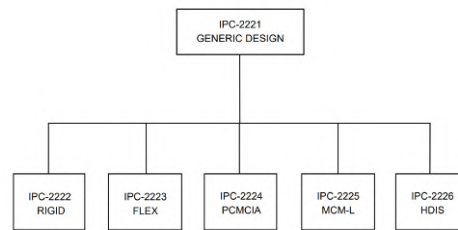


Fig. A-2.5 IPC-2221 standard tree

- **Dielectric/material selection:**

The typical material in PCBs is FR-4 (NEMA designator for glass-reinforced epoxy laminate material) (FR stands for Flame Retardant). For space applications, it depends on the entity, NASA for instance uses a poliamida Arlon 33N as a substitute due its resistance to radiation and high temperatures. Moreover, all material and varnishes shall not contaminate or outgas in space conditions.

This material should have T_g above 170°C (Glass transition Temperature), to avoid the CTE expansion (Coefficient of Thermal Expansion), and prevent delamination. Furthermore, the copper profile has to be smooth, the material thermal conductivity above 0.4 W/mK for dissipation. For frequency applications the dielectric constant is less than 4. Lastly it should be CAF-resistant and RoHS compliant to resist corrosion like lead, cadmium, mercury, etc. CAF (Conductive Anodic Filament) problem is usually formed when oxide or any material grows in the PCB creating a short-circuit.

- **Stack-up rules:**

Symmetric stackup, with power/ground planes to minimize EMI (Electromagnetic Interference). For GND it is recommended 1 oz while for high current zones 2-3 oz. Between coppers 2 plies of prepregs and the minimum dielectric thickness of 3.54 mil.

- **Quality for vias:**

It is not acceptable to have a plating void. Laminate cracks up to 3.14 mil are acceptable. At least 10 mils of separation between drill-to-copper. Large annular rings and vias with teardrops are encouraged to relieve material stress.

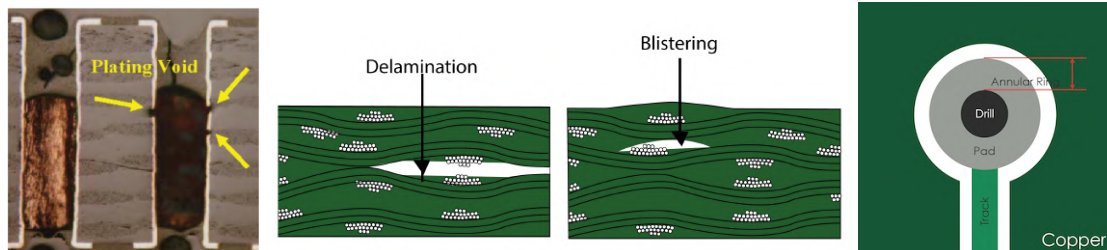


Fig. A-2.6 Via and delamination

Copper wrap plating on vias to increase reliability around 0.472 mil with protrusion of less than 1.96 mil. The minimum plating thickness for through, blind and buried hole is 0.98 mil while blind (<2 layers) and buried microvias is 0.472 mil.

In Class 3 boards positive etch back is recommended (especially for space applications, where high temperature conditions are reached). When temperature rises, the expansion of dielectric material can produce cracks on vias. To avoid this, a plasma etcher is used to burn/reduce the sidewalls of the vias. After that, the vias are plated with copper, wrapping the layers with 3 interconnection joints, making a reliable connection resistant to thermal stress.

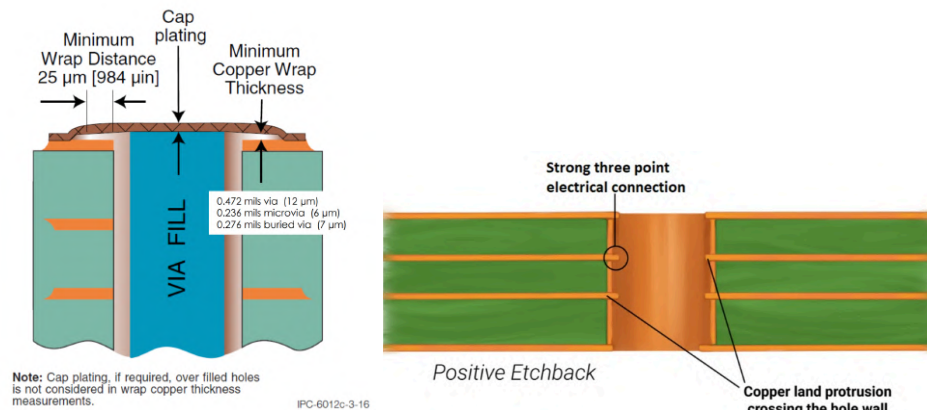


Fig. A-2.7 Via plating and positive etch back treatment

- **Minimum annular ring diameters:**

For pads with holes, the minimum value of the annular ring to the hole is 2 mil while for internal layers is 1 mil. IPC-6012 shows that no fractured/lifted rings and no drill breakouts are allowed.

For Pad size = hole diameter + 2(minimum annular ring size) + fabrication allowance.

The drill should be 4-5 mil greater than the finished hole due to obtaining a proper plating.

- **Soldermask and silkscreen:**

Soldermask between 0.6-1.4 mil with an adhesion of 100%. Soldermask bridge should be a minimum 4-5 mil. Connectors should be plated to avoid chipping off the soldermask.

Silkscreen clearance should be 5 mil with a minimum size of height 25 mil and line width of 4 mil.

Additionally, Class 3 boards shall use IPC-SM-840, Class H solder resist coating (soldermask). In order to maintain electrical integrity a total encapsulation of all conductors is mandatory, tented vias should be covered with permanent solder resist. Devoted for military and aerospace products where failure is critical, withstanding 500 VDC per 1 mil of thickness, zero peeling/flaking at 260°C during 10 s, it should survive a minimum of rapid

cycling between -65°C and $+125^{\circ}\text{C}$ without cracking, tested for 168 h at 85°C with 85% humidity with no dendrite grow.

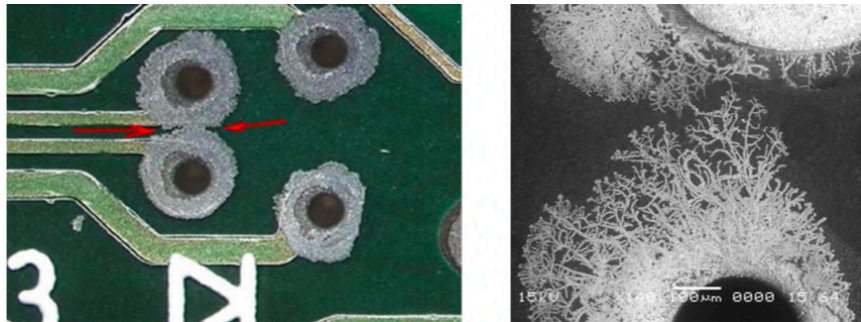


Fig. A-2.8 Dendrite grow

- **Soldering standard:**

Regarding J-STD-001, no defects like barrel crack are permitted. Through hole lead should be above 75% in connection. PTH (Plating Through Hole) at least 15 mil in diameter. Defects on path should be lower than 10% on external while internal is 20%. Lastly the boards should maintain at least 4 excursions of thermal and mechanical stress during solder reflow.

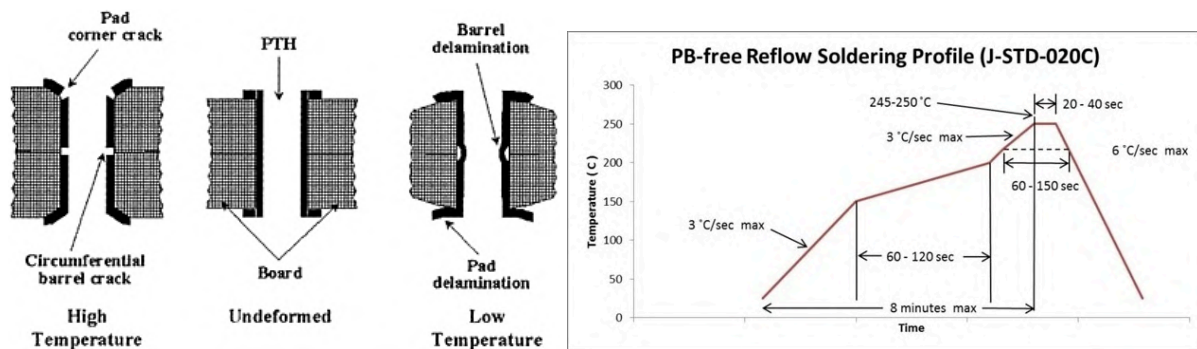


Fig. A-2.9 Via-Pad errors and solder reflow steps [72]

Regarding the standard on assembly acceptance IPC-A-610, [73]:

In the development of a PCBA, multiple stages take action. For all boards a visual inspection is required, usually with a microscope. When mounting components, pick-and-place (PnP) takes visual inspection of all mounted components. Then AOI (Automated Optical Inspection) machines verifies that all mounted components and boards comply with the standard and requirements. Other important inspections that are important too are:

1. X-ray inspection for capsuled components, mandatory in Class 3.
2. SPI (Solder Paste Inspection).
3. ICT (In-Circuit Testing) inspection on short circuits, opens, continuity and passive values.
4. FCT (Functional Test) not IPC but still required in aerospace products.

Factors to inspect:

- **Defects on leads and terminations:** It can be considered as a defect when +10% of lead is not connected. If it can present any careless bending or repeated deformation or heavy marks such as due instrumentation.
- **Bent or warped leads:** The system ensures IPC-A-610 compliance by automatically detecting lead damage, such as heavy tool marks or excessive bending. A defect is officially recorded if the damage exceeds 10% of the lead's thickness or if poor coplanarity (misalignment) prevents a proper solder connection.

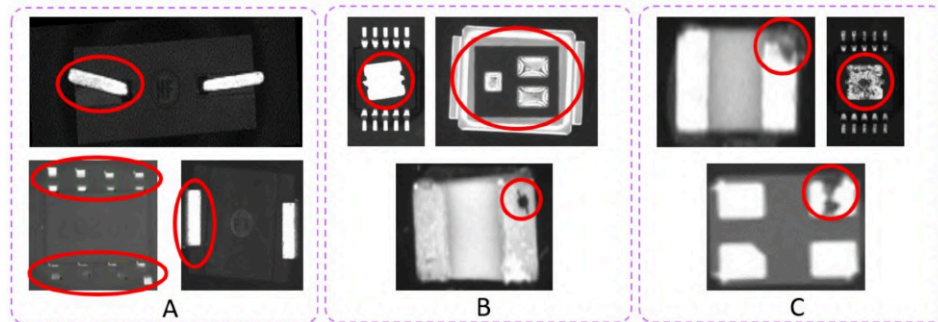


Fig. A-2.10 (A) bent leads, (B) deformed leads, (C) damaged leads

- **Corrosion and cleanliness:** Colored residues or rusty appearance on metallic surfaces or hardware is considered a defect.
- **FOD (Foreign Object Debris):** Critical failures like FOD, violations of minimum electrical clearance, and the presence of white residues or crystalline deposits on the board and terminations.

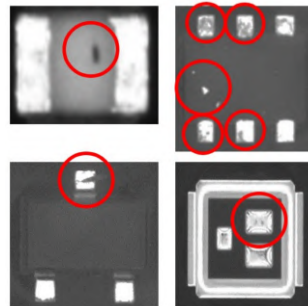


Fig. A-2.11 FOD in PnP process

- **Loss of metalization:** Exceeding 25% of the termination's metalization width or height is considered a defect. This effect can impact conductivity, with the possibility of underperforming the system. Irregular shapes exceeding max/min dimensions or a metalization loss of more than 50% is considered as a defect. Components and connections with cracks, flaking, peeling, blistering shall be revised.
- **Mounting side and Miss-marking:** Need of evaluation for correct usage.

A-2.4. NASA Electronic Standards

During the 1960s, NASA relied on internal handbooks (NHB/NHBK), which provided extensive guidance on reliability, workmanship and electronic hardware practices, including PCB-related aspects (NHB 5300.4 series).

From the late 1980s onward, NASA progressively formalized workmanship requirements through the NASA-STD-8739 series, which standardized soldering, PCB assembly, cabling and fiber-optic practices. Following the Perry Memorandum and the broader shift toward commercial standards (COTS), NASA increasingly adopted IPC standards for space applications, such as IPC-2221/2222, J-STD-001S, IPC-6012S and IPC-A-610S, while retaining some series of NASA-STD-8739 as the governing framework. The introduction of IPC was difficult especially because these standards are RoHS compliant. RoHS establishes the non-usage of SnPb, which can lead to the apparition over the time of “tin whiskers” connections, creating short circuits. Nowadays the solution still is a field of research, mainly for commercial PCBs mitigating with special coatings. NASA uses SnPb solder.

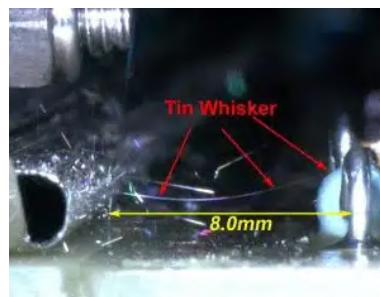


Fig. A-2.12 Tin Whiskers short circuit

Regarding Integrated Circuits (ICs), MIL-STD-883 (test methods) and MIL-M-38510 (general specifications) were used as the primary standards during the 70s and 80s. Following the Perry Memorandum, NASA migrated to MIL-PRF-38535/34, toward performance-based requirements and the QML (Qualified Manufacturer List) system. To manage modern mission risks, NASA introduced EEE-INST-002, which governs the selection, screening and qualification, including the use of COTS components.

Other components like resistances (MIL-PRF-55342 and MIL-PRF-39007), connectors (MIL-DTL-38999 and MIL-DTL-83513), capacitors (MIL-PRF-123 and MIL-PRF-39006), optoelectronics (MIL-PRF-19500), etc.

For radiation measurement and testing NASA uses MIL-STD-883 (specially method 2007, 1019 and 1010), ASTM F1192/JESD57 and NASA-HDBK-4002A (handbook). For outgassing ASTM E595 and plasma NASA-STD-4005 (for LEO orbit). Interestingly, these are only a few of them, there are many more standards.

A-2.4.1. NASA-STD-8739

Apart from the IPC standards, NASA-STD-8739 provides a series of documents. Some of them are not used anymore like 8739.2 (SMT) or 8739.3 (solder on connections), both replaced by IPC J-STD-001S. In the case of the document 8739.7 (ESD Control), it was replaced by ANSI/ESD S20.20. NASA-STD-8739.4 (wiring) is

still under revision but mostly IPC is preferred. And documents like versions 1, 5 and 6 are still under usage. Some open source and relevant data coming from these documents can be, [74]:

- **NASA-STD-8739.1B (Polymeric Application):**

- **Material selection:** Not every adhesive can be used. It must comply with ASTM E595 normative for outgassing (less than 1% total mass loss TML and less than 0.1% of collected volatile condensable material CVCM). Previous to the application of the material (1% of required mass precision), a degassing is required to eliminate trapped bubbles.
- **Cleanliness requirements:** At the void a dirty surface can create material delamination on PCBs. Immersion, ultrasonic, solvent, plasma cleaning and microscope verification is necessary. Cleaning methods depend on material usage. If a solvent is used, this can be extracted after the cleaning. Measuring the resistivity of this solvent can provide a value of how clean the board is.
- **Staking:** If a component weighs typically more than 7 g, shocks and vibrations can break its connections to the board. It shall be fixed to the board with adhesive with thermal resistance on non-soldered joints and cured. Fastener staking can be used, but bolts and screws shall contain staking material. Solder joints must not be covered as they can affect connections.
- **Conformal coating:** A coating is required to protect the PCBA from humidity, handling, debris, and contamination. If it is too thin, it won't protect. If it is thick, it can suffer cracks due to thermal stress. Coating bridging is forbidden. Preferable vacuum deposition in order to apply coating, though other methods like brushing or spray can be used.

Type of Coating	Cured Coating [mm (inch)]
ACRYLIC	0.025 – 0.127 (0.001 to 0.005)
URETHANE	0.025 – 0.127 (0.001 to 0.005)
EPOXY	0.025 – 0.127 (0.001 to 0.005)
SILICONE	0.051 – 0.203 (0.002 to 0.008)
PARAXYLENE	0.013 – 0.051 (0.0005 to 0.002)

Fig. A-2.13 Coating thickness

- **Bonding:** Defects can be bubbles, delamination, mealy appearance, cracks, etc. Marking shall be visible with stress relief without creating any excess of material.
- **Encapsulation:** Potting and encapsulation processes require strict management. Encapsulated bubbles next to high voltage connectors can ionize and create electrical arcs. Encapsulated

PCBA zones shall contain a good conduction of thermal relief, and they shall be flexible. Lack of adhesion, spillage or improper filling is considered a defect. Flexible bubbles or cavities less than 0.635mm in diameter can be allowed after inspection.

- **NASA-STD-8739.5A (Fiber Optic Terminations):**

- **Optical fiber:** Materials shall comply with the normative ASTM E595. Optical fiber adhesives shall be space graded (not containing silicones or lubricants and with glass transition temperature above the maximum found during test/mission). Degassed or centrifuged prior to the application to eliminate bubbles and voids as they can create a fiber breakage.

Change of temperature in space can drastically affect the optic fiber. Pistoning is an effect that makes the fiber move forwards or backwards the connector. To retain the fiber in the connector, adhesive like space graded epoxy has to be cured. It is mandatory to keep more optic fiber than needed on connections, because at low temperature it can tense and stress the fiber breaking it.

- **Integrity:** Bend radius shall be 10 times the cable diameter to avoid cracks. All cables shall be secured with braces due to high accelerations at launch. Visual inspection at terminations is mandatory; X-rays inspection and other means are permissible (not detrimental to the product). No chipping or crack is allowed on the fiber core.

- **NASA-STD-8739.6B and ANSI/ESD S20.20 (Requirements/ESD):**

- **ESD (ElectroStatic Discharge):** Sometimes ESD does not lead to catastrophic failure on components, sometimes it weakens the internal oxide layers of the semiconductor, creating a latent defect named "Walking Wounded". If thermal cycling or radiation treats these zones, it can even destroy them. Standard S20.20 classifies rupture voltage sensitivity with HBM (Human Body Model) and CDM (Charged Device Model), usually <250 V.
- **Working conditions:** Usually these rooms are clean room classified ISO-7 $\frac{1}{8}$. Temperature around 18-30°C with RH below 30-70% (modified in case of ESD control). All materials shall meet project contamination, outgassing and offgassing requirements (NASA-STD-6016).

Deionized water, ethyl alcohol, and isopropyl alcohol are considered standard solvents to use for cleaning PCBs. Moreover it is mandatory to ionize air to maintain neutral air charge to reduce static.

During the management of ESD special equipment, surfaces shall be between 10^6 to $10^9 \Omega$ and the operator should wear a wrist strap system to reduce ESD (all equipment connected to GND). All material and equipment shall be traceable from origin “cradle-to-grave”.

A-2.4.2. MIL-PRF-38535M

This standard is the main standard of IC manufacturing for military, space and critical applications of USA DoD, [75]. It is a large document, but let's keep it simple. This standard bases the manufacturing on the QML (Qualified Manufacturer List). Instead of qualifying a single product it qualifies and certificates the complete fabrication line. This is done by the TRB (Technology Review Board) operators, monitoring the manufacturing. This provides homogeneity in the final product, where the first and the last IC are identical. Usually when a manufacturing line is considered in the QML is when technology is statistically proven and it won't have flaws.

This standard provides different screening levels:

- **Class V:** Space grade hermetic product (usually metal or ceramic encapsulation). All ICs are tested. Including PIND (Particle Impact Noise Detection) and X-ray test to mitigate launch and microgravity risks.
- **Class Y:** Space grade non-hermetic. Similar to class V, though, high-performance processors (FPGAs, CPUs, ASICs, etc) that require flip-chip or organic packaging. Includes specific tests for outgassing and moisture sensitivity.
- **Class Q:** Military grade. It does not require outgassing or PIND tests. But sometimes used in LEO orbits.
- **Class M:** Military grade, but certified at the end of the fabrication.
- **Class N:** Military grade. Plastic based.
- **Class T:** Space grade. Not as resistant Class V or Y. Used for LEO orbits.
- **Class K:** Similar to Class V but on hybrid ICs (multiple microcircuits in a common package). Explained in MIL-PRF-38534.
- **Class H:** Similar to Class Q but on hybrid ICs. Explained in MIL-PRF-38534.
- **Others:** Class-G,D,E

Currently the most used classes for space applications are Class Y and K. Where cost and production capabilities are increasingly important.

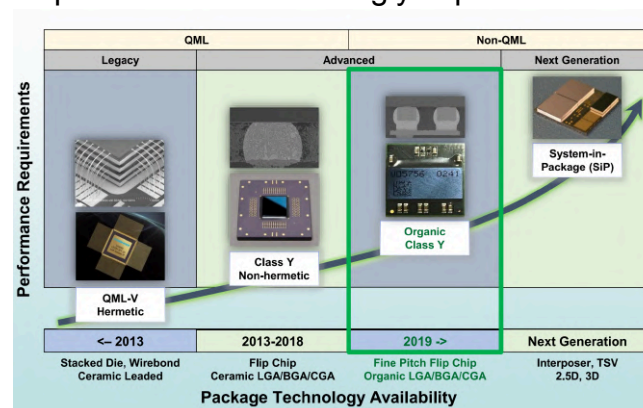


Fig. A-2.14 NASA MIL-PRF-38535M revision, [76]

Every IC, before leaving the manufacturing facility, shall be processed by the screening flow (MIL-PRF-38535M, [75], in Table AI). Summary:

1. Wafer lot acceptance: chip wafer is tested in order to see if it can bear the TID (Total Ionizing Dose).
2. Inspection: visual inspection.
3. Temperature cycling: Temperature cycling between -65°C and +150°C.
4. Acceleration tests: Acceleration is tested up to 30000 Gs.
5. PIND (Particle Impact Noise Detection): If a particle is inside the chip during microgravity it can produce a short-circuit. This method searches for any possible flaw.
6. Burn-in: Test at high temperature and at a maximum voltage for a long period of time, usually at +125°C.
7. Electrical test at different temperatures.
8. Sealing test (Hermeticity).

Moreover this standard defines multiple levels of RHA (Radiation Hardness Assurance) depending on the TID (Total Ionizing Dose). Usually level R for GEO and level F for long missions.

RHA level designator (see 3.6.2.1)	Total ionizing dose (TID) level in Rad (Si)
/ or -	No RHA
M	3 krad(Si)
D	10 krad(Si)
P	30 krad(Si)
L	50 krad(Si)
R	100 krad(Si)
F	300 krad(Si)
G	500 krad(Si)
H	1 Mrad(Si)

Fig. A-2.15 RHA levels

Single event effects (SEE) testing shall be performed for all RHA. Not limited to: single-event latchup (SEL), single-event burnout (SEB), single-event gate rupture (SEGR), single-event upset (SEU), single-event transient (SET), single-event dielectric rupture (SEDR) and single-event functional interrupt (SEFI).

SEL, SEB and SEGR are considered destructive effects while SEU and SET are considered non-destructive effects. SEFI is considered to be nondestructive unless accelerated life testing. These tests are performed according to JESD57 on heavy ion radiation and according JESD234 for proton radiation.

Additionally, this standard provides a IC nomenclature for SMD:

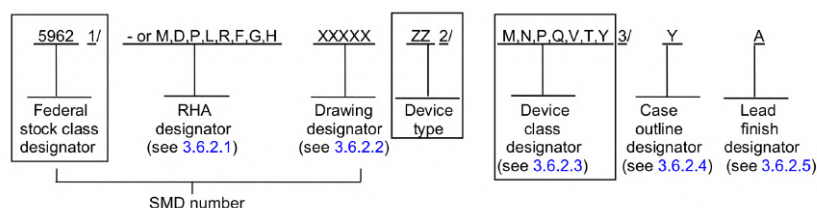


Fig. A-2.16 Component designator in MIL-PRF-38535M

A-2.4.3. EEE-INST-002

This standard is titled: Instructions for EEE Parts Selection, Screening, Qualification, and Derating, [77].

Screening as mentioned before, stands for tests intended to remove nonconforming parts. While derating is the reduction of electrical and thermal stresses applied to a part during normal operation in order to decrease the degradation rate and prolong its expected life.

It exposes 3 quality levels:

- **Level 1:** High reliability, deep space products, (>5 years). Only class V is allowed, where rigorous screening and destructive physical analysis on every lot.
- **Level 2:** Standard reliability, (1-5 years). Class Q is acceptable, where reduced screening tests have been performed.
- **Level 3:** High risk, (<1 year). Not as reliable as level 2. Used for cubesats and demos. Permits the use of COTS and plastics (tested for space applications).

Why derating is important: In an environment where vacuum is present, and conduction is the only method for thermal relief. Any electronic component running at its 100% capacity will overheat, leading to accelerated degradation and reduced operational lifespan. Some derating methods are:

- Resistors shall operate at less than 60% of their rated power.
- Ceramic capacitor powered with less than 50% of its rated voltage.
- Tantalum capacitor powered with less than 50% of its rated voltage. Prone to produce catastrophic explosions.
- Semiconductors shall be working below 110°C.

A-2.4.4. MIL-STD-883E (a few methods)

This standard contains how to replicate and develop test methods for microcircuits, [78]. It comprises multiple methods, many of which are mentioned in EEE-INST-002 and MIL-PRF-38535M.

- **Method 1010, Temperature cycling test:** The microcircuit is placed in a dual-chamber air-to-air system. A minimum of 10 cycles between chambers to replicate the eclipse cycle. In each chamber it must stay at least 10 min. Typical temperatures are -65°C and +150°C. After the process visual inspection is required (usually delaminations, cracks or wire-bonds).

		Test condition temperature (°C)					
Step	Minutes	A	B	C	D	E	F
1 Cold	≥10	-55 +0 -10	-55 +0 -10	-65 +0 -10	-65 +0 -10	-65 +0 -10	-65 +0 -10
2 Hot	≥10	85 +10 -0	125 +15 -0	150 +15 -0	200 +15 -0	300 +15 -0	175 +15 -0

Fig. A-2.17 Test temperatures

- Method 1019, TID (Total Ionizing Dose) test procedure:** Samples shall be exposed to Cobalt-60 gamma ray source. The device must be tested inside a Pb/Al container (0.7 mm Al inner layer and 1.5 mm Pb outer layer). To calculate the dosimetry, use standards ASDTM E (666, 668, 1249, etc). Electrical parameters are measured before and after to check any possible damage. It is crucial to mention ELDRS (Enhanced Low Dose Rate Sensitivity), this phenomena applies on BiCMOS and Bipolar transistors (op-amps, regulators, etc). To check this phenomenon the device has to be tested at low radiation levels for a long period of time. This is because when it is exposed at high radiation levels it creates an electric field that protects the transistors from incoming radiation (secondary radiation indeed). While during periods of low radiation no field is generated, leaving the most critical part of the transistor exposed.

What's more, for CMOS ICs, it is mandatory to age the device (perform a burn-in test) due to the fact that it can modify radiation tests.

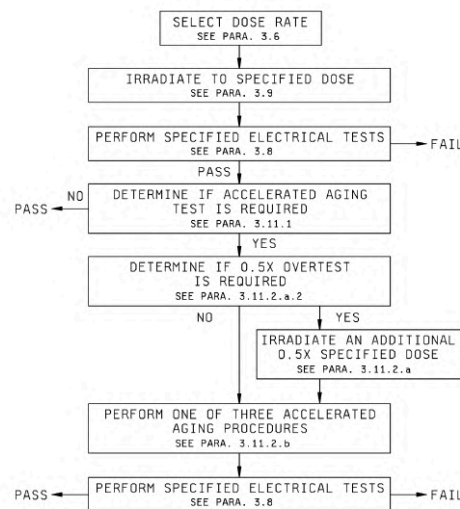


Fig. A-2.18 Flow diagram for ionizing radiation test procedure

- Method 2007, Vibration test:** Simple test where the component is fastened to a vibration platform. Subjected to a simple harmonic motion but logarithmically variable in frequency between 20-2000 Hz. Finally returning to 20 Hz in not less than 4 min (complete test). Performed 4 times in each of the 3 axes of the component (X, Y, Z). After the test visual inspection is required, specially on wire bonds and die-attachment.

A-2.4.5. NASA-STD-4005

Its title is Low Earth Orbit Spacecraft Charging Design Standard, [79]. Plasma is another environmental constraint that electronics can find in space. High voltages can produce aging, power drains, power disruptions and loss of spacecraft coatings.

LEO (Low Earth Orbits) ranges from 200 km to 1000 km. It contains the ionosphere composed of light molecules and ions and the inner Van Allen belt, composed mainly by protons, both in the state of quasi-neutral plasma.

This standard exposes 2 main charging mechanisms.

- When the spacecraft crosses polar regions, it can encounter high energy electron flux. This flux can be deposited on dielectric surfaces like solar cells coverglass and blankets creating a huge differential of potential between surfaces.
- Wake effect: Due to orbital velocity, the spacecraft creates a plasma wake behind it. This results in an ion void that becomes highly negatively charged relative to the surrounding plasma (electrons are faster than ions, ions can't be deposited due to the wake).

The main hazard produced by these mechanisms is the ESD (Electrostatic Discharge). When 2 surfaces have different potential a discharge arc can occur under gas/plasma influence (commonly by local outgassing products), following the Paschen equation. Creating physical damage to surfaces, short circuits, and EMI injection capable of corrupting logic signals or resetting the On-Board Computer (OBC).

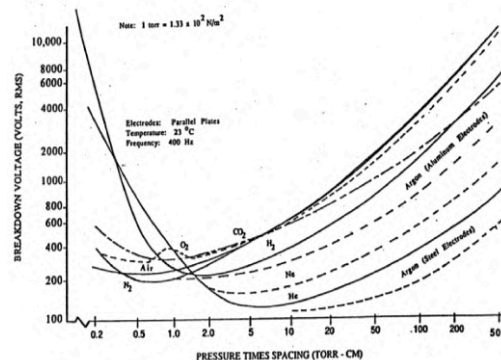


Fig. A-2.19 Paschen curves for different gases, from Dunbar, 1988

There are several mitigation techniques:

- Use system voltages lower than the trigger arc threshold.
- Encapsulating exposed electron-collecting conductors.
- Eliminate floating conductors, all metallic components grounded with a resistance less than 1 Ω .
- If possible external dielectrics replaced with conductive material.
- Triple junction components are the primary initiator. They shall be shielded.
- Geometry designed to not exceed 100 V on surface potential.
- Prevent sharp triple-points.
- Grouting the edges of cells or any electrical component.
- Use of thick coverslides on solar arrays.
- Creating physical barriers between if no other choice.
- Use of blocking diodes.
- Mitigation of sustained arcs via RC time constants: The circuit impedance is designed to ensure that the response time exceeds the arc discharge duration. This delay prevents the stored energy from the rest of the array from feeding the local discharge.

- Open-circuiting strings where arcs can occur, this can prevent the arc from extending in duration, usually $<200 \mu\text{s}$.
- Use of plasma contactor with a grounded neutralizer.
- Using thick dielectric coatings with a high breakdown voltage.
- Place vents and nozzles far away from conductors.
- Vent gases with high Paschen minimum voltages and create void.
- Fill enclosures with electron-sponge gas (SF_6).
- Limiting circuit capacitance.

A-2.5. ECSS-ESA Standards

The European side is much simpler indeed. European system divides in:

- ECSS-P: Standardization objectives, policies and organization.
- ECSS-S: System description.
- ECSS-D: Configuration and information management.
- ECSS-M: Management
- ECSS-U: Sustainability
- ECSS-E: Engineering (important on electronics)
- ECSS-Q: Product Assurance (important on electronics)

There are multiple standards but only a few of them are important to mention:

A-2.5.1. ECSS-E-ST-20C

Entitled as Electrical and Electronic, [80]. This standard establishes the fundamental rules and general principles applicable to electrical, electronic, electromagnetic, and microwave engineering processes. This section focuses specifically on the requirements relevant to electronics and the space environment.

The design philosophy mandates a system that is SPF (Single Point Failure) free. A single failure must not propagate to neighboring components or interfaces. For instance, power lines must ensure redundancy through physical separation. The spacecraft electrical system generally requires single-failure tolerance for unmanned missions and double-failure tolerance for manned missions (human-rated safety).

Protection and autonomy:

- All units dissipating more than 20 W shall include a temperature monitoring function.
- Any autonomous protection function must be capable of being isolated, reset, or inhibited/bypassed by external ground commands.
- For hot redundant functions, latch redundancy shall not apply unless a periodic reset is performed.
- To mitigate radiation-induced SEL (Single Event Latch-ups), it is mandatory to incorporate LCLs (Latching Current Limiters). Upon detecting a high-current event, the LCL isolates the load to prevent thermal damage and allows a reset after recovery.
- SEE (Single Event Effects) shall not trigger protection circuits of essential functions; passive mitigation techniques, such as filtering, should be employed.

Insulation and environmental constraints: Electrical insulation materials must possess a dielectric strength exceeding 2 times the peak operating voltage in the actual application context. Manufacturing processes must be conducted in cleanrooms with controlled contamination levels and a maximum 65% RH. Finally, to mitigate the risks of ESD in space, strict grounding and bonding is mandatory.

A-2.5.2. ECSS-Q-ST-70-12C

This standard defines the design rules for rigid, flexible, and rigid-flex PCBs to ensure survivability under the harsh thermo-mechanical stresses of the space environment, [81]. It is not based on ESA data alone, it uses IPC standards too. This section focuses on rigid PCBs, as they constitute the vast majority of manufactured products.

For the design of a PCB, it is important to decide values such as T_g (Glass Transition Temperature), Young's modulus, dielectric constant, CTE (Coefficient of Thermal Expansion), and hygroscopy. Epoxy and polyamide are the most common materials for PCBs (common FR-4 is not used because it has a low T_g). Polyamide has the advantage of being more thermally stable, for instance under thermal cycling, but it is more hygroscopic. However, epoxy has multiple variants available, including grades with high T_g values. For high frequency applications, typically PTFE (Teflon) is used.

It is crucial to select a material with a low Z-axis CTE. Materials with high Z-axis expansion during thermal cycling cause stress on PTH (Plated Through Holes) and vias, leading to barrel cracking or delamination.

All materials shall pass outgassing requirements as mentioned previously of CVCM <0.1% and TML <1%.

Blind and buried vias must be filled with prepreg resin or plugging paste to prevent contaminants. Through-going vias may be open or filled. PCB copper balance should be symmetrical to prevent warping. Typical copper thicknesses are 70 µm, 35 µm, or 17 µm. A minimum of two sheets of prepreg shall be used for insulation between 2 layers of copper; in the case of glass-reinforced cores, 2 sheets of glass are required.

For HDI (High Density Interconnect) PCBs, microvias can be produced on <17 µm copper layers, and they must never be stacked directly above a buried via. All microvias using laser drill.

For polyimide PCBs, the typical manufacturing thickness shall be less or equal to 4 mm (designed 3.6 mm). For epoxy, less or equal to 2.4 mm (designed 2.2 mm). The manufacturing tolerance shall be less than 10%. The maximum number of copper layers is typically restricted (26 for epoxy and 20 for polyimide) depending on the technology qualification.

The aspect ratio (ratio of board thickness to hole diameter) for vias must be less than or equal to 7:1. The minimum track width is typically around 4.7-6.3 mil (approx. 120-160 µm), with a relative tolerance of 20%. Tracks should generally not be routed on external layers, particularly under components, to avoid trapping contaminants and prevent damages.

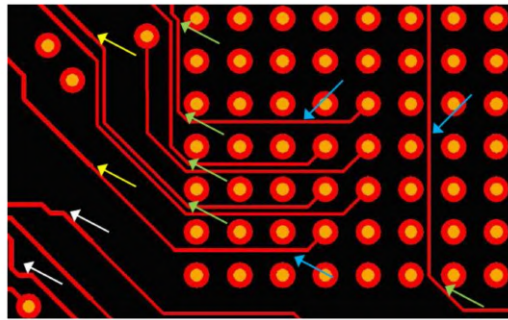


Fig. A-2.20 Example of automated fine pitch routing and possible improvements

Non-functional pads on PTHs shall be removed if they reduce electrical clearance or pose an environmental risk. Pads must be designed to ensure a minimum annular ring. The standard requires the pad diameter to exceed the finished hole diameter by a specific margin (typically a minimum 150 μm to 200 μm external annular ring) to guarantee connection integrity after drilling tolerances. The use of teardrop shapes for junctions is recommended to increase robustness against thermal stress and defects. Peeling and silvering of traces shall be prevented by design.

Spacing between tracks is increased compared to terrestrial standards to avoid arcing and to prevent metal migration, such as dendrite growth. Problems associated with standard solder masks in space include poor adhesion to tin-lead finishes and poor outgassing performance, special qualification is required

Heat sinks shall be made of copper or aluminum cores bonded to the PCB layers, manufactured by milling or etching (milling is preferred due to accuracy).

Finally, ESA projects are not subjected to RoHS restrictions; therefore, the use of SnPb solder is mandatory to mitigate the risk of tin whiskers.

A-2.5.3. ECSS-Q-ST-60C and ECSS-Q-ST-60-15C

1. **ECSS-Q-ST-60C:** Electrical, electronic and electromechanical (EEE) components, [82].

This standard is the European counterpart to EEE-INST-002. It establishes the requirements for the selection, control, procurement, and usage of EEE components for space missions. It references and mandates the application of ECSS and MIL standards for verification and qualification activities.

It defines 3 classes that provide 3 levels of trade-off between assurance and risk:

- **Class 1:** Applicable to critical missions where failure is catastrophic. It requires 100% screening, extensive qualification, and full traceability.
- **Class 2:** A compromise between cost and risk, typically employed in scientific constellations with some redundancy.
- **Class 3:** Lowest assurance. Used on low cost missions like Cubesats, accepting COTS (Commercial Off-The-Shelf) components with reduced testing.

When selecting an EEE component, it should be a qualified part (listed in the European Space Components Coordination QPL or NASA QML). Otherwise a CCP (Component Control Plan) must be generated. This plan is composed of multiple stages where testing, justification and qualification have to demonstrate the component's suitability for the specific mission environment.

Furthermore this standard specifically prohibits the use of pure tin. As pure tin may create "tin whiskers" that can induce electrical instantaneous or permanent short-circuit, metal vapour arc in reduced pressure conditions (when +12 V tin whiskers vaporize generating an arc) or contamination that can interfere with mechanical and optical elements.

2. **ECSS-Q-ST-60-15C:** Radiation hardness assurance for EEE, [83].

This standard is used in parallel with the ECSS-E-ST-10-04 (radiation environments models and rules), the ECSS-E-ST-10-12 (sector analysis or Monte Carlo analysis for radiation). It defines the methodology to ensure that EEE can withstand the specific radiation environment of the mission.

A few concepts are mentioned:

- TID (Total Ionizing Radiation): cumulative energy deposited mainly by protons and electrons mainly over the mission duration. Causing long-term degradation of semiconductor parameters. The standard requires calculating the mission dose and comparing it to the component's sensitivity. Always using the WCA (Worst Case analysis) considering the device EOL (End Of Lifetime).
- TNID (Total Non-Ionizing Radiation or Displacement Damage): damage mechanism is caused by the physical displacement of atoms in the crystal lattice, primarily due to protons and neutrons. Its impact can be seen in optoelectronics reducing efficiency. It is mandatory to obtain degradation curves under proton irradiation.
- SEE (Single Event Effects): Stochastic events caused by a high-energy particle striking sensitive component part. These impacts can cause non-destructive effects such as SEU (Single Event Upsets) or SET (Single Event Transients). Or destructive effects such as SEL (Single Event Latch-up), SEGR (Single Event Gate Rupture) or SEB (Single Event Burn-out). Destructive events shall be prevented using hardened technology or system protection like LCLs.
- To account for EEE uncertainties, RDM (Radiation Design Margin) enforces a safety margin. Always considering components at EOL (End Of Lifetime), RDM should be between +1.2 for tested components and +2 for non tested components. The typical value at geostationary orbits is +1.2 on tested devices.

$$RDM = \frac{\text{Component Failure Level}}{\text{Mission Environment Level}} = \frac{TID}{TIDL} = \frac{TNID}{TNIDL} \quad (\text{A-2.1})$$

A-2.6. SEE and TID calculation (ECSS-E-ST-10-12C)

1. SEE (Singe Event Effect):

These effects are induced by a high energy ionizing particle (MeV-GeV) that disrupts electronic devices. The primary sources are galactic cosmic rays, solar energetic particles, trapped radiation in the Van Allen belts, and secondary radiation generated by primary radiation sources such as Bremsstrahlung radiation. More information can be found in: [84], [85].

- **Destructives:**

- **Single event latch-up (SEL):** Occurs when a parasitic PNP thyristor structure (CMOS technology) is triggered, creating a low impedance path (short-circuit) between power and ground. This leads to overheating and potential melting of the silicon. Typically the use of LCL (Latching Current Limiter) can avoid this problem.
- **Single event gate rupture (SEGR):** Primarily affects high-voltage MOSFETs. When a heavy ion crosses the gate oxide while a high voltage is applied, it creates a plasma conductor track breaking the oxide (dielectric-SiO₂); destroying the MOSFET.
- **Single event burn-out (SEB):** Affects BJT and MOSFETs devices. When the ion crosses the base of the transistor, it creates an electrical avalanche burning the source junction. In comparison the SEGR breaks the dielectric gate while SEB burns the union of the semiconductor by the current/thermal.
- **Single event snap-back (SESB):** Similar to SEL but in a NMOS device. The transistor locks into a state of high current and low voltage. While it may not destroy the device instantly, it causes high current consumption and loss of functionality.

- **Non-destructives:**

- **Single event upset (SEU):** The most common effect. When the ion impacts a sensitive part of cell memory, changing the bit state from 0 to 1 or vice-versa. Rewriting the data solves the problem.
- **Multiple-cell upset (MCU) or Multiple-bit upset (MBU):** When the ion impacts multiple memory cells due to an angular impact path. This impact changes the state of the memory. Rewriting the data solves the problem.
- **Single-word multiple-bit upsets (SMU):** A critical variation of MCU where multiple bits within the same data word are corrupted. When EDAC (Error Detection And Correction) tries to correct the bit it can create a fatal error as it is only able to

correct 1 bit per word. To mitigate this problem, bit-interleaving is used, in other words, bits of a word are separated physically.

- **Single event functional interrupt (SEFI):** A specific type of SEU affecting critical control logic. This can lead the system into an unknown state. An external soft reset or power cycle is used to eliminate this problem.
- **Single event hard error (SEHE):** This effect can be destructive or non-destructive. The impact of the ion leaves the memory between the state of 0 and 1. The memory where the bit is affected shall try to restore the memory or establish the direction of the memory as unusable.
- **Single event disturb (SED):** Typical in Flash memories or FPGAs. When impact, it creates a perturbation that unables a good lecture of data or a wrong operation, without permanently damaging the cell.
- **Single event transient (SET):** The impact creates a peak of voltage in combinational logic during a clock edge. This effect can lead into the apparition of a SEU or it may disappear.

Component type	Technology	Family	Function	SEL	SESB	SEGR	SEB	SEU	MCU/SMU	SEDR	SEHE	SEFI	SET	SED
Transistors	Power MOS					X	X							
ICs	CMOS or BiCMOS or SOI	Digital	SRAM	X*				X	X		X			
			DRAM/SDRAM	X*	X			X	X		X	X		
			FPGA	X*				X		X		X		X
			EEPROM/Flash EEPROM	X*						X		X		X
			μP/μcontroller	X				X			X	X		X
		Mixed	ADC	X*				X				X	X	X
		Signal	DAC	X*				X				X	X	X
		Linear		X*						X			X	
	Bipolar	Digital						X					X	
		Linear						X					X	
Opto-electronics			Opto-couplers										X	
			CCD										X	
			APS (CMOS)	X								X	X	

*except SOI

Fig. A-2.21 Single event effects as a function of component technology and family

To predict the probability of SEE, it is important to understand a few things. More details can be found in ECSS-E-ST-10-12C, [86].

- **LET (Linear Energy Transfer):** This value measures the mass stopping power or ionizing capability of the particle. It provides how

much energy the particle loses by distance. Usually in $MeV \cdot cm^2/mg$. The E variable is the kinetic energy loss of the incident particle while the x variable is the path length.

$$LET = \frac{dE}{dx} \quad \text{or} \quad LET_{norm} = \frac{1}{\rho} \cdot \frac{dE}{dx} \quad (\text{A-2.2})$$

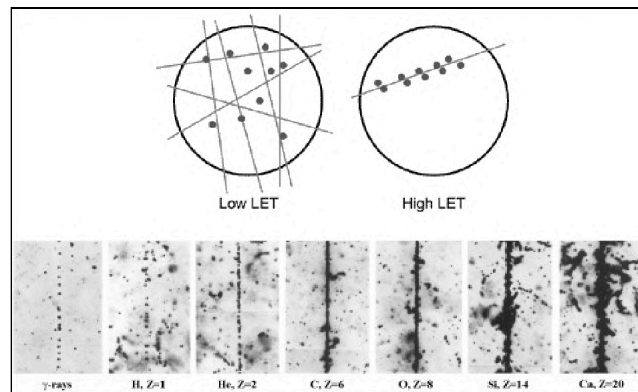


Fig. A-2.22 LET at the microscope

- **Weibull distribution:** The sensitivity of the device to radiation is quantified by the cross-section, which represents the effective sensitive area of the component for a given particle flux, [87].

$$\sigma_{SEE} = \frac{\text{Number of events}}{\text{Particles per } cm^2} \quad (\text{A-2.3})$$

Once the cross-section data is obtained experimentally (cross-section vs LET) in particle accelerators, the ECSS standard requires fitting these points to a continuous curve. The standard model used is the Weibull cumulative distribution:

$$\sigma(L) = \sigma_{sat} \left(1 - \exp \left[- \left(\frac{L - L_{th}}{W} \right)^S \right] \right) \quad (\text{A-2.4})$$

Saturation cross-section is the limiting value of the cross-section at high LET, L is the actual LET and L-threshold is the minimum LET that can affect the device. Finally W (width) determines the slope of the transition region and S determines the shape of the plot.

- **Methods for heavy ion prediction:** There are several methods available to calculate the SEE error rate. The standard approach by ECSS is the IRPP (Integral Rectangular Parallelepiped) method. This method accounts for an isotropic radiation environment and the statistical distribution of path lengths within the volume.

While particles with low LET might be insufficient to cause an upset at normal incidence, the IRPP method considers that they can trigger a SEE if they traverse a long diagonal path through the device, depositing

more energy. This method can be calculated with the following formula, for more information ECSS-E-ST-10-12C and ECSS-E-HB-10-12.

$$N = \frac{A}{4S} \int_{LET_{i,Min}}^{LET_{i,Max}} \left\{ \frac{d\sigma_{ion}}{d(LET)}(LET_i) \int_{\frac{h}{D_{Max}} LET_i}^{LET_{Max}} \frac{d\Phi}{d(LET)}(LET) \cdot P_{CL}(> D(LET)) d(LET) \right\} d(LET_i) \quad (\text{A-2.5})$$

- **Methods for proton/neutron prediction:** Proton and neutron to produce a SEE shall impact directly to the SiO₂ nucleus, LET is not used anymore. To calculate SEE rates the following formula is used:

$$R_{nucleon} = \int_{E_{min}}^{E_{max}} \frac{d\Phi}{dE}(E) \cdot \sigma_{nucleon}(E) dE \quad (\text{A-2.6})$$

Where:

- $\frac{d\Phi}{dE}$ is the particle flux spectrum obtained from models like AP-8.
- $\sigma_{nucleus}(E)$ is the cross-section curve as a function of particle energy (similar to Weibull distribution).

2. TID (Total Ionizing Dose) calculation:

To evaluate the TID in a device is quite different from a biological perspective (which is much simpler). This accumulative degradation phenomena consist mainly of trapped electrons and protons in the device over its lifetime. ECSS uses the SHIELDOSE-2 algorithm to compute this TID. To define this TID several steps shall be performed.

- **Mission environment:** It is important to define the mission environment to determine the total fluence (particles per square cm). The main sources are protons and electrons for instance in the Van Allen belts (AP-9/AE-9 models). Radiation sources to be considered are: solar energetic particles from solar flares or coronas mass ejections (protons and other heavier particles), deep space radiation (gamma, X-rays, electrons, protons, etc) and secondary radiation like Bremsstrahlung (X-rays), neutrons, protons, alpha particles, etc.
- **Dose-depth curve:** The deposited dose is calculated via SHIELDOSE-2 algorithm which is adopted by ESA. It calculates the energy deposited in a target material (SiO₂, GaAs, etc) shielded by a material like aluminium. The result is a dose-depth curve in krad or gray vs mm of aluminium.
- **Shielding analysis:** Since real components are embedded within complex satellite structures, sector analysis (ray tracing) is used. For each ray the software calculates the mass of material transversed, calculating the equivalent aluminium thickness. This provides the specific dose received by that component.

- **RDM (Radiation Design Margin):** To account for uncertainties in the environment and models, the RDM parameter is applied. For standard missions, the RDM is typically a factor of 2.

$$Dose_{design} = Dose_{calculated} \cdot RDM \quad (\text{A-2.7})$$

ANNEX-3. TESTING, PROJECT ISSUES, COMPARISON AND COST

This annex begins by describing the steps followed for testing. Exposing testing results, resolved and unresolved issues, comparison between the developed flight controller and commercial boards, and possible future improvements with board production costs.

A-3.1. Testing with Mission Planner

In order to test the developed board a few things have to be done beforehand. Such as a PCB cover and drone set up.

A-3.1.1. Board case

A simple 3D case has been designed using Solidworks with a tolerance of 0.1 mm, 2 piece case. The holes are sized for M2.5 screws. It is a quite simple structure with 2 mm wall thickness.

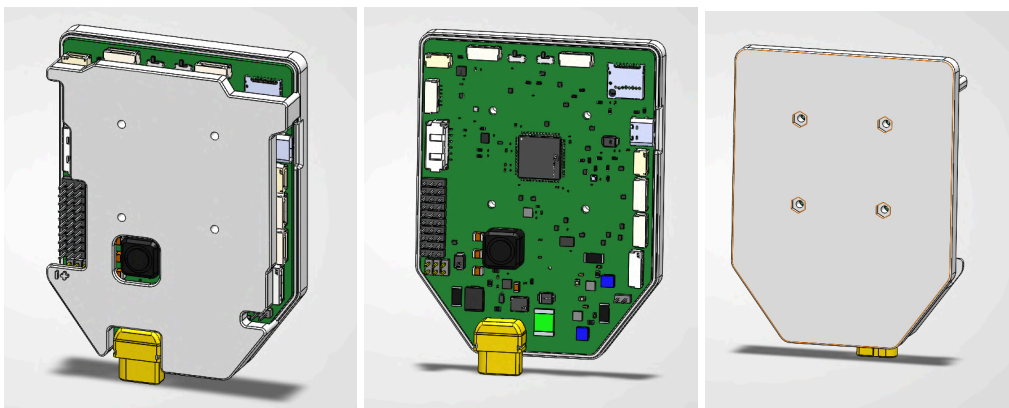


Fig. A-3.1 Board case

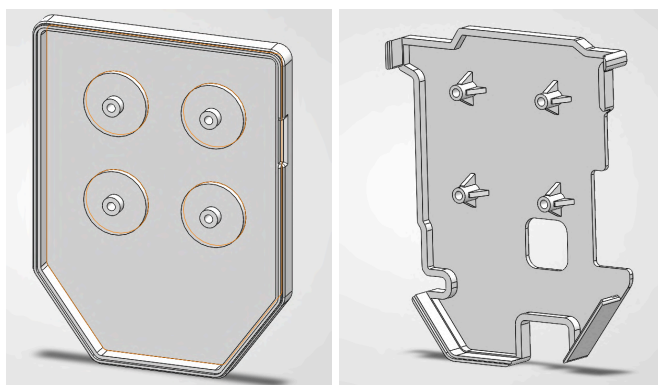


Fig. A-3.2 3D printed case components

A-3.2.1. Drone construction and mission planner configuration

Drone material construction is composed of: Hexsoon EDU 450 frame, a kit of 4 motors + ESCs + propellers, PDB (Power Distribution Board), a GNSS receiver and compass named Here4, a telemetry radio, a radio receiver and a battery 4S.

Here4 was connected to CAN bus, telemetry radio to Telem port and the radio receiver to RCIN port. The battery powers the PDB. This PDB powers using a pixhawk power connector, the flight controller. The order of the motor connectors is shown in the following image.

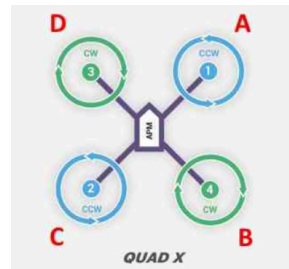


Fig. A-3.3 Motor PWM connector order

Assembling a drone is quite simple. I will focus on the followed configuration for testing:

- **First step:** Connect the telemetry radio to the computer. Go to the SETUP section. In the Sik Radio section, load settings. Both radios must contain the same parameters and net ID. After this configuration, the telemetry radio can be installed within the drone.
- **Second step:** In SETUP, enter in the Failsafe section, enable RTL (Return To Launch); on Config Extended tuning RC7: set ArmDisarm with Air...
- **Third Step:** In SETUP, Test motors. Use this window for motors testing and order checking.
- **Fourth Step:** In the Config section, on the Full parameter list. Apply hexsoon frame parameter file and the following parameters:
 - RTL_ALT = 700 (7 m height during RTL).
 - PILOT_SPEED_UP = 70 (70 cm/s speed).
 - CAN_D1_PROTOCOL = 1
 - CAN_P1_DRIVER = 1
 - GPS1_TYPE = 9 (in case of error 1 which is auto mode).
 - NTF_LED_TYPES = 231
 - BATT_VOLT_PIN = 11 (ADC2_INP11, check pin name in STM32IDE)
 - BATT_CURR_PIN = 10 (ADC1_INP10, check pin name in STM32IDE)
 - BATT_AMP_OFFSET = 0.5 or 1 (depends on the battery)
 - BATT_VOLT_MULT = 12.15 (depends on the battery)
 - FS_THR_VALUE = 975 (radio thrust PWM is 982, so 975 is fine)
 - LOG_BACKEND_TYPE = 0 (disables the use of MicroSD)
 - LOIT_SPEED = 300 (less 300 cm/s during loiter mode)
 - FENCE_TOTAL = 15 (15 geofence waypoints maximum)
- **Fifth Step:** In SETUP, Flight Modes. Loiter and Land were selected for this testing setup.
- **Sixth Step:** In SETUP, Geofence section, enable geofence and Brake or Land were selected.

The mandatory calibrations required to fly the drone are the compass, accelerometer, ESCs, and radio calibrations. Compass calibration was not possible using the BMM150; although the sensor was detected, external noise made it unusable. Only the Here4 compass was calibrated. The accelerometer calibration proceeded without issues, and the ESCs calibration did not require interaction with Mission Planner.



Fig. A-3.4 Mounted drone with designed flight controller and connections

A-3.1.1. Tests

For drone testing, the main flight mode used was Loiter, flying via GNSS positioning. In case of a malfunction or if the drone reaches the mapped boundaries, it will stop at the limit or proceed to land.

3 main tests were conducted. The first test was performed with the drone anchored to a weight to prevent it from flying into non-testing zones. The second test consisted of a free flight within a predefined geofence. The third test involved a geofence and a predefined autonomous mission with 4 different waypoints. Tests 2 and 3 were performed without the anchor.

All tests were successful and completed without any issues. The geofence and mission planning had to be uploaded directly connected to the flight controller, as it was not possible to upload them via telemetry. **Consequently, the designed board is fully functional for the use in drones.**

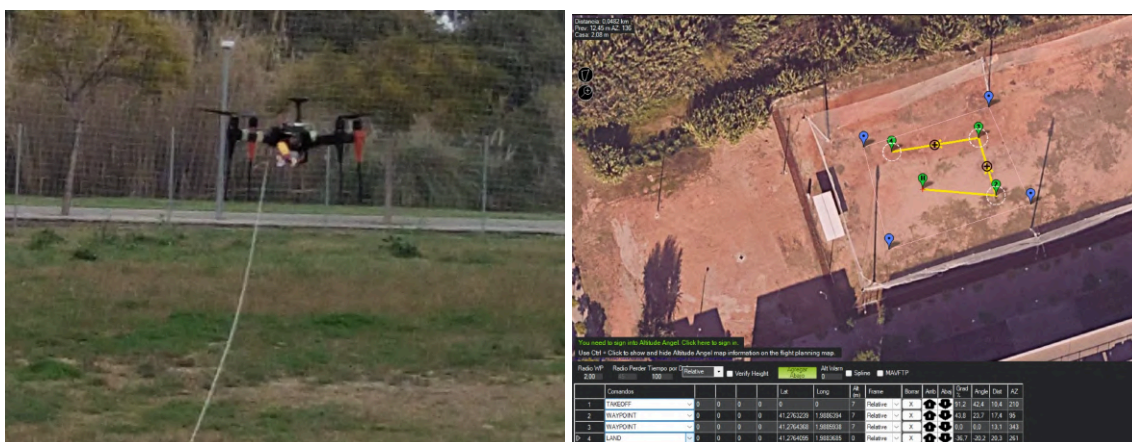


Fig. A-3.5 Left→First test, drone anchored; Right→ Example of performed autonomous mission

During testing, the drone demonstrated a really good horizontal stability; however, while the vertical performance is fine, it could be improved by upgrading the sensor or implementing differential measurements with redundant sensors. Further tests need to be carried out to fully evaluate the board, such as using real time MAVLINK commands via telemetry.

Aside from the flight tests, the board was tested using a power supply at 24 V to simulate a direct 6S battery source. The board operated without issues. When powering the system with a real battery, it is recommended to connect a low-ESR electrolytic capacitor (around 470 μ F) at the power input.

A-3.2. Resolved and remaining Issues

During the development of this project multiple issues appeared, some remarkable ones are:

A-3.2.1. Configuration, clock and ESD protection

- **Configuration and clock:** The board was designed to measure battery voltage and current via the Pixhawk connector, as well as through a direct battery connection. Despite this, only the Pixhawk connector values could be measured due to constraints within the hwdef.dat file. A proper configuration of this file should enable the use of direct voltage measurements. Furthermore, the initial 26 MHz oscillator was not supported by the ArduPilot firmware via software configuration. Substituting the oscillator from 26 MHz to 24 MHz was essential for ArduPilot implementation. Note: This initial 26 MHz clock is supported by the STM32 programme though it is not within ArduPilot.
- **ESD protection:** During development, the board was subject to constant handling and manual manipulation during long testing periods of time. Due to static electricity, the high speed oscillator suffered an electrostatic discharge. This required the replacement of the component. Moreover, the motor outputs and ADCs currently lack external hardware protection (only MCU protection). Implementing TVS diodes is necessary for future versions.

A-3.2.2. Storage and compass

- **MicroSD storage:** Communication between the MCU and the MicroSD card could not be established. Several factors may be responsible, such as crosstalk caused by routing or, most probably, an initialization failure due to DMA priority conflicts within the hwdef.dat file. Due to the limited development time, resolving this problem has been postponed for future hardware versions.
- **Compass/magnetometer:** This sensor is detected and data can be obtained from it; however, it is impossible to calibrate due to a noisy environment. A better isolation is required, such as no copper pour around and being distanced from other planes and signals.

A-3.3. Comparison

To compare the designed flight controller the best option is to take a look the most popular boards in the market:

1. SpeedyBee F405 V4 for FPV drones.
2. Matek H743 Wing V3 for medium complexity drones.
3. Holybro Pixhawk 6C for high end drones.
4. Cube Orange + ADS-B (Newest version) for high end drones.

Important to mention that magnetometer and microSD have to be modified to be operational in the CustomH7 (created board).

Comparison and ranking (1→best, 5→worst) considering that if every implemented component works:

- **MCU, storage and frequency:**
 1. **Cube Orange:** STM32H757ZI at 400 MHz (IC with 2 cores), 2 MB of Flash + STM32F103 at 24 MHz.
 2. **Holybro 6C:** STM32H743 at 480 MHz, 2 MB of Flash + STM32F103 at 72 MHz.
 3. **Matek:** STM32H743VIT6 at 480 MHz, 2 MB of Flash.
 4. **CustomH7:** STM32H743VIT6 at 400 MHz, 2 MB of Flash. (Changing hwdef.dat file it is possible to reach 480 MHz easily).
 5. **SpeedyBee:** STM32F405 at 168 MHz, 1 MB of Flash.
- **IMUs:**
 1. **Cube Orange:** ICM-42688-P, ICM-20948, ICM-20649
 2. **Holybro 6C:** ICM-42688-P and BMI088
 3. **Matek:** ICM-42688-P and ICM-42605
 4. **CustomH7:** ICM-20602 and BMI088 (ICM-20602 is a legacy sensor).
 5. **SpeedyBee:** ICM-42688-P
- **Barometer:**
 1. **Cube Orange:** 2 of MS5611
 2. **Matek:** DPS310
 3. **SpeedyBee:** DPS310
 4. **Holybro 6C:** MS5611
 5. **CustomH7:** BMP388 (Robust but not as accurate as DPS310 or MS5611).
- **Magnetometer:**
 1. **CustomH7:** BMM150 (Best sensor but difficult to isolate). (It needs to be modified to be operational).
 2. **Holybro 6C:** IST-8310 (Good sensor)
 3. **Cube Orange:** ICM-20948 (Inside the IMU, not very effective).
 4. **SpeedyBee:** No
 5. **Matek:** No

- **Case and isolation:**
 1. **Cube Orange:** Aluminum case with special damping foam.
 2. **Holybro 6C:** Plastic or aluminum case with special damping foam.
 3. **CustomH7:** 3D plastic case with the possibility to add foam.
 4. **SpeedyBee:** Integrated silicone dampers
 5. **Matek:** Includes external silicone dampers
- **Max Input voltages:**
 1. **Matek:** 36 Vmax
 2. **CustomH7:** 36 Vmax on battery, 5.4 Vmax on UBEC and Pixhawk connection.
 3. **SpeedyBee:** 25.2 Vmax
 4. **Holybro 6C:** 5.7 Vmax
 5. **Cube Orange:** 5.7 Vmax
- **External BEC:**
 1. **Matek:** 5 V (System), 9 V/12 V switchable (VTX/Cam), and 5 V/6 V/7.2 V switchable (+5 A for servos).
 2. **SpeedyBee:** 5 V at 2 A, 9 V at 2 A for DJI/VTX, and 3.3 V.
 3. **CustomH7:** 5 V at 3 A and 3.3 V at 2 A (3.3 V BEC external dedicated).
 4. **Holybro 6C:** Only on data connections 5 V/3.3 V.
 5. **Cube Orange:** Only on data connections 5 V/3.3 V.
- **Electrical protection:**
 1. **Cube Orange:** Aerospace grade. Full signal isolation (via buffers), triple power redundancy, and IEC 61000-4-2 ESD certification.
 2. **CustomH7:** Contains dedicated low capacitance TVS diodes on data lines (full ESD protection on external communication interfaces), main battery input fuse, reverse polarity protection, and industrial self-protecting regulators (Texas Instrument) with Schottky diode. Lack of tolerance against short-circuits, it needs active protection and TVS on PWM-ADCs lines.
 3. **Holybro 6C:** Offers redundant power architecture, active overcurrent protection on individual peripheral ports (limits current per port) and common TVS protection on peripherals.
 4. **Matek:** Contains a few common TVS on input lines, reverse polarity protection, and common regulators.
 5. **SpeedyBee:** Basic protection.
- **External buses (not debug and not radio connectors):**
 1. **Holybro 6C:** 1-USB-C, 2-CAN, 1-I2C, 1-SPI, 5-ADCs, 3-Telem (UART+ Flow control), 2-GPS (UART + I2C + Flow control), 1-DSM.
 2. **Cube Orange:** 2-USB (via pins and microUSB), 2-CAN, 2-I2C, 5-ADCs, 2-Telem (UART + Flow control), 2-GPS, 1-DSM, 1-UART.
 3. **Matek:** 1-USB-C, 1-CAN, 7-UART, 2-I2C, 1-shared SPI, 6-ADCs.
 4. **CustomH7:** 1-USB-C, 1-CAN, 2-Telem (UART), 2-GPS (UART + I2C), 1-SPI, 1-I2C, 3-ADCs.
 5. **SpeedyBee:** 1-USB-C, 6-UART, 1-I2C, 2-ADCs.

- **PWM for motors:**
 1. **Holybro 6C:** 16 channels (full protected with resistances)
 2. **Cube Orange:** 14 channels (full protected with resistances)
 3. **Matek:** 13 channels (with resistances)
 4. **SpeedyBee:** 8 channels (no protection)
 5. **CustomH7:** 8 channels (no protection)
- **FPV OSD:**
 1. **Matek:** AT7456E IC
 2. **SpeedyBee:** AT7456E IC
 3. **CustomH7:** No
 4. **Holybro 6C:** No
 5. **Cube Orange:** No
- **External storage: No difference**
 - **SpeedyBee:** MicroSD
 - **Matek:** MicroSD
 - **CustomH7:** MicroSD (It needs to be modified to be operational).
 - **Holybro 6C:** MicroSD
 - **Cube Orange:** MicroSD
- **PCB layers and size (considering a cube form):**
 - **SpeedyBee:** 4 layers PCB, $41.6 \times 39.4 \times 7.8 \text{ mm}$ or 12784 mm^3
 - **Matek:** 2 boards of high density of layers $54 \times 36 \times 13 \text{ mm}$ or 25272 mm^3
 - **CustomH7:** 4 layers, $104 \times 79 \times 18.4 \text{ mm}$ or 151174 mm^3
 - **Holybro 6C:** high density of layers, $84.8 \times 44 \times 12.5 \text{ mm}$ or 46750 mm^3
 - **Cube Orange:** 2 interconnected pieces, high density of layers, with size approx $94.5 \times 44.3 \times 38.4 \text{ mm}$ or 160755 mm^3
- **Compatible firmware:**
 1. **Holybro 6C:** PX4, ArduPilot
 2. **Cube Orange:** PX4, ArduPilot
 3. **Matek:** ArduPilot, iNAV, Betaflight
 4. **CustomH7:** ArduPilot already implemented, port-ready for PX4
 5. **SpeedyBee:** Betaflight, iNAV
- **Weight:**
 1. **SpeedyBee:** 10.5 g
 2. **Matek:** 30 g
 3. **Holybro 6C:** 35.6 g - 59.3 g
 4. **Cube Orange:** 73 g
 5. **CustomH7:** printed case 37.9 g + PCB 35.5 g + screws 4 g = 77.4 g
- **Operational temperature:**
 1. **Cube Orange:** -10°C to 55°C (in reality it is a wider range) (heaters implemented)
 2. **CustomH7:** -25°C to 85°C (approx considering components and PCB materials) (USB, MicroSd and JST connectors are critical components) Test should be required.

3. **Holybro 6C:** -25°C to 85°C (Same level as CustomH7)
 4. **Matek:** -20°C to 80°C
 5. **SpeedyBee:** -10°C to 60°C
- **Additional considerations:**
 1. **Cube Orange:** Contains DSM, switchable PWM voltage, ADS-B (uAvionix), heater against cold temperatures, redundant power connectors and aerospace grade certifications.
 2. **Holybro 6C:** Contains DSM, switchable PWM voltage, redundant power connectors.
 3. **CustomH7:** 3 power connection (Pixhawk power, UBEC 5V and direct from battery) and dedicated programmable oscillators: MEMS against vibrations and TCXO against temperature deviations.
 4. **Matek:** Designed for rc-planes with high power servos and contains a buzzer.
 5. **SpeedyBee:** Bluetooth module integrated and 4 level LED battery indicator.
 - **Cost:**
 1. **SpeedyBee:** 30 - 60 €
 2. **Matek:** 90 - 120 €
 3. **Custom:** 180 € prototype (cost can be reduced drastically to even 65 - 70 € the production).
 4. **Holybro 6C:** 140 - 200 €
 5. **Cube Orange:** 350 - 450 €
 - **Use of a drone standard:**
 1. **Holybro 6C:** Pixhawk Standard specifically FMUv6
 2. **Cube Orange:** Pixhawk Standard modified FMUv6
 3. **CustomH7:** Pixhawk Connectors/Bus
 4. **SpeedyBee:** No
 5. **Matek:** No
 - **Plug and play:**
 1. **Cube Orange:** Yes
 2. **Holybro 6C:** Yes
 3. **CustomH7:** Yes
 4. **SpeedyBee:** No, it needs to be soldered.
 5. **Matek:** No, it needs to be soldered.

The **CustomH7** prototype is an open source bridge between professional and hobbyist hardware. Integrating **Pixhawk standard connectivity** and timing components (**MEMS and TCXO**) typically found in specialized boards.

Developed in just **5 months**, this project demonstrates that it is possible to build a powerful, standardized, and open-source flight controller.

A-3.4. Improvements

There are multiple possible improvements, these have been divided into 3 different types:

A-3.4.1. Easy short term improvements

Short term improvements require short development time and can be implemented and made operational quickly.

- Sensors: The first change that I would consider drastically beneficial and easy to implement, is replacing the IMU ICM-20602 with the newer ICM-42688-P. Additionally adding a second barometer to the same bus I2C or changing the current barometer to the MS561 or DPS310 would improve the vertical stability.
- Add low value resistors to the PWM lines.
- Add ESD protection with industrial grade TVS diodes to the PWM and ADCs lines. Internal buses can be considered too. Changing current TVS diodes with a lower break voltage is considered.
- Add mechanical protection and damping foam.
- Check and replace any non-industrial grade IC implemented in the board.
- Change the actual programmable oscillators with a better and more accurate option. If possible MEMS oscillators.
- Add a buzzer. This can be implemented using a simple transistor circuit or via bus connectivity.
- Make the MicroSD connection operative.

A-3.4.2. Medium long term improvements

Medium term improvements require significant modifications in the prototype design.

- Increase the number of PWM signals. Maybe adding 2 PWM would not require any modification; however, adding 4 or more PWM lines would require changing the MCU to a larger package, instead of LQFP-100, using a LQFP-144 or LQFP-176.
- Increasing the number of ADCs is questionable. They are required to measure battery voltage and current. Nowadays applications like peripheral analog evaluation are not used anymore, it is an old practice. In case of adding more ADCs, the use of a new MCU package is required.
- The prototype is designed with a 4-layer PCB. Increasing the layer number to 6 or 8 layers would significantly reduce the board area, enabling a smaller flight controller. Despite this, several aspects must be considered. The layer stack-up must be symmetrical to avoid bending moments and Z-axis displacement during thermal cycling. Additionally, power and control sections must be clearly separated and isolated to minimize crosstalk and EMI.
- The use of laser, buried or blind vias would reduce the board area, though it isn't really recommended; the manufacturing cost will increase drastically.
- Mounting components on both the top and bottom sides of the PCB would reduce both the area and volume of the flight controller. This approach involves only a minimal increase in manufacturing cost.

- Add IC memory with an SPI bus.
- Change the MCU or its package. Changing the package may expose additional pins, such as VDD33USB, which it is used for USB functionality and detection.
- Implement flow control protocol on Telem ports, in other words, RTC and CTS lines. This will require additional UART/USART connections.

A-3.4.3. Complex improvements

Implementing these improvements would require significant modifications or specialized technical knowledge. These are ordered from easiest to most difficult.

1. Development of a protective case with integrated damping structures, manufactured with plastic injection or CNC machining.
2. Implement active protection on data and power lines against short circuits.
3. Implement power redundancy with parallel power lines and additional protection measures. Thermal management with vias has an important role.
4. Proper board conditioning for the magnetometer to minimize noise interference.
5. Implement a GNSS receiver or a radio receiver.
6. Implementation of a main MCU as the FMU and a secondary MCU for I/O tasks. The main FMU would run ArduPilot, while the secondary processor would require custom firmware developed in C or a reduced ChibiOS firmware. The communication between both MCUs must be on SPI or via MAVLINK UART protocol.

A-3.5. Cost evaluation

The cost of PCB fabrication and PCBA is dependent on manufacturer pricing, production volume, tariffs, shipping fees, customs duties, and geopolitical factors.

For instance, the initial price for 2 prototype boards, a stencil and 3 non-mounted boards was 227.29 €. However, due to tariffs and international trade regulations, the final cost reached 344.04 € (a 51.36% increase with respect to 227.29 €).

A breakdown of this cost shows: 5.50 € for X-ray inspection and 7 € for stencil creation, feeder loading fees (from 55 € to 70 €, unique per component type) and the component cost for two boards, which was around 85 €.

If the production volume increases to 10 mounted boards, the total price at JLCPCB would be 500€ approx. The reduction in cost is due to volume discounts on components and the distribution of fixed fees over a larger production. After including customs and tariffs (typically 30-40% price increase), the final estimated cost would be around 650 - 700 €. Dropping the cost per unit around 65 - 70 €.

It is important to mention that the cost of a 3D printed case and the mounting screws would be less than 2 - 3 €, slightly increasing the final price.

ACRONYMS

A

- **AC:** Alternating Current.
- **ADC:** Analog-to-Digital Converter.
- **ANSI:** American National Standards Institute.
- **AOI:** Automated Optical Inspection.
- **ARM:** Advanced RISC Machines.
- **ASIC:** Application-Specific Integrated Circuit.

B

- **BEC:** Battery Elimination Circuit.
- **BIOS:** Basic Input/Output System.
- **BOM:** Bill of Materials.
- **BPF:** Band Pass Filter.

C

- **CAF:** Conductive Anodic Filament.
- **CAN:** Controller Area Network.
- **CDM:** Charged Device Model.
- **CIPO:** Controller In Peripheral Out.
- **CNC:** Computer Numerical Control.
- **COTS:** Commercial Off-The-Shelf.
- **CPL:** Component Placement.
- **CRC:** Cyclic Redundancy Check.
- **CRSF:** Crossfire Serial Protocol.
- **CS:** Chip Select.
- **CSMA/CR:** Carrier Sense Multiple Access with Collision Resolution.
- **CTE:** Coefficient of Thermal Expansion.
- **CTS:** Clear To Send.
- **CVCM:** Collected Volatile Condensable Material.

D

- **DC:** Direct Current.
- **DFM:** Design for Manufacturability.
- **DFU:** Device Firmware Upgrade.
- **DIN:** Data IN
- **DK:** Dielectric Constant.
- **DMA:** Direct Memory Access.
- **DoD:** Department of Defense.
- **DoF:** Degrees of Freedom.

- **DSM:** Digital Spectrum Modulation

E

- **ECC:** Error Correction Code.
- **ECSS:** European Cooperation for Space Standardization.
- **EEE:** Electrical, Electronic, and Electromechanical.
- **EEPROM:** Electrically Erasable Programmable Read-Only Memory.
- **ELDRS:** Enhanced Low Dose Rate Sensitivity.
- **EMI:** Electromagnetic Interference.
- **ENIG:** Electroless Nickel Immersion Gold.
- **EOL:** End Of Lifetime / End Of Life.
- **EOP:** End Of Packet.
- **ESC:** Electronic Speed Controller.
- **ESD:** Electrostatic Discharge.
- **ESR:** Equivalent Series Resistance.
- **EXTI:** External Interrupt.

F

- **FC:** Flight Controller.
- **FCT:** Functional Circuit Test.
- **FMU:** Flight Management Unit.
- **FOD:** Foreign Object Debris.
- **FPGA:** Field-Programmable Gate Array.
- **FPU:** Floating Point Unit.
- **FPV:** First-Person View.
- **FR-4:** Flame Retardant 4.
- **FRAM:** Ferroelectric Random Access Memory.

G

- **GCR:** Galactic Cosmic Ray.
- **GCS:** Ground Control Station.
- **GND:** Ground.
- **GNSS:** Global Navigation Satellite System.
- **GPIO:** General Purpose Input/Output.
- **GPS:** Global Positioning System.
- **GPSDO:** GPS Disciplined Oscillator.

H

- **HAL:** Hardware Abstraction Layer.
- **HASL:** Hot Air Solder Leveling.
- **HBM:** Human Body Model.
- **HDI:** High Density Interconnect.

- **HPF:** High Pass Filter.
- **HSE:** High Speed External oscillator.

I

- **I/O:** Input/Output.
- **IC:** Integrated Circuit.
- **ICT:** In-Circuit Testing.
- **IDT:** Interdigital Transducers.
- **IMU:** Inertial Measurement Unit.
- **IPC:** Association Connecting Electronics Industries.
- **IR:** Infrared.
- **IRPP:** Integral Rectangular Parallelepiped.
- **IWDG:** Independent Watchdog.

J

- **JTAG:** Joint Test Action Group.

L

- **LCL:** Latching Current Limiter.
- **LDO:** Low Dropout (Regulator).
- **LEO:** Low Earth Orbit.
- **LET:** Linear Energy Transfer.
- **LGA:** Land Grid Array.
- **LPF:** Low Pass Filter.
- **LSB:** Least Significant Bit.
- **LSE:** Low Speed External oscillator.

M

- **MBU:** Multiple-Bit Upset.
- **MCU:** Microcontroller Unit or Multiple-Cell Upset .
- **MEMS:** Micro-Electro-Mechanical System.
- **MIL-STD:** Military Standard (Defense Standard).
- **MISO:** Master Input Slave Output.
- **MPN:** Manufacturer Part Number.
- **MPU:** Microprocessor Unit or Memory Protection Unit.
- **MRO:** Maintenance, Repair, and Overhaul.
- **MSP:** MultiWii Serial Protocol.

N

- **NEMA:** National Electrical Manufacturers Association.
- **NHB:** NASA Handbook.

- **NPTH:** Non-Plated Through Hole.
- **NRZI-S:** Non-Return-To-Zero Inverted Space.

O

- **OBC:** On-Board Computer.
- **OCXO:** Oven Controlled Crystal Oscillator.
- **OSD:** On-Screen Display.
- **OSP:** Organic Solderability Preservative.
- **OTG:** On-The-Go.

P

- **PCB:** Printed Circuit Board.
- **PCBA:** Printed Circuit Board Assembly.
- **PDB:** Power Distribution Board.
- **PIND:** Particle Impact Noise Detection.
- **PLL:** Phase-Locked Loop.
- **PnP:** Pick and Place.
- **PPM:** Pulse Position Modulation.
- **PSS:** Procedures, Specifications, and Standards.
- **PTFE:** Polytetrafluoroethylene (Teflon).
- **PTH:** Plated Through Hole.
- **PWM:** Pulse Width Modulation.

Q

- **QML:** Qualified Manufacturer List.
- **QPL:** Qualified Part List.

R

- **RAM:** Random Access Memory.
- **RCIN:** Radio Control Input.
- **RDM:** Radiation Design Margin.
- **RF:** Radio Frequency.
- **RH:** Relative Humidity.
- **RHA:** Radiation Hardness Assurance.
- **RSSI:** Received Signal Strength Indicator
- **RTC:** Real Time Clock.
- **RTOS:** Real-Time Operating System.
- **RTL:** Return To Launch.
- **RTS:** Request To Send.

S

- **SBUS:** Serial Bus.
- **SDG:** Sustainable Development Goals.
- **SDI:** Serial Data Input.
- **SDMMC:** Secure Digital MultiMediaCard.
- **SDO:** Serial Data Output.
- **SEB:** Single-Event Burnout.
- **SED:** Single-Event Disturb.
- **SEDR:** Single-Event Dielectric Rupture.
- **SEE:** Single-Event Effect.
- **SEFI:** Single-Event Functional Interrupt.
- **SEGR:** Single-Event Gate Rupture.
- **SEHE:** Single-Event Hard Error.
- **SEL:** Single-Event Latchup.
- **SET:** Single-Event Transient.
- **SEU:** Single-Event Upset.
- **SMD:** Surface Mount Device.
- **SMT:** Surface Mount Technology.
- **SMU:** Single-Word Multiple-Bit Upset.
- **SOF:** Start Of Frame.
- **SPE:** Solar Particle Events.
- **SPF:** Single Point Failure.
- **SPI:** Serial Peripheral Interface o Solder Paste Inspection.
- **SPXO:** Simple Package Crystal Oscillator.
- **SS:** Slave Select.
- **SSO:** Sun-Synchronous Orbit.
- **STM:** STMicroelectronics.
- **SWD:** Serial Wire Debug.

T

- **TCXO:** Temperature Compensated Crystal Oscillator.
- **TFM:** Master's Thesis Project.
- **Tg:** Glass Transition Temperature.
- **TID:** Total Ionizing Dose.
- **TML:** Total Mass Loss.
- **TNID:** Total Non-Ionizing Dose (Displacement Damage).
- **TRB:** Technology Review Board.
- **TVS:** Transient Voltage Suppression.

U

- **UART:** Universal Asynchronous Receiver-Transmitter.
- **UAV:** Unmanned Aerial Vehicle.

- **UBEC:** Universal Battery Elimination Circuit.
- **UEFI:** Unified Extensible Firmware Interface.
- **USB FS:** Universal Serial Bus Full Speed.
- **UV:** Ultraviolet.

V

- **VCXO:** Voltage Controlled Crystal Oscillator.
- **VTX:** Video Transmitter.

W

- **WWDG:** Window Watchdog.

X

- **XO:** Crystal Oscillator.

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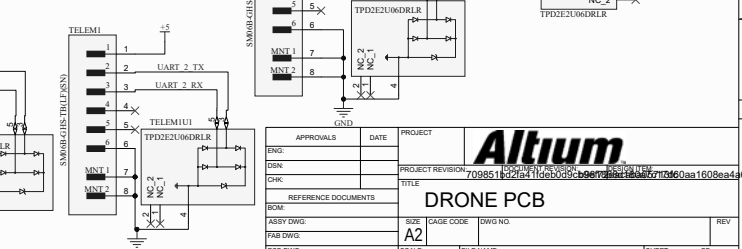
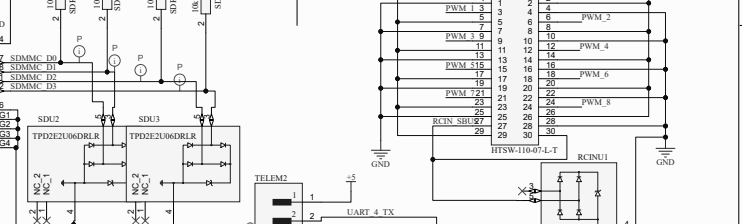
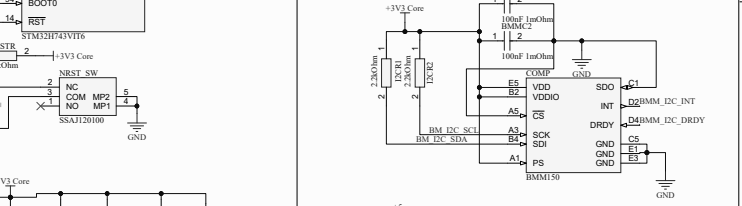
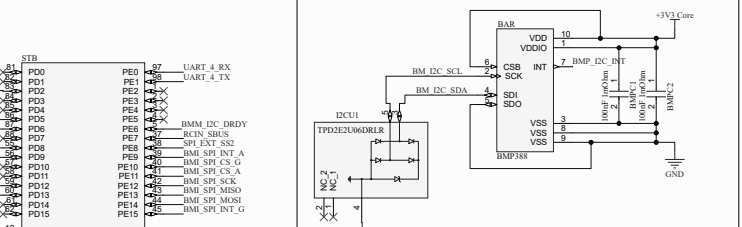
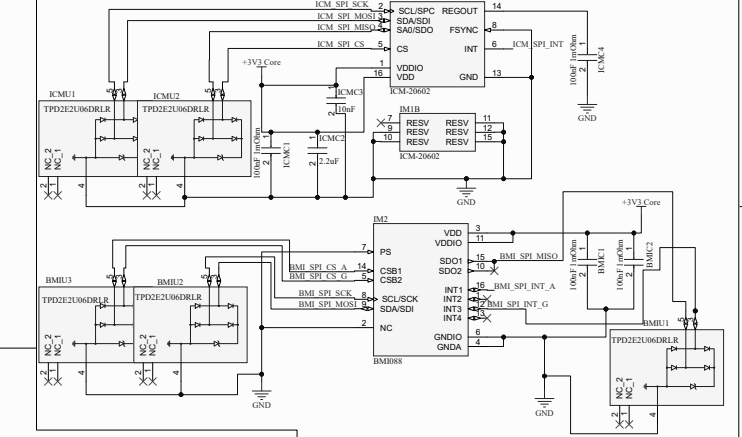
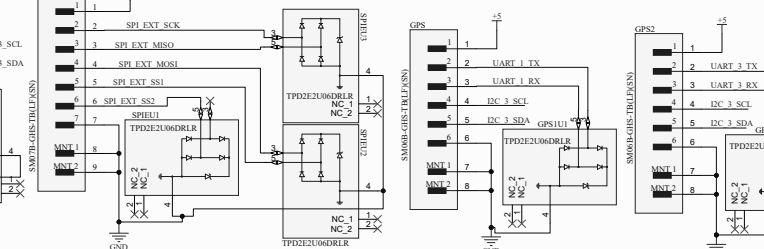
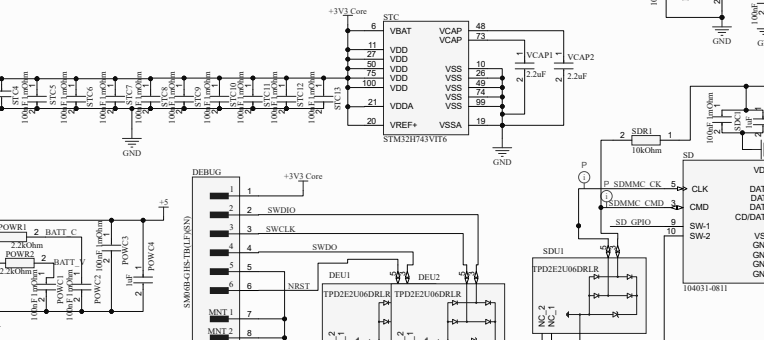
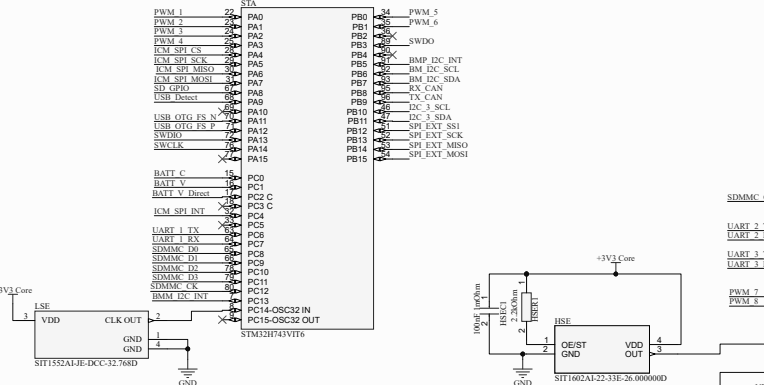
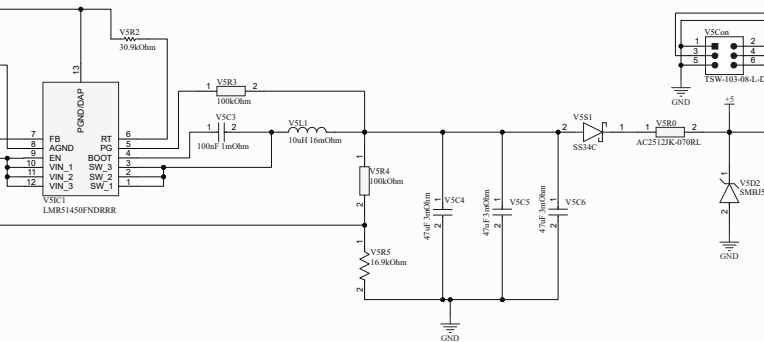
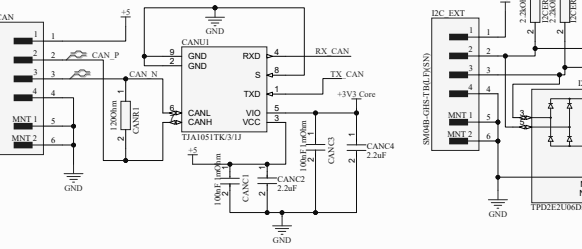
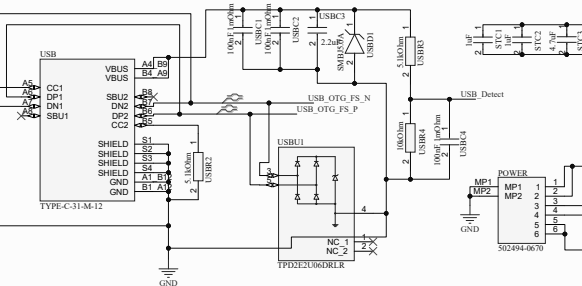
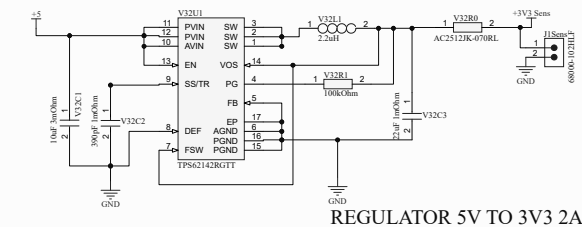
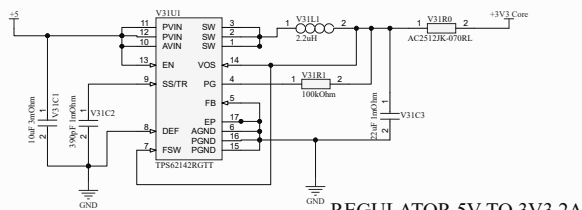
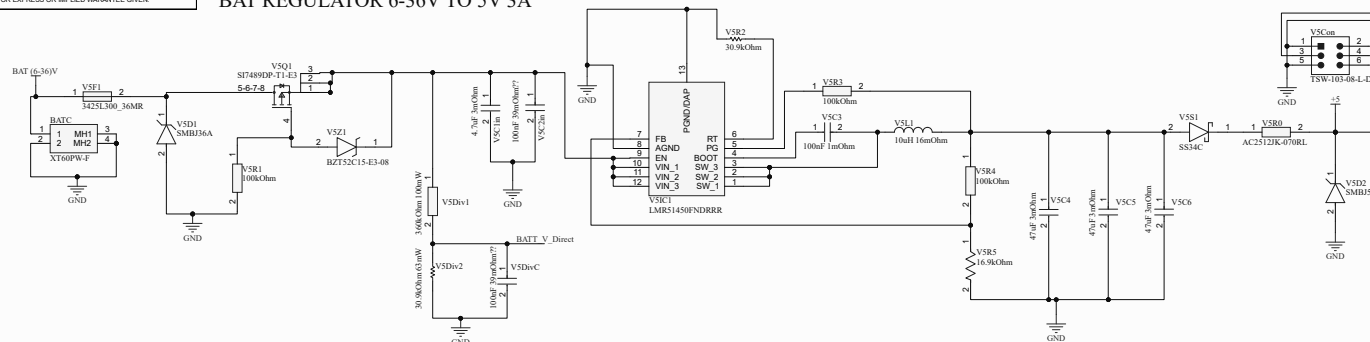
ADDITIONAL DOCUMENTATION

PIN MAPPING: Altium Schematics to real MCU pins

GROUP	NET	PIN STM32
USB	USB_OTG_FS_N	PA11
	USB_OTG_FS_P	PA12
	USB_Detect (from VBUS)	PA9
	VDD_33_USB	No pin in VIT version
	VDD_50_USB	No pin in VIT version
CAN	RX CAN	PB8 (FDCAN1_RX)
	TX CAN	PB9 (FDCAN1_TX)
POWER MEASUREMENT	BATT Current (measure current)	PC0 (ADC1_INP10)
	BATT Voltage (measure voltage)	PC1 (ADC1_INP11)
	BATT Voltage 2 (measure voltage 2) (NOT USED)	PC2 (ADC3_INP0)
ICM-20602 (IMU)	ICM_SPI_SCK	PA5 (SPI1_SCK)
	ICM_SPI_MOSI	PA7 (SPI1_MOSI)
	ICM_SPI_MISO	PA6 (SPI1_MISO)
	ICM_SPI_CS	PA4
	ICM_SPI_INT (interruptor)	PC4
BMI088 (IMU)	BMI_SPI_SCK	PE12 (SPI4_SCK)
	BMI_SPI_MOSI	PE14 (SPI4_MOSI)
	BMI_SPI_MISO	PE13 (SPI4_MISO)
	BMI_SPI_CS_A (accelerometer)	PE11
	BMI_SPI_CS_G (gyroscope)	PE10
	BMI_SPI_INT_A (accelerometer interruptor)	PE9
	BMI_SPI_INT_G (gyroscope interruptor)	PE15
BMP388 (barometer)	BMP_I2C_SCL	PB6 (I2C1_SCL)
	BMP_I2C_SDA	PB7 (I2C1_SDA)
	BMP_I2C_INT (interruptor)	PB5
BMM150 (magnetometer)	BMM_I2C_SCL	PB6 (I2C1_SCL)
	BMM_I2C_SDA	PB7 (I2C1_SDA)
	BMM_I2C_INT	PC13
	BMM_I2C_DRDY	PE6

DEBUG	SWO	PB3
	SWDIO	PA13
	SWCLK	PA14
MicroSD	SDMMC_D0	PC8
	SDMMC_D1	PC9
	SDMMC_D2	PC10
	SDMMC_D3	PC11
	SDMMC_CK	PC12
	SDMMC_CMD	PD2
	SD_GPIO (to detect SD)	PA8
MOTORS/SERVOS, AND COMM BUSES	PWM_1	PA0 (TIM2_CH1)
	PWM_2	PA1 (TIM2_CH2)
	PWM_3	PA2 (TIM2_CH3)
	PWM_4	PA3 (TIM2_CH4)
	PWM_5	PB0 (TIM3_CH3)
	PWM_6	PB1 (TIM3_CH4)
	PWM_7	PD12 (TIM4_CH1)
	PWM_8	PD13 (TIM4_CH2)
	RCIN-SBUS	PE7 (UART7_RX)
EXTERNAL CONNECTORS	I2C_3_SCL	PB10 (I2C2_SCL)
	I2C_3_SDA	PB11 (I2C2_SDA)
	SPI_EXT_SCK	PB13 (SPI2_SCK)
	SPI_EXT_MISO	PB14 (SPI2_MISO)
	SPI_EXT_MOSI	PB15 (SPI2_MOSI)
	SPI_EXT_SS1 or (CS_1)	PE8
	SPI_EXT_SS2 or (CS_2)	PB12
	UART_1_RX	PC7 (USART6_RX)
	UART_1_TX	PC6 (USART6_TX)
	UART_2_RX	PD6 (USART2_RX)
	UART_2_TX	PD5 (USART2_TX)
	UART_3_RX	PD9 (USART3_RX)
	UART_3_TX	PD8 (USART3_TX)
	UART_4_RX	PE0 (UART8_RX)

	UART_4_TX	PE1 (UART8_TX)
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REVISION

DESCRIPTION

DATE

APPROVED

Transmission Line Structure Table

Impedance Id	Transmission Line	Target Impedance	Calculated Impedance	Trace layer	Wide Trace Width	Narrow Trace Width	Gap	Reference layers	Substack	Clearance	Target Tolerance
1	Edge-Coupled Coated Microstrip	90	89.32	Top Layer	0.29mm	0.29mm	0.20mm	Int1 (GND)	Board Layer Stack	0.13mm	10%
2	Edge-Coupled Coated Microstrip	120	119.12	Top Layer	0.18mm	0.18mm	0.36mm	Int1 (GND)	Board Layer Stack	0.13mm	10%
3	Coated Microstrip	50	49.93	Top Layer	0.35mm	0.35mm		Int1 (GND)	Board Layer Stack	0.13mm	10%
4	Edge-Coupled Coated Microstrip	90	89.32	Bottom Layer	0.29mm	0.29mm	0.20mm	Int2 (PWR)	Board Layer Stack	0.13mm	10%
5	Edge-Coupled Coated Microstrip	120	119.12	Bottom Layer	0.18mm	0.18mm	0.36mm	Int2 (PWR)	Board Layer Stack	0.13mm	10%
6	Coated Microstrip	50	49.93	Bottom Layer	0.35mm	0.35mm		Int2 (PWR)	Board Layer Stack	0.13mm	10%

Top Layer (Scale 1:1)

Int1 (GND) (Scale 1:1)

Int2 (PWR) (Scale 1:1)

Bottom Layer (Scale 1:1)

Top Layer (Scale 1:1)

Drill Drawing View (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Tolerance
◇	374	0.30mm	Plated	
☆	2	0.60mm	Non-Plated	
⊕	4	0.60mm	Plated	
▣	6	0.71mm	Plated	
⊗	8	1.02mm	Plated	
□	31	1.10mm	Plated	
▽	2	1.80mm	Plated	
▽	4	2.60mm	Plated	
○	2	2.70mm	Plated	
433 Total				

APPROVALS

DATE

PROJECT

ENG:

DSN:

CHK:

REFERENCE DOCUMENTS

BOM:

ASSY DWG:

FAB DWG:

PCB DWG:

DATE

PROJECT

PROJECT REVISION

TITLE

SIZE

CAGE CODE

DWG NO.

REV

Altium

VersionControl

ProjectReference

27a41e6b9b0

ItemR

DOC NO SCH_DWG

ItemRevision

2

OF 2